Euler's Cabinet of Curiosities

Physics fidelity report: how each of the 21 exhibits is simulated, and how it was tested against theory

Bullet Physics Playground (bpp), demo/WyomingWill

Every exhibit run afresh on 15 October 2026, and run again after the fixes described in section 2

21 exhibits	**272** checks against theory, tabulated
15 agree	**6** agree, with a numerical or open caveat

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1. Summary

Euler's Cabinet of Curiosities is a gallery of physics exhibits in the Bullet Physics Playground. Its point is to be faithful to the physics: each exhibit should do what the real thing does, for the reason the real thing does it, and say how it knows. This report collects, exhibit by exhibit, how each one is simulated and how it was tested against theory, with the numbers.

Every simulated number here comes from a run made for this report, on 15 October 2026, or from the exhibit's README, and the exhibit sections say which. Most theory values are worked out by the exhibits themselves from their own parameters; where a theory value, a fit or a comparison was worked out for this report instead (for example the Dzhanibekov flip period, the gyroscope's formula values, the fits to Galileo's ramp), the section says so. Each exhibit was run headless, from its own folder, with the start pause off (HOLD_S=0, which changes nothing in the physics; see section 3), and its console output was read against its README and its code.

A first pass of this report found several exhibits with real problems (section 2.1 lists them) and many places where the documentation said something the code didn't do. Those were fixed the same day, and every exhibit that changed was run again: the exhibit sections describe the gallery as it now is, and section 2 lists what was found, what was done about it, and what is still open.

Wing A: Turning

Exhibit	Checks	Headline result	Status
Gömböc variety	9	One stable point: Gömböc-C ends upright in 11 of 11 random drops; righting from a headstand 8.75 and 13.03 s against the earlier study's 8.8 and 13.2 s. Sloan shapes 0 of 16 upright in one run, 3 of 9 in a seeded re-run (study: 9%; small samples, and the bodies knock into each other).	**Agrees**
Bille and the polyhedra	11	Bille ends on its one stable face in 27 of 27 drops and 6 of 6 set-downs, by the predicted face paths (a quick crossing of a face is now logged, in brackets).	**Agrees**
Galileo's ramp	10	Finish times within -0.013% of sqrt(2d(1+k)/(g sin θ)); distances in equal times 1 : 2.998 : 4.997 (odd numbers; fitted for this report); first-order convergence in step size.	**Agrees**
The double cone	9	Rolls up or down exactly as the rail geometry predicts on all three tracks; centre's drop within 1 μm of the exact geometry; energy loss 0.1-0.3% (numerical).	**Agrees**
The oloid and the sphericon	8	Oloid's centre stays within 4.2426-4.5000 cm, the roller and sphericon level at 4.2426 cm, exactly as theory says; side-to-side weaves 3.6, 5.1 and 5.8 cm, now stated correctly.	**Agrees**
Euler's disk	12	Precession rate within 0.08-0.18% of theory from 25° down to 1°; with losses off, energy steady to the eighth figure. Its finite-time end follows its rolling-resistance law (now said), not Moffatt's air.	**Agrees**
The rattleback	11	Spun the wrong way it reverses (0.67 s), the right way never; the reference model's growth rates (8.9/s and 1.7/s at 1 turn a second) and the rolling-resistance window now stated as measured. New README.	**Agrees**
The spinning eggs	21	The left egg rises at 0.30 s and stands at 0.68 s, then rolls and leans as spin friction slows it, back on its side at 24.40 s, at every step rate from 2400 to 19,200; an exact rolling integration gives 11.97° at 3 s and 24.39 s (the exhibit: 11.99°, 24.40 s). Rise from 60° to 30° within 9% of Moffatt and Shimomura (2.5% started in balance); spin falls at 0.934 turns/s² against 0.928 from Jellett's quantity; energy budget closes exactly.	**Agrees**
The tippe top	11	Inversion criterion right for all three tops; centre of mass rises 1.057 to 2.545 cm as theory says; stem threshold about 80 rad/s, within the sleeping-top rule's bounds. Open: the green top's spin collapses after 40 s without rolling resistance.	**Agrees, with caveats**

Exhibit	Checks	Headline result	Status
The Dzhanibekov effect	12	Flip period 3.3129 s against 3.3131 s from the exact elliptic-function solution (-0.01%), 36 flips in 120 s. Energy and angular momentum are held by projection, and the README now says so. New README.	**Agrees**
The gyroscope	13	Precession 2.58 rad/s against m g r/(I ω) = 2.59 at the spin it has, and to 0.0-0.5% with the tilt term as it sags; README now right (2.56 rad/s at 100 turns/s). Rocked 1.5° it slides further than aimed for, cause not separated.	**Agrees, with caveats**
The bicycle wheel	8	Stool's turning matches angular-momentum conservation exactly with friction off; L kept to 0.093% over 120 s. A guard now reports and recovers from a NaN blow-up, whose cause (seen once) is still unknown.	**Agrees, with caveats**

Wing B: Swinging

Exhibit	Checks	Headline result	Status
Newton's cradle	13	First strike the same at every step rate from 1200 to 9600 steps/s (was: a third of the energy lost at 9600); momentum to 0.00%; ball 4 gets 13.6% (Herrmann & Seitz: ~12%); the row settles over minutes, as real cradles do.	**Agrees**
Huygens' clocks	8	All three starts lock in anti-phase at 180.0°, at 0.9985-0.9988 Hz against 0.9984 Hz from theory; at 9600 steps/s the same to 1e-4 Hz (one start, 45 s).	**Agrees**
Kapitza's pendulum	9	Stability threshold 0.783 cm bracketed (0.76 falls, 0.80 stands); rocking period 1.133 s against 1.135 s for its 7.6° swing (-0.2%), now at a converged 19,200 steps/s (was 4800, 1.7% short).	**Agrees**
The double pendulum	16	Energy to 2e-11; half-step twin 64 fm off at 1 s; Lyapunov 3.42/s; normal modes to 0.000% and 0.135%; Floquet window 0.728-1.356 cm. The throw-over time of the too-hard chain (not the fact) isn't converged, and the README says so.	**Agrees, with caveats**
Pendulums with a twist	12	No flip below 70.53°; at μ = 3 the hidden integral stays 0.7500000000 (wanders at 2.8 and 3.2); spring's growth 0.526/s against 0.527 (Floquet).	**Agrees**
The chain fountain	35	Now starts from a poured heap of ball chain of Biggins and Warner's size: h2/h1 = 0.140 against 0.12-0.14 measured, matching every link simulated within the pour-to-pour spread (an independent bead-chain model agrees), at 14 ms a frame at 1/8 speed. The old laid coil (H) stands at 0.06-0.07: the pile's arrangement was what was missing.	**Agrees, with caveats**
The falling Slinky	13	Bottom stays put 0.2038 s against sqrt(M/3k) = 0.2018 s: 0.6% from having 86 separate coils (an exact 86-coil chain waits 0.2030 s) and 0.4% from Bullet's step, both now explained in the README.	**Agrees**
The Brazil-nut effect	14	Rebuilt in real glass (bounce 0.95, friction 0.1), by a validated bead model ported into bpp bit for bit. Glued-bead walls give the right rolls (up the middle 5.9-7.5, down the walls 11.7-13.8 mm/s) and the solid big bead rises 7.6-10.6 s into the hard shaking and stays up; smooth glass walls reverse the rolls and it stays down, in all four pours. Floor push at Γ = 0.8: 0.19 of the weight against 0.20 rigid. A glass lid closes the box, as in real cells (open, beads flew to 67-84 cm over 44 cm walls). Beads 1/100 as stiff as glass (results unchanged 10x stiffer); about 20 ms a frame against 16.7 at 1/4 speed on the test machine.	**Agrees, with caveats**
The solar system	17	Planets within 5 km of JPL's DE421 after 10 years, the Moon 39 km; Mercury's perihelion 42.981″/century against 42.981″; rotation checks hold. The checks now stop at their exact dates, and give the same lines at any rate.	**Agrees**

2. What was found, and what was done

A first pass of this report, earlier on 15 October 2026, ran every exhibit afresh and read its output against its README and code. What it found was fixed the same day, and the exhibits that changed were run again; the numbers in sections 4 and 5 are from those runs. First the fixes to simulations and their checks, then to the documentation, then what is still open.

2.1 Fixed: the simulations and their checks

- **Newton's cradle.** At 9600 physics steps a second the first strike lost a third of its energy. The cause: before each step, only contacts that would still be squeezed at the step's end if the balls moved freely were worked out, so a contact that was squeezed but already opening was skipped, and Bullet let the balls pass through, losing the stored compression. Now any contact squeezed now or touching within the step is worked out, and each ball is handed back on its hinge's circle, turned with its threads. The first strike is the same at every step rate from 1200 to 9600. The fast merging of the row (in about 15 s) was the same energy leak: now, with no damping the row never merges, and with the default damping it settles over minutes (its relative swing halves in about 80 s), as real cradles do (Hutzler et al.: 1-2 minutes).
- **The chain fountain.** Links were woken and frozen once per drawn frame, so the physics depended on the display speed. All of that now runs per physics step, with a fixed 1/1800 s step; the output is identical at 0.25x, 0.5x and 1x. The 60° stall: links woken ahead of the pick-up had free ends and were dragged off, until the whole coil was loose in the jar and the hanging chain rested on the frozen links below; the last woken link is now held to the unmoved coil, and landed links stop colliding. It empties at every bend. A theory check against Biggins and Warner (2014) is now printed, and the README's height table is re-measured. A second pass found the coil laid with joints bent past their limits (304 of 1054 at 20°, plus upright links between layers), which Bullet snapped back when woken, kicking the chain: that, not the stiffness, was why the stiff chain's fountain was higher. The coil is now laid within the limits, the speed is the rate links land, and the pile's push can be measured directly (CF_PUSH=1). A third pass found why its fountain stood at half the measured height: the chain was laid as a coil. Poured into the jar as in the experiments, a ball chain of their size gives their fountain, in Bullet and in an independent bead-chain model; a live zone round the pick-up reproduces the fully simulated heap at 14 ms a frame at 1/8 speed, and the poured heap is now the exhibit (H switches to the old coil). Full account: ChainFountain/FINDINGS.md.
- **The spinning eggs.** From 3° the left egg sat at the edge where friction holds its contact still, and the step's small errors decided when it broke free: 0.50 s at 4800 steps a second, 2.45 s at 19,200, never at 28,800 and 38,400. It now starts at 6°, twice that threshold, and rises at 0.30 s at every step rate; a new console check compares its rise with Moffatt and Shimomura's rate.
- **The spinning eggs (standing).** Once standing, the left egg's tilt and fall-back depended on the step (tilt at 3 s 4.5°, 7.3°, 9.3° at 4800, 9600, 19,200 steps a second). The egg's own contact step imposed no-slip at the start-of-step contact point, though Bullet's update centres the velocities at mid-step; the gyroscopic term and the friction direction were first order too. The step is now second order and time-symmetric (no-slip half a step on, an implicit-midpoint gyroscopic term, Coulomb friction against the mean slip): 11.97-12.01° at 3 s and back on its side at 24.40 s at every rate, matching an exact rolling integration (reconstruction/rolling-check.py) to 0.04° and 0.01 s.
- **The solar system.** The yearly and monthly checks ran on the first frame after their date, and at high rates reached back to it inaccurately. The integration now stops at each check's exact date, runs the check, and goes on; and a second rate dependence in the moons' stepping was found and fixed. At 5, 60 and 100 days a second the check lines are now identical; at 100 days a second, Mimas on 2026-02-01 went from 4688 km to 41, Mercury in 2036 from 1220 km to 1. The README's samples are fresh.
- **Kapitza's pendulum.** At its old default of 4800 steps a second the rocking period was 1.7% short of converged, and that error happened to cancel the large-swing correction. The default is now 19,200

steps a second (converged to 0.1%), the console also compares with the period for the actual swing (1.133 s against 1.135 s), and a restart that misreported the size of the rocking (8.0° instead of 7.6°) is fixed.

- **The gömböcs and the Billes.** Their attendant's lift ran during the start pause, so with the pause on their output differed. The lift now waits for the pause to end, and the pause changes nothing in them either (checked with fixed seeds).
- **The bicycle wheel.** A guard now checks every body's state each frame; if anything is not a number it prints the full state, the time and the move under way, and restarts the run. Pressing R mid-turn no longer prints a stray event.
- **The start pause (gyroscope, rattleback, tippe top).** They timed their console lines by bpp's frame count, which includes the pause, so with the pause on their lines came at other moments (the physics was the same). They now count only the frames they run; all 21 exhibits pass the pause check. (Exhibits that start themselves again pause again each time, as the hall's README says, so their later lines come 0.5 s later per restart, unchanged.)
- **The Brazil-nut effect (measurement).** The floor's push was taken from the 10,000 kg box's change of velocity, which single-precision Bullet quantises to about 0.85 of the bed's weight per step; it is now the bed's own change of momentum, the big or coloured bead counted in the bed, and averaged over the shakes the text says (5-32; the old report covered 5-9.75). The 46-48° lift-off it reported was compared with the wrong theory: see section 2.3.
- **The Brazil-nut effect (rebuilt).** An independent bead model (reconstruction/bead-bed.c, checked against a single bead's bounce, sliding on a slope and gentle shaking, and converged in step and stiffness) showed the Bullet version's pace was not settled (about 2x too fast at its default step) and its beads were not glass (friction 0.64 and 0.8, bounce 0.01). With real glass the rolls reverse and the big bead never rises in a shallow bed with smooth walls; a design study found a buildable one that works: a deeper bed, walls lined with glued glass beads, shaking at 5 Hz. Bullet can't run bouncy glass (its bounce boils a packed bed at any step), so the exhibit now runs the bead model itself through a new bpp hook (DemBed), which reproduces it bit for bit. Two boxes differ only in their walls: rough walls lift the big bead, smooth glass keeps it down. The old version's story is in BrazilNut/reconstruction/FINDINGS.md.

2.2 Fixed: the documentation

- **Friction and bounce, gallery-wide.** Bullet multiplies the two bodies' friction and bounce at each contact, and several exhibits stated one body's value. Now stated at the contact: Galileo's ramp 0.64 (was '0.8'), the double cone 0.81 ('0.9'; a realistic 0.4 changes nothing measurable), the gömböcs' and Billes' bounce at the table 0.02 ('0.1'), the Brazil nut's beads 0.64 and walls 0.8, bounce 0.01. The gyroscope's foot on the wire is 0.0025, not the stated 0.005 (so it slides from 0.14°, not 0.3°); making 0.005 true would change the validated tightrope times (rocked 1°: 62 to 54 s), and 0.005 was a chosen value, so the physics is kept and the text corrected. The other exhibits were right or work out their contacts themselves; bpp never changes Bullet's rules, and rolling and spinning friction are 0 everywhere (the exhibits' own Lua supplies them).
- **The gyroscope.** Precession at 100 turns a second now 2.56 rad/s (was 2.67, not what its own parameters give); off-wire times re-measured (rocked 1°: 62.0 s; 1.5°: 49.0 s; 2°: 12.7 s; 3°: 5.9 s; still wire 55.0 s, or 48.2 s on a rough foot); the turns-a-second it comes off at stated from runs (38, 45, 52).
- **The oloid.** The weaves are now 3.6, 5.1 and 5.8 cm (was '4 to 6 cm' for all three).
- **The rattleback and the Dzhanibekov effect.** Each has a README now. The rattleback's rolling-resistance window and growth rates, and the Dzhanibekov header's figures for Bullet's own modes (-31.9% energy, -17.5% angular momentum), are as measured; both run from the repository root.

- **The tippe top.** Rolling resistance never stops the red top turning over (0-300 μm); what it does do is now stated. The stem threshold is about 80 rad/s; the README's example is a fresh run.
- **The falling Slinky.** Its 1% offset is explained: 0.6% from 86 separate coils (an exact chain of 86 point masses waits 0.2030 s; its excess over the continuum's 0.2018 s halves as the coils double) and 0.4% from Bullet's step. On the stairs, nearly one walk in two goes wrong at the normal speed (12 of 21 clean).
- **Euler's disk, the spinning eggs, the double pendulum, Brazil nut.** Euler's disk says its end follows its rolling-resistance law, not Moffatt's; the eggs' README gives the energy budget by step size, egg by egg; the double pendulum's says the throw-over time isn't converged and the Lyapunov comparison is an order-of-magnitude one; the Brazil-nut's says its late lift-off is mostly soft contacts.
- **Comments and help text.** Huygens' clocks (the in-phase mode is slower), Kapitza (env options), double pendulum and pendulums with a twist (stale figures), chain fountain help, Gömböc comments and reconstruction notes (resting tilt, where a second basin first appears; the reconstruction README notes the findings PDF's own inconsistency), Bille (its step depends on the speed setting).

2.3 Still open

- **The bicycle wheel.** The cause of the NaN blow-up seen once in the GUI is unknown; the guard will leave a full diagnostic if it happens again.
- **The chain fountain.** The heap's theory window is short (9.8 m is what the jar holds) and the chain slows in it, so the speed reading of α (0.22) disagrees with the height's (0.11); the console says so. The live zone's pile push is about a quarter below the full heap's; about one run in ten catches the rim; the bending is held at 30° in heap mode (the heap was poured at 30°; other bendings would need heaps of their own, each checked). A faithful run at 1/4 speed would need cheaper per-body work in bpp or Bullet. The coil (H) keeps its documented faults: 0.06-0.07, its shortcut 1.2-2x too tall.
- **The tippe top.** With no rolling resistance, the green top's spin collapses from 25.6 to 3.2 turns a second between 38 and 46 s, far faster than its spin friction allows. Not investigated.
- **The spinning eggs.** Minor: the middle (jumping) egg's fall-back spreads from 25.8 to 27.2 s over 2400-38,400 steps a second; above about 20,000 steps a second Bullet's single-precision rounding shows (-0.05° at 38,400); the rim contact is still first order (the default eggs never touch it); from a 3.5° start the rise onset sits at the stick/slip edge and is step-sensitive (the default starts at 6°).
- **The gyroscope.** Rocked 1.5°, the tightrope gyroscope wobbles wider (9-17°) and slides out to 14-15 cm against the 9 cm aimed for; the wobble pushing the foot along is the likely cause, not yet separated from the slope's push.
- **The Brazil-nut effect.** The beads are 1/100 as stiff as glass, for speed (rolls, rise and height unchanged at 1/10; the gentle push 0.19 against 0.199). The model has no front or back panes and no air, and the glued wall beads are assumed to bounce like free glass. No experiment with 1 cm glass beads in a one-bead-thick cell was found to compare with, so the comparison is with the reference model, the mechanism in the literature, and rigid-bed theory. At the default 1/4 speed a frame takes about 20 ms against 16.7 on the test machine (two threads), so it runs at about 0.8 of its nominal speed there.
- **The double pendulum.** The too-hard chain's throw-over time isn't converged (3.96 s at 4800 steps/s, 4.46 s at 9600); only that it is thrown over is.
- **The falling Slinky.** Kept at 4800 steps a second for speed, so its 0.4% step error is documented rather than removed.

3. How the gallery is simulated and checked

How the exhibits are simulated

Bullet (the Bullet Physics library, through bpp) carries the rigid bodies, joints and ordinary contacts. Where Bullet's contacts can't do the physics faithfully, the exhibit works it out itself and hands the result to Bullet each step: Newton's cradle integrates Hertz's law between the balls in quarter-microsecond sub-steps; the oloid and the spinning eggs work out their own contacts; the tippe top and the rattleback advance their spin by Euler's equations (RK4) with Bullet's gyroscopic term off; the gyroscope applies its own gyroscopic torque (its rotor is a number, a gyrostat, not a body spun at 100 turns a second). Some exhibits don't use Bullet's dynamics at all, and Bullet only draws them: Euler's disk integrates the exact rolling equations; the double pendulum, pendulums with a twist and the solar system integrate their exact equations of motion by Bulirsch-Stoer (Gragg's modified midpoint, extrapolated, tenth order), with RK4 for the shaken pendulums and Wisdom-Holman for the moons. The Brazil-nut effect's glass beads are simulated by a soft-sphere bead model in C++ (DemBed: Hertz-Mindlin contacts with Coulomb friction), not by Bullet, which can't handle bouncy glass in a pile. Each exhibit's section gives its integrator, its step and its parameters.

What 'tested against theory' means here

Most exhibits print, as they run, the quantities theory predicts, next to what the simulation did: periods, rates, thresholds, conserved quantities, positions against an ephemeris. The theory values are worked out by the exhibit from its own parameters (so a change of slider changes both), or, for the solar system, taken from JPL. Some don't, or only partly: the rattleback's console is qualitative (its numbers are in a separate reference model), and the Brazil-nut effect is mostly qualitative; for these, and wherever else it helped, the comparison was made for this report and is marked so. Where possible a check is also run at a different step size, to see that the agreement isn't a numerical accident; the sections say where that was done and where it wasn't.

The hall and the exhibits on their own

Each exhibit runs in its own physics world inside the hall, and the hall only runs the one you're at. Every exhibit's console output in the hall is compared, line for line, with its output when run on its own, and they are identical. This was checked for each exhibit as it was added, and again, for this report, for every exhibit whose console changed in the fixes (Newton's cradle, the chain fountain, the solar system, Kapitza's pendulum, the Brazil-nut effect, the bicycle wheel, and the exhibits whose text changed). The exceptions are the Gömböc and Bille drops, which are seeded from the clock and so differ from run to run anywhere.

The start pause

Each exhibit stands still for half a second whenever it starts, so you see how it begins. During the pause bpp's own time step is zero and the exhibit's step is skipped, so the pause changes nothing in the physics: with the pause on, the output is the output without it, delayed, line for line. This was checked for all twenty exhibits that have it when it was added (at 2 s), and at 0.5 s for every exhibit changed in the fixes, including the gömböcs and the Billes with fixed seeds (their attendant's lift now waits for the pause to end). The runs in this report were made with it off.

Rendering and sound

Bodies the physics moves are drawn where the physics has them. Some things are drawn without being simulated, and the sections say so: the chain fountain's heap away from the pick-up (fixed until links come within 4 cm of it) and its links once they land, the falling Slinky's pile once it has landed (laid to rest), the cradle's threads, the bicycle figure's arms, the cords of the swinging Atwood machines and springs (drawn as beads). The solar system's maps of each surface are drawn, not photographs (their markings placed where they are on the real body; the turning itself is the IAU's model, and is checked),

and the gyroscope's tilt gauge turns twenty times as far as the wire does. Newton's cradle's clack is computed from the same Hertz strike, as the sound a jerked steel sphere sends into the air.

4. Wing A: Turning

4.1 Gömböc variety

Agrees with theory. One stable point: Gömböc-C ends upright in 11 of 11 random drops; righting from a headstand 8.75 and 13.03 s against the earlier study's 8.8 and 13.2 s. Sloan shapes 0 of 16 upright in one run, 3 of 9 in a seeded re-run (study: 9%; small samples, and the bodies knock into each other).
Folder: demo/WyomingWill/Gomboc/

What it shows

A Gömböc is a convex body of uniform density with exactly one stable and one unstable equilibrium, so however it is set down it rolls back to the same upright pose. The table drops up to 12 bodies at random orientations, cycling through three shapes: the real Gömböc (Gomboc-C, at 9 cm and at 75% and 125% size) and two of M. L. Sloan's analytic Gömböcs (β = 0.07 and 0.08). The lesson is that having one resting point in theory is not enough. Gomboc-C rights itself every time, while the gently sloped Sloan shapes usually stall on their sides under ordinary rolling resistance.

Model

Bullet handles contact. Each body is a `btConvexHullShape`:

- **Gomboc-C:** the convex hull of a 1.2 mm tessellation of the GOMBOC900 CAD B-spline surfaces (13,476 points, per gomboc-drop-c.lua).
- **Sloan shapes:** hull points of $r^4 = 1 + 4\beta \sin\varphi \cos(\theta - 5\varphi)$.
- **Margin:** 0.04 cm, scaled with size.
- **Mass and inertia:** exact (CAD or analytic) at resin density 1.2 g/cm³. That gives 323 g at 9 cm, 136 g at 75%, 630 g at 125%, and 407 g / 401 g for β = 0.07 / 0.08.

The script steps the world itself at 1200 steps per simulated second (20 substeps of 1/1200 s per 1/60 s frame), using Bullet's semi-implicit Euler. Before every substep it applies two losses as constant moments at the contact:

- rolling resistance $M = \delta \cdot m \cdot g$, with lever arm δ = 150 µm by default;
- spin friction of 0.7 rad/s².

Other settings: friction 0.5 (Bullet multiplies it by the table's 0.8: 0.4 at the table, 0.25 body on body), restitution 0.1 (table 0.2: 0.02 at the table, 0.01 body on body), air drag 0, no taps. The tray is flat with 8° ramps at the edges and a rim. Status thresholds: "Upright" means within 15° of the resting normal for 2 s. "At rest" means less than 0.2° of turn and 0.02 cm of movement over 2 s. Five seconds after a body comes to rest it is lifted away and the next shape is dropped (GV_REDROP).

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Gomboc-C rests only on its one stable point	1 stable + 1 unstable equilibrium (CAD surface checked, code comment)	random drops: 11 of 11 runs that came to rest were upright, tilt 0.4–7.9°; 60 s tally: 8 Gomboc-C runs, 0 not upright	100% upright
Gomboc-C from a headstand (balanced on the unstable point, 1–3° nudge)	must leave the unstable point and right itself	3 of 3 at rest upright: 0.3°, 4.0°, 12.5° (the last propped against a neighbour)	consistent
Sloan β 0.07 / 0.08 self-righting at 150 µm rolling resistance	one basin (mathematically monostatic); findings PDF Table 5b: 14 of 157 (9%) upright	drops: 13 of 13 runs at rest NOT upright (18–142°); headstand 0 of 3 (71–138°)	0/16 vs 9% expected (about 1.4 of 16); small sample

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Residual tilt of Gomboc-C at rest	0° ideal; with a constant rolling moment it stops short. gomboc-drop-c.lua comment, corrected 15 Oct 2026: "0.3-8 deg ... (7.9 deg the most seen, on the 75% one in Gomboc Variety: the smaller, the further off)" (was "0.3-7 deg")	report run 0.3–7.9° (excluding the propped 12.5°); seeded re-run 15 Oct 2026: 0.3–4.6°	within the corrected range
Time to right from a headstand, 75% / 100% / 125%	findings PDF Table 6: 8.8 / 13.2 / 14.3 s	8.75 / 13.03 / 2.73 s (125% was knocked by a neighbour)	75% and 100% within 0.2 s
Mass at 100 / 75 / 125%	V·ρ = 268.851 cm³ × 1.2 g/cm³ = 322.6 g; × s³: 136.1 / 630.1 g	323 / 136 / 630 g	exact
Hull's false resting faces (Gomboc-C)	ideally none	code comment: 34 hull resting faces, all within 16° of true rest, deepest needs a 0.53 µm lift	— (documented)
Height of balancing point above resting point	findings Table 5: Gomboc-C 11.3 mm; Sloan β 0.07 / 0.08: 4.8 / 5.1 mm	— (documented)	—
Energy drift with all losses off (step-size check)	no gain	gomboc-drop-c.lua comment: at 1200 steps/s "energy stays level, with no steady gain"; 2400 "no better"; 600 also no steady gain	documented, not rerun

The 60-second tally printed by this run (time to rest, median and range): Gomboc-C 2 runs, 1.2 s (1.2–6.4). At 125%, 3 runs, 17.2 s (6.4–22.0). At 75%, 3 runs, 8.3 s (5.0–11.0). β 0.07: 5 runs, 3.1 s, 5 not upright. β 0.08: 5 runs, 5.7 s, 5 not upright.

Run used (random seed from the clock; one run each, so the numbers vary)

```
cd demo/WyomingWill/Gomboc && HOLD_S=0 stdbuf -oL timeout 3000 bpp -f gomboc-variety.lua -n 4200   # 70 s, 6 bodies,
    redrop 5
HOLD_S=0 GV_START=headstand GV_REDROP=0 timeout 3000 bpp -f gomboc-variety.lua -n 3600             # headstands
```

(A 9000-frame run was started but stopped: on the shared 2-core machine it ran at well under 1/5 of real time.)

```
  Orange  at rest after  17.15 s, 0.5 deg off upright -- Orange (Gomboc-C at 125%)
  Yellow  at rest after   8.30 s, 7.9 deg off upright -- Yellow (Gomboc-C at 75%)
  Purple  at rest after   2.25 s, 133.6 deg off upright  ** NOT UPRIGHT ** -- Purple (Sloan beta 0.07)
=== 60 s: 18 finished runs ===
    Gomboc-C at 125%             3 runs, time to rest median  17.2 s (6.4-22.0)
    Gomboc-C at 75%              3 runs, time to rest median   8.3 s (5.0-11.0)
    Sloan beta 0.07              5 runs, time to rest median   3.1 s (1.9-11.1), 5 not upright
    Sloan beta 0.08              5 runs, time to rest median   5.7 s (1.9-13.1), 5 not upright
  Yellow  upright after   8.75 s -- Yellow (Gomboc-C at 75%)
  Red     upright after  13.03 s -- Red (Gomboc-C)
  Red     at rest after  16.05 s, 0.3 deg off upright -- Red (Gomboc-C)
  Green   at rest after  16.22 s, 137.5 deg off upright  ** NOT UPRIGHT ** -- Green (Sloan beta 0.07)
```

Seeded re-run after the fixes (15 Oct 2026):

```
GV_SEED=3 GV_COUNT=4 GV_REDROP=1 HOLD_S=0 bpp -f gomboc-variety.lua -n 3700
=== 60 s: 13 finished runs ===
    Gomboc-C                     3 runs, time to rest median  12.8 s (11.8-14.0)
    Gomboc-C at 125%             2 runs, time to rest median   6.4 s (2.6-10.3)
    Gomboc-C at 75%              2 runs, time to rest median   5.1 s (3.0-7.1)
    Sloan beta 0.07              3 runs, time to rest median   3.5 s (1.6-5.1), 2 not upright
    Sloan beta 0.08              3 runs, time to rest median  10.4 s (7.2-20.3), 1 not upright
```

In the 60 s tally 1 of 3 β 0.07 and 2 of 3 β 0.08 runs ended upright; over the whole run (every body that came to rest, first report each) 1 of 4 β 0.07 and 2 of 5 β 0.08. So in this seeded run the Sloan shapes sometimes righted themselves, unlike the report's 0 of 16. 3 of 9 is more than the PDF's 9% (Table 5b) would make likely (a 4% chance), 0 of 16 less (22%); the samples are small and bodies here knock into each other, so neither run tests the 9% well. One β 0.07 body (Green, run 1) was re-reported at rest five times (4.4 to 51.6 s, creep) and so never entered the tally.

Pause and hall checks after the lift fix (15 Oct 2026): with `GV_SEED=3 GV_COUNT=4 GV_REDROP=1`, HOLD_S=0 for 900 frames is line for line a prefix of the default half-second pause for 930 frames (31 lines, lifts and redrops inside the window); hall against standalone (cmp2.sh, 1200 frames): IDENTICAL (32 run lines).

Known departures and honest caveats

- Fixed 15 Oct 2026: the attendant's lift ran during the start pause (a `holdTick` that called `liftStep`), so with the pause on the output differed. `holdTick` is removed; the lift waits for the pause to end and the pause changes nothing (checks above).
- Found 16 Oct 2026, physics not changed: **the restitution slider's 0.1 is 0.02 at the table.** Bullet multiplies the two bodies' restitution (and friction), and the table's is 0.2. The slider's help said only "bounciness"; it now gives the product, and a comment at the table gives both products (friction 0.4, restitution 0.02 at the table; 0.25 and 0.01 body on body). No theory check uses the restitution; the value was chosen, not measured, and a resin body on a wooden table bounces more than 0.02, so the bounce here is on the low side. It is left as it is, since raising it would change the validated statistics.
- **Rolling resistance and spin friction are script-side models.** They are constant moments, not Bullet physics. δ = 150 μm and 0.7 rad/s² were chosen, not measured. The findings PDF (Table 5b) shows that the outcome depends strongly on δ: without rolling resistance nothing ever settles, Gomboc-C included.
- **Gomboc-C stops a few degrees short of upright** (here up to 7.9°). The constant moment holds it wherever gravity's restoring lever is smaller than δ, and its valley is flat. The code says this is real behaviour.
- **The Sloan shapes are not strictly convex at these β.** They roll on their convex hull. The findings PDF cites arXiv:2604.17120, which finds strict convexity only below β ≈ 0.036. The hull gains a second basin at higher β. Re-checked 15 Oct 2026 by re-running the reconstruction's basin scan (scanb.py's method, form 1, 25000 surface points) at finer β: deepest extra basin 5.9 μm at 0.08 (the hull facets' own noise is ~5 μm), 11.9 μm at 0.085, 25.3 at 0.09, 43.0 at 0.095, 66.7 at 0.10 (the PDF's Table 5 gives 67 at 0.10 and none at 0.08). The lua comment ("first found at beta 0.085") was consistent with this; it now gives the depths and notes that the PDF rounds the limit to "above β ≈ 0.09". The scan result is recorded in variety-reconstruction/README.txt. The PDF was not edited.
- **The β 0.08 resting point:** the lua says its lowest point is ~35° from the formula's minimum; computed 15 Oct 2026 from the `down` and `top` vectors in gomboc-meshes/beta1-08.lua: 34.6° (β 0.07: 4.6°). The findings PDF §4 says the Sloan shapes' "true resting orientation lies 4–5° from the formula's minimum", which holds for β 0.07 only; its own §5 says the balancing point sits ~35° off the top on β 0.08. This inconsistency inside the PDF is now noted in variety-reconstruction/README.txt (there is no Gomboc README.md). The PDF was not edited.
- **Re-reported rests (creep).** Several bodies were reported "at rest" again and again: Yellow #3 (125%) five times between 19.6 and 32.4 s, and one β 0.07 three times. Each time the body had moved more than 0.2° or 0.2 mm within 2 s after being declared at rest, which matches the contact-solver creep admitted in gomboc-drop-c.lua ("up to ~0.3 deg and ~0.7 mm per minute"). The tally uses the last of these times.
- Fixed on 15 Oct 2026, tally median: with an even number of runs the "median" used to print the lower middle value; it is now the mean of the two middle values, in gomboc-variety.lua and gomboc-drop-c.lua. Seeded re-run: "Gomboc-C at 125% 2 runs, time to rest median 6.4 s (2.6-10.3)" = mean of 2.58 and 10.27 s.
- **Collisions between bodies** (props, knocks) affect individual results. The console flags them as "touching".
- Fixed on 15 Oct 2026: variety-reconstruction/README.txt listed meshes beta1-06 and beta2-04, which are not in gomboc-meshes/. It now names the two kept (beta1-07, beta1-08), says gen.py's __main__ makes a different larger-β set, and that the kept files' `down` is the lowest face of the hull's one basin, not the formula minimum gen.py writes. Sloan's form 2 is left out by design.

- Statistics are from one short run. The Sloan upright fraction (0/16) is compatible with the 9% in the findings PDF, given the small sample.

References

- Kiss Attila, GOMBOC900 SolidWorks CAD model (2013), as cited in code.
- M. L. Sloan, "An Analytical Gomboc", arXiv:2306.14914 (2023); MathWorld, "Gömböc".
- "Sloan's Analytical Gömböc at Published β: A Strict-Convexity-Constrained Reanalysis", arXiv:2604.17120 (2026), as cited in the findings PDF.
- P. L. Várkonyi, G. Domokos, "Mono-monostatic bodies: the answer to Arnold's question", Math. Intelligencer 28 (2006).
- G. Domokos, P. L. Várkonyi, "Geometry and self-righting of turtles", Proc. R. Soc. B 275 (2008).
- W. M. Spears with Claude, "One resting point is not enough" (gomboc-findings.pdf, 9 Oct 2026, working notes).

4.2 Bille and the polyhedra

Agrees with theory. Bille ends on its one stable face in 27 of 27 drops and 6 of 6 set-downs, by the predicted face paths (a quick crossing of a face is now logged, in brackets).
Folder: demo/WyomingWill/Bille/

What it shows

These are bodies that rest on only one face because of where their mass is, not because of their shape alone.

- **Bille** is the monostable tetrahedron of Almadi, Dawson & Domokos (2025): a hollow carbon-tube frame with a dense tungsten-carbide wedge, so that set down on any face it tips, face to face, onto face D. It has one stable face and two unstable corners.
- **The spiral polyhedra** (after Domokos & Kovács 2023) have 21, 26 and 37 corners, each with equal weights at the corners. Each has one stable face (the base) and one unstable vertex (the apex). Their bands of side faces tip one onto the next: … → S2 → S1 → base.

Bodies are dropped at random orientations or set down on a wrong face, and the console prints the faces each one passed through.

Model

Bullet handles contact through each body's convex hull: the 4 vertices for Bille, the corner points for the polyhedra. Mass and principal inertia are taken from the data files:

- **Bille at size 1:** 120 g, I = (0.219, 1.225, 1.321) kg·cm². The collision margin is the real tube radius, 0.05 cm.
- **Polyhedra:** scaled 4.5× to 40 cm tall, with mass scaling as size³.
- **Per-colour sizes:** 1.0, 0.6, 1.3, 0.8, 1.15 and 0.7.

The losses are applied by the script before every physics step, as constant moments: rolling resistance with lever arm δ = 50 μm (not scaled with size) and spin friction of 0.7 rad/s². Other settings: friction 0.5 (× table 0.8 = 0.4 at the table, 0.25 body on body), restitution 0.1 (× table 0.2 = 0.02 at the table, 0.01 body on body), no air drag.

Time step. Requested: 1200 steps per simulated second (slider), but at least one step per frame. Actual, from the code `NSTEPS = max(1, floor(steps·speed/60 + 0.5))`:

- At the default 1/50 slow motion (speed 0.02), NSTEPS rounds to 1, so the step is 1/3000 s.
- At speed 0.1 it is 1/1200 s.

Detection thresholds. A face counts as "lain on" when its normal is within 2° of straight down for 0.03 s. "At rest" means less than 0.2° and 0.03 cm of movement over 0.4 s. An edge counts as "rolled on" when exactly two of its corners are within 0.15% of the body's size of the table.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Bille rests only on face D (random drops)	monostable on D (paper; reconstruction puts the COM 0.81 mm inside the face-D loading zone, which is at most 1.48 mm deep: README.txt)	27 of 27 drops on D (6 at speed 0.02, 21 at speed 0.1); tallies "16 runs, 16 on the stable face"	100%
Bille set down on a wrong face	ends on D	6 of 6 on D	100%
Falling pattern from C	C → D (quasi-static rule)	2 of 2: C → D, over edge AB (= C∩D)	matches
Falling pattern from A	A → D	1 of 1: A → D, over edge BC (= A∩D)	matches
Falling pattern from B	B → A → D	report run: 3 of 3 logged as "B → D" (edges CD = B∩A then BC = A∩D). Since 15 Oct 2026 a face crossed in under 0.03 s is logged in brackets: re-run (BD_SEED=5 BD_BODIES=1 BD_COUNT=3 BD_START=setdown) logs Green "B -> (A) -> D; edges it rolled on: CD, BC"	path now matches in the face log too
Polyhedra rest only on the base (drops)	one stable face	21-corner 4/4 (incl. one in the mixed run), 26-corner 2/2, 37-corner 4/4	10/10
Polyhedra set down on a band	S_n → ... → S1 → base	37-corner: S6→S5→S4→S3→S2→S1 (then rocks S2/S1/base); 26-corner from S4: S4 → base; 21-corner from S1: S1 → base	all end on base; order never reversed except brief rocking (e.g. 26-corner drop S1→S2→S1→base)
Bille time to rest after drop	header: a single face-to-face tip takes ~1/20 s	median 0.39–0.41 s (tallies), range 0.36–0.69 s	—
Polyhedra time to rest (40 cm scale)	findings Table 6 at 9 cm: 4.3 / 2.3 / 2.5 s (21/26/37)	21: 11.15–16.22 s; 26: 4.09–5.16 s; 37: 4.33–6.01 s	not comparable directly (scaled 4.5×, sizes 0.6–1.3)
Masses	120 g × s³; polyhedra 109 / 142 / 201 g at 9 cm (findings) × (4.5 s)³	Bille 120 / 264 / 183 g at s = 1 / 1.3 / 1.15; 21-corner 9951 g at s=1 (109.2 g at 9 cm), 26: 28362 g (141.7 g), 37: 27878 g (201.1 g)	exact
Earlier statistics (README)	—	code header: Bille "re-dropped 346 times, always ended on face D"; findings: polyhedra 116/117 on base (26-corner: 37/38)	documented

Run used (random seeds from the clock; one run each)

```
cd demo/WyomingWill/Bille && HOLD_S=0 timeout 1500 bpp -f bille-drop.lua -n 9000                    # speed 0.02, 3
    s simulated
HOLD_S=0 BD_SPEED=0.1 bpp -f bille-drop.lua -n 9000                                                 # 15 s simulated
HOLD_S=0 BD_START=setdown BD_COUNT=6 BD_REDROP=0 bpp ... -n 3600   (speed 0.02, and again with BD_SPEED=0.1)
HOLD_S=0 BD_SPEED=0.1 BD_BODIES=2 BD_COUNT=6 BD_REDROP=1 bpp ... -n 12000                           # polyhedra only
```

Re-run after the 15 Oct 2026 logging change (seeded):

```
BD_SEED=5 BD_BODIES=1 BD_COUNT=3 BD_START=setdown BD_REDROP=0 HOLD_S=0 bpp -f bille-drop.lua -n 2500
  Green   set down on B; predicted: B -> A -> D
  Yellow  at rest on face D after  0.23 s; faces it lay on: A -> D; edges it rolled on: BC -- Yellow (Bille)
  Red     at rest on face D after  0.25 s; faces it lay on: A -> D; edges it rolled on: BC, AB -- Red (Bille)
  Green   at rest on face D after  0.34 s; faces it lay on: B -> (A) -> D; edges it rolled on: CD, BC -- Green (Bille)
BD_SEED=5 BD_BODIES=1 BD_COUNT=3 BD_REDROP=1 HOLD_S=0 bpp -f bille-drop.lua -n 4000
=== 1 s: 3 finished runs ===
    Bille                                3 runs, 3 on the stable face, time to rest median 0.33 s; paths: D x3
```

(Bille seeds from the clock unless BD_SEED is set; seeded, it is checked for hall identity below. The tally's new median was not exercised with an even count in this short run; the formula is (t[floor((n+1)/2)] + t[floor(n/2)+1]) / 2.)

```
  Yellow  set down on B; predicted: B -> A -> D
  Yellow  at rest on face D after  0.45 s; faces it lay on: B -> D; edges it rolled on: CD, BC -- Yellow (Bille)
  Red     at rest on face D after  0.19 s; faces it lay on: C -> D; edges it rolled on: AB -- Red (Bille)
  Orange  at rest on face D after  0.21 s; faces it lay on: A -> D; edges it rolled on: BC -- Orange (Bille)
=== 12 s: 16 finished runs ===
    Bille                              16 runs, 16 on the stable face, time to rest median 0.41 s; paths: D x16
  Purple  set down on S6; predicted: S6 -> S5 -> S4 -> S3 -> S2 -> S1 -> base
  Purple  at rest on face base after  5.04 s; faces it lay on: S6 -> S5 -> S4 -> S3 -> S2 -> S1 -> S2 -> S1 -> base
    -> S1 -> base -- Purple (37-corner)
    21-corner polyhedron (ideal)           3 runs, 3 on the stable face, time to rest median 14.98 s; paths: S1 ->
    base x2, S3 -> S2 -> S1 -> base -> S1 -> base x1
    37-corner polyhedron (shell + weights)   4 runs, 4 on the stable face, time to rest median 4.58 s; ...
```

Pause and hall checks after the lift fix (15 Oct 2026): with `BD_SEED=5 BD_BODIES=1 BD_COUNT=3 BD_REDROP=1`, HOLD_S=0 for 3000 frames is a prefix of the default pause for 3030 frames (36 lines); hall against standalone (cmp2.sh, 1500 frames): IDENTICAL (12 run lines). With a fixed seed the hall-identity check applies like any other exhibit's.

Known departures and honest caveats

- Fixed 15 Oct 2026: the attendant's lift ran during the start pause (a `holdTick` that called `liftStep`), so with the pause on the output differed. `holdTick` is removed; the lift waits for the pause to end and the pause changes nothing (checks above).
- Fixed on 15 Oct 2026: the tally's "median" with an even number of runs printed the lower middle value; it is now the mean of the two middle values (same fix in ../Gomboc).
- **The Bille shape is a reconstruction, not the authors' drawing.** The paper gives vertex coordinates only in a figure. The shape was fitted to the published longest edge (500 mm), volume (668.624 cm³) and the four loading-zone volumes, all matched exactly. The mirror image fits equally well. The WC wedge mass design is the author's own, not the paper's.
- **Bille has a tiny design margin.** Its centre of mass sits 0.81 mm inside a zone at most 1.48 mm deep, and "a 0.1 g glue blob at a far vertex moves it ~0.25 mm" (README.txt).
- **The 21-corner body is the paper's idealisation** (point masses, weightless skin). The findings PDF says it is "essentially a mathematical object". Domokos & Kovács' Table 1 centre-of-mass values could not be reproduced from the stated construction, so the shapes were refitted (opt21/optnk). The 26- and 37-corner bodies are refitted for a 0.5 mm polycarbonate shell with tungsten weights.
- **Logged paths are dynamic, not quasi-static.** Bodies dropped or set down carry momentum and can cross a face in under the 0.03 s that counts as lying on it, or rock back up a band (S1 → S2). Fixed on 15 Oct 2026: such a quickly crossed face is no longer dropped from the log; it is printed in brackets, e.g. "B -> (A) -> D"; a bracketed face later lain on properly is replaced by the plain name.
- Found 16 Oct 2026, physics not changed: **the restitution slider's 0.1 is 0.02 at the table.** Bullet multiplies the two bodies' restitution (and friction), and the table's is 0.2. The slider's help said only "bounciness"; it now gives the product, and a comment at the table gives both products (friction 0.4, restitution 0.02 at the table; 0.25 and 0.01 body on body). No theory check uses the restitution; the value was chosen, not measured, and a resin body on a wooden table bounces more than 0.02, so the bounce here is on the low side. It is left as it is, since raising it would change the validated statistics.
- **Rolling resistance and spin friction are script-side constant moments** (δ = 50 µm chosen, not measured).
- **Time step depends on speed.** The effective step is 1/3000 s at the default speed and 1/1200 s at speed 0.1, despite the "steps" slider reading 1200 (there is always at least one step per frame). Documented on 15 Oct 2026 in the lua header and the slider's description ("at least one a frame (at 0.02 speed: 3000 a second)"); not changed, since a step independent of speed would need fractional steps per frame. No convergence study is documented for this exhibit.

- **Not fully tested.** The polyhedra were tested in only 10 drops and 3 set-downs here, and the 21-corner body takes 11–16 s to settle at this scale. All results come from one short session.

References

- G. Almadi, R. J. MacG. Dawson, G. Domokos, "Building a monostable tetrahedron", arXiv:2506.19244 (2025); Quanta Magazine, 2025-06-25.
- G. Domokos, G. Almadi, K. Regős, R. Dawson, "On Equilibria of Tetrahedra", Math. Intelligencer (2023).
- G. Domokos, F. Kovács, "Conway's spiral and a discrete Gömböc with 21 point masses", Amer. Math. Monthly 130 (2023); arXiv:2103.13727.
- S. Bozóki, G. Domokos, F. Kovács, K. Regős, Int. J. Solids Struct. (2021) and Eur. J. Oper. Res. 310 (2023), as cited in the findings PDF.
- W. M. Spears with Claude, "One resting point is not enough" (../Gomboc/gomboc-findings.pdf, 2026).

4.3 Galileo's ramp

Agrees with theory. Finish times within -0.013% of sqrt(2d(1+k)/(g sin θ)); distances in equal times 1 : 2.998 : 4.997 (odd numbers; fitted for this report); first-order convergence in step size.
Folder: demo/WyomingWill/GalileoRamp/

What it shows

Five rollers are let go together on a 5-degree ramp: a wooden ball, a small steel ball, a wooden disc, a brass hoop and a can of soup. The order at the finish line depends only on how much of each body's mass sits out at its rim ($k = I/(m\ r^2)$), not on its mass or size. The soup can wins because the liquid inside hardly turns, the two balls tie, and the hoop comes last.

Model

Bullet does the contacts. The balls are `btSphereShape`. The disc, hoop and can also collide as spheres of their own radius, but they are held to their lanes (linear factor (1,1,0), angular factor (0,0,1)) and given their own inertia tensors. The README says Bullet's cylinders, touching along a line, "jitter and lose 2% of their speed", which is why they are not used. The racers (code and README):

- wooden ball: r = 2.5 cm, 0.7 g/cm^3, 45.8 g, k = 0.4
- steel ball: r = 1.0 cm, 7.85 g/cm^3, 32.9 g, k = 0.4
- wooden disc: r = 2.5 cm, 2 cm thick, 27.5 g, k = 0.5
- brass hoop: 3 mm thick, 1.5 cm wide, 56.5 g; $k = (r^2 + r_i^2)/(2\ r^2) = 0.887$
- can: 40 g of tin plus 330 g of soup. Only the tin counts towards the spin inertia, which gives k = 0.094.

Physics settings: friction 0.8 on both ramp and bodies, which Bullet multiplies to 0.64 at each contact (the README and lua said "friction 0.8" until 16 Oct 2026; they now give 0.64), restitution 0, damping 0, g = 981 cm/s^2, 30 solver iterations, 2400 steps per simulated second by default (GR_STEPS). Bullet's integrator is its usual semi-implicit (symplectic) Euler. The ramp is 120 cm long, and the finish line is 100 cm below the start line. The script interpolates each finish crossing to within one physics step.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Soup time to 100 cm	t = sqrt(2d(1+k)/(g sinθ)), k=0.094: 1.5994 s	1.5992 s	-0.013%
Wooden ball time	k=0.4: 1.8097 s	1.8095 s	-0.011%
Steel ball time (ties wooden ball)	k=0.4: 1.8097 s	1.8095 s	-0.011%; ties ball to 0.1 ms
Disc time	k=0.5: 1.8732 s	1.8730 s	-0.011%

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Hoop time	k=0.887: 2.1011 s	2.1009 s	-0.010%
Speed at line, soup / ball / disc / hoop	v = sqrt(2 d g sinθ/(1+k)): 125.0 / 110.5 / 106.8 / 95.2 cm/s	125.1 / 110.5 / 106.8 / 95.2 cm/s	≤0.1 cm/s
Acceleration (least-squares fit of d = ½at² + v0 t + d0 to GR_LOG, 0.2 s to finish)	a = g sinθ/(1+k): ball 61.07, disc 57.00, hoop 45.30, soup 78.18 cm/s²	61.071 / 56.999 / 45.305 / 78.182 cm/s²	<0.01%
Galileo's odd-number rule (distance in successive 0.5 s intervals)	1 : 3 : 5	1 : 2.998 : 4.997 (all five racers)	0.1%
Step-size convergence, soup time error	error → 0 as dt → 0	600 st/s -0.052%; 2400 -0.013%; 9600 -0.003%	error ∝ dt (first order, ~4× per 4× steps)
Finish order	soup < balls (tie) < disc < hoop	Soup, Ball, Steel, Disc, Hoop	matches

The theory values for time and speed are printed by the script. The fit and odd-number rows I computed from the GR_LOG CSV of this run. Times in the README are identical to this run's, to every printed digit.

Run used

```
cd demo/WyomingWill/GalileoRamp && HOLD_S=0 timeout 1500 bpp -f galileo-ramp.lua -n 600            # 2400 steps/s
HOLD_S=0 GR_STEPS=600  ... -n 600 ;  HOLD_S=0 GR_STEPS=9600 ... -n 600                            # convergence
HOLD_S=0 GR_LOG=gr.csv ... -n 150                                                                   # distance log
```

```
  Soup   the can of soup         370.0 g, k = I/(m r^2) = 0.094: a = g sin(theta)/(1 + k) =  78.2 cm/s^2; theory
    1.599 s
  Hoop   the brass hoop           56.5 g, k = I/(m r^2) = 0.887: a = g sin(theta)/(1 + k) =  45.3 cm/s^2; theory
    2.101 s
  Soup   1st across the line at 1.5992 s (theory 1.5994, -0.013%), at 125.1 cm/s (theory 125.0)
  Ball   2nd across the line at 1.8095 s (theory 1.8097, -0.011%), at 110.5 cm/s (theory 110.5)
  Steel  3rd across the line at 1.8095 s (theory 1.8097, -0.011%), at 110.5 cm/s (theory 110.5)
  Disc   4th across the line at 1.8730 s (theory 1.8732, -0.011%), at 106.8 cm/s (theory 106.8)
  Hoop   5th across the line at 2.1009 s (theory 2.1011, -0.010%), at 95.2 cm/s (theory 95.2)
GR_STEPS=600:  Soup   1st across the line at 1.5986 s (theory 1.5994, -0.052%) ...
GR_STEPS=9600: Soup   1st across the line at 1.5994 s (theory 1.5994, -0.003%) ...
```

Known departures and honest caveats

- The disc, hoop and can are spheres to the collision detector and are locked to planar motion with spin about their axes only. That is exact for an ideal line contact, but the model cannot show wobble, steering or edge effects. Collision shape and inertia are chosen separately, on purpose.
- The can is an idealisation: the soup is treated as not turning at all, so it counts only towards translation. Real soup is viscous and slowly spins up, so a real can would sit somewhere between k = 0.094 and that of a solid cylinder (0.5).
- All times are about 0.2 ms early (-0.01%). The error halves roughly with each doubling of the step rate (first-order integrator). At the default 2400 steps/s it is far below anything visible.
- Friction (0.64 at the contact) was chosen so that nothing slips. At 5° the friction needed is small (μ ≥ k tanθ/(1+k), at most 0.041 for the hoop), so this is comfortable. Rolling resistance and air drag are absent: the ramp is ideal.
- The odd-number rule and acceleration fit are not printed by the exhibit. I computed them from the optional log. In the log, the ball passes the 114 cm end of the ramp before t = 2.0 s, so only the first three intervals (up to 1.5 s) were used.
- The README's console excerpt matches this run exactly.

References

- Galileo Galilei, inclined-plane experiments (named in the exhibit title; no specific citation in code or README).

- Standard rolling-without-slipping result $a = g \sin\theta/(1 + I/(m r^2))$ (stated in code and README; no source cited).

4.4 The double cone

Agrees with theory. Rolls up or down exactly as the rail geometry predicts on all three tracks; centre's drop within 1 µm of the exact geometry; energy loss 0.1-0.3% (numerical).
Folder: demo/WyomingWill/DoubleCone/

What it shows

Two cones are joined base to base and set on a pair of rails that rise and spread apart in a V. Let go at the low, narrow end, the double cone rolls up to the high, wide end, so it seems to roll uphill. It does not really: the spreading rails meet it nearer its points, where it is thinner, so its centre of mass sinks faster than the rails rise. Two control tracks sit beside it. On the same rails a plain cylinder rolls down. On rails rising 5 cm instead of 1.5 cm (too steep), the same cone rolls down.

Model

Bullet does the contacts:

- **Double cone:** a `btCompoundShape` of two `btConeShapeZ` ($R = 5$ cm, $H = 12$ cm each), collision margin 0.01 cm, 400 g, solid. Inertia is $I_{axis} = 3/10\ m R^2 = 3.0$ kg·cm² and $I_{perp} = m(3/20\ R^2 + H^2/10)$.
- **Cylinder:** a `btCylinderShapeZ` with $r = 3$ cm, length 26 cm, 400 g.
- **Rails:** 8 mm brass rods (radius 0.4 cm), 50 cm long, 2.5 cm from the centre line at the low end and 9.5 cm at the high end, so $\tan\varphi = 0.14$ per side. They rise 1.5 cm ($\tan\theta = 0.03$) at the back and middle tracks and 5 cm ($\tan\theta = 0.10$) at the front. For collision, each rail is a compound of 10 cm capsules laid end to end. The README says a single 54 cm cylinder gave an "invisible bump halfway up", and 3 cm pieces lost "a few per cent" of energy at the joins.
- **Settings:** friction 0.9 on rails and rollers, 0.81 at the contact (Bullet multiplies the two; the README and lua said 0.9 until 16 Oct 2026, they now give 0.81), restitution 0, damping 0, 30 solver iterations, 2400 steps per simulated second (DC_STEPS), integrated by Bullet's semi-implicit Euler.

The script itself works out the exact predicted centre height at every x. It finds by bisection the height at which the rail's centre line just clears a cone enlarged by $(\rho + \text{margin})/\cos(\text{taper})$, allowing for the off-centre contact point on sloping rails. It also tracks the total energy, kinetic (translational plus rotational) plus potential.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Rolls up or down (back)	rolls up iff $\tan\theta < \tan\varphi \cdot R/H$; $0.030 < 0.14 \cdot 5/12 = 0.0583$	rolls up 36.8 cm to the high end	correct
Rolls up or down (front)	$0.100 > 0.0583$, so rolls down	rolls down 23.9 cm	correct
Back cone: centre drop while rolling up 36.8 cm	exact geometry (script): 1.045 cm; thin-rail estimate $(\tan\varphi\ R/H - \tan\theta)\cdot\Delta x = 0.0283 \cdot 36.8 = 1.04$ cm (my arithmetic)	1.045 cm down	0.000 cm (exact); 0.2% (estimate)
Front cone: centre drop rolling down 23.9 cm	geometry 0.994 cm; estimate $(0.100 - 0.0583)\cdot 23.9 = 1.00$ cm	0.994 cm	exact; 0.6% (estimate)
Cylinder: centre drop rolling down 31.8 cm	geometry 0.955 cm; $\tan\theta\cdot\Delta x = 0.954$ cm	0.955 cm	exact
Largest deviation from predicted centre height along the run	0	back 0.0000 cm, front 0.0001 cm, middle 0.0001 cm (4 cycles)	≤1 µm

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
End-to-end centre change (48 cm), printed prediction	back −1.36 cm, cylinder +1.44 cm, front +2.00 cm	(prediction only)	—
Energy conservation (rolling, no slip)	ΔE = 0	loss as % of centre-fall energy, over 4 cycles: back 0.14–0.17%, front 0.09–0.12%, middle 0.20–0.26%	≤0.26% lost
Step-size check (DC_STEPS=600)	—	deviation ≤0.0004 cm; energy loss back 0.42–0.50%, front 0.48–0.57%, middle 0.62–0.65%	loss about 3× larger at 4× the step

Run used

```
cd demo/WyomingWill/DoubleCone && HOLD_S=0 timeout 1500 bpp -f double-cone.lua -n 1200     # 2400 steps/s, 4 cycles
HOLD_S=0 DC_STEPS=600 timeout 1500 bpp -f double-cone.lua -n 1200                          # step-size check
```

```
  Back   rails rise 1.5 cm over 50 (tan 0.030); the cone's centre would go down by 1.36 cm from end to end:
         tan(theta) 0.030 < tan(phi) R/H = 0.140 5.000 / 12.000 = 0.0583, so it rolls UP the rails (its centre going
    down)
  Middle a cylinder on the same rails: its centre rises 1.44 cm with them, so it rolls down
  Front  rails rise 5.0 cm over 50 (tan 0.100); the cone's centre would go up by 2.00 cm from end to end:
  Front  nears the low, narrow end at 1.55 s, having rolled 23.9 cm down the rails (the rails falling 2.39 cm): its
    centre went down 0.994 cm (the geometry: 0.994),
         off the geometry by 0.0001 cm at worst on the way; it lost 0.12% of the energy its centre's fall gave it
  Middle nears the low, narrow end at 1.80 s, having rolled 31.8 cm down the rails (the rails falling 0.95 cm): its
    centre went down 0.955 cm (the geometry: 0.955),
         off the geometry by 0.0001 cm at worst on the way; it lost 0.20% of the energy its centre's fall gave it
  Back   nears the high, wide end at 2.32 s, having rolled 36.8 cm up the rails (the rails rising 1.10 cm): its
    centre went down 1.045 cm (the geometry: 1.045),
         off the geometry by 0.0000 cm at worst on the way; it lost 0.14% of the energy its centre's fall gave it
```

The README's back-cone line (2.32 s, 36.8 cm, 1.045 cm, 0.0000 cm, 0.14%) matches this run's first cycle exactly. The README's "0.1 to 0.3%" energy loss matches the 0.09–0.26% range seen here.

Known departures and honest caveats

- The "geometry" check is the script's own exact contact geometry, which includes the rail thickness and Bullet's collision margin. It shows that Bullet holds the body on the ideal surface; it is not an independent physical measurement. The simple thin-rail formula agrees to within 0.6%.
- Energy is not exactly conserved. 0.1–0.3% of the energy released by the centre's fall is lost, and the loss roughly triples at 600 steps/s, so it is numerical (contact and solver dissipation), not physical rolling resistance. The losses also vary slightly from cycle to cycle (e.g. back 0.14–0.17%), because each restart begins from Bullet's leftover contact state.
- The rails were segmented into 10 cm capsules specifically to avoid Bullet contact artefacts: a bump with one long cylinder, energy loss at joins with 3 cm pieces (README). This was tuned, not derived.
- Friction 0.81 at the contact is high, chosen so that nothing slips; brass on brass is nearer 0.4. Tried 16 Oct 2026 (not adopted): with 0.4 at the contact (1800 frames) the times are the same, the heights within 0.001 cm, and each roll's energy loss moves by -0.03 to +0.06 points (0.11-0.25% of the centre's fall energy, against 0.09-0.26%), so the friction hardly matters here. Rolling resistance and air drag are absent.
- The checks are reported only between 6.5 cm from each end (away from the stops). Behaviour at the stops is not checked.
- Only the outcome (direction and centre descent) is checked. Times to reach the end (e.g. 2.32 s) are not compared with an analytic prediction.

References

- The 18th-century "mechanical paradox" lecture-room instrument (mentioned in code and README; no specific author or paper cited).

4.5 The oloid and the sphericon

Agrees with theory. Oloid's centre stays within 4.2426-4.5000 cm, the roller and sphericon level at 4.2426 cm, exactly as theory says; side-to-side weaves 3.6, 5.1 and 5.8 cm, now stated correctly.
Folder: demo/WyomingWill/Oloid/

What it shows

Three wooden rollers, each the convex hull of two circular arcs (radius 6 cm), are pushed along a level board at 15 cm/s. None is round, yet each rolls and lays its whole surface on the board while weaving from side to side. The oloid's (Schatz) centre of mass rises and falls a little as it rolls. The two-circle roller's (Stewart) and the sphericon's stay at one height, R/√2.

Model

Each roller is a Bullet rigid body (mass and principal moments from `reconstruction/gen-meshes.py`). Bullet's own collision is switched off (`collides=false`), and Bullet's gravity and gyroscopic term are off too (`setFlags(1)`). Before every physics step the script does three things. (1) It applies gravity (g = 981 cm/s²). (2) It advances the spin with Euler's torque-free equations, using RK4 in the body frame. (3) It finds the exact contact, the lowest point of each of the two arcs, from the analytic arc support function. Normal and Coulomb-friction impulses (μ = 0.6, a 2D friction cone) are solved together by sequential impulses, 8 iterations, with Baumgarte bias 0.2 on penetration. Rolling resistance is a moment δ·N with δ = 20 μm (`DELTA = 0.002` cm), applied against the rolling but not about the board normal. Bullet then only integrates positions and orientations. The run uses 2400 physics steps/s (`OL_STEPS`) and no damping. Theory for the height range is computed when the script loads: it scans 720 × 180 "down" directions and bisects for directions where both arcs reach equally far. Masses (this run, matching README): oloid 461.5 g, two-circle 496.2 g, sphericon 316.7 g (wood 0.7 g/cm³). Setting-down yaws (−14.7°, 12.0°, −25.6°) were "found by trying" so each rolls straight on average.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Oloid COM height range while rolling	between R/√2 and 3R/4 = 4.2426 to 4.5000 cm (script's own scan of the shape)	4.2426 to 4.5000 cm	exact to 4 decimals (0.1 μm)
Two-circle roller COM height	constant, 4.2426 cm (scan; Stewart roller)	4.2426 to 4.2426 cm	constant to 4 decimals
Sphericon COM height	constant R/√2 = 4.2426 cm	4.2426 to 4.2426 cm	constant to 4 decimals
Energy loss besides rolling resistance (should be 0 for rolling without slipping), 2400 steps/s	0	sphericon 0.229 %, two-circle 0.681 %, oloid 0.227 % of push KE	README: "0.2 to 0.7 %" — matches
Same, 9600 steps/s (step-size convergence)	0; README: "about a fifth" of the 2400 value	0.046 %, 0.126 %, 0.054 %	ratios 0.20, 0.19, 0.24 — matches README
Share of push KE taken by rolling resistance (δ = 20 μm)	(bookkeeping, no theory value)	6.6 %, 18.9 %, 15.9 %	—
Side-to-side weave	README/header now: two-circle 3.6, oloid 5.1, sphericon 5.8 cm; help text "3.6 to 5.8 cm" (fixed 15 Oct 2026; was "4 to 6 cm")	sphericon 5.8, two-circle 3.6, oloid 5.1 cm (same on every push in a 25 s run)	matches
Oloid volume	3.0524 R³ (README, from gen-meshes.py)	not re-run	README only

Run used

```
cd demo/WyomingWill/Oloid && HOLD_S=0 timeout 1500 bpp -f oloid.lua -n 720
cd demo/WyomingWill/Oloid && HOLD_S=0 OL_STEPS=9600 timeout 1500 bpp -f oloid.lua -n 720
cd demo/WyomingWill/Oloid && HOLD_S=0 bpp -f oloid.lua -n 1500      # 15 Oct 2026 re-run, weave figures, identical to
    the above
```

Default (2400 steps/s):

```
  Oloid     461.5 g; lying along a line on the board, its centre is between 4.2426 and 4.5000 cm up (theory)
  Sphericon reaches the far end at 3.93 s, 108.3 cm along the board, weaving 5.8 cm side to side:
            its centre 4.2426 to 4.2426 cm up (theory: 4.2426 to 4.2426);
            energy: of what the push gave it, 6.6% went to rolling resistance, and 0.229% otherwise (rolling without
    slipping, it should lose none)
  2-circle  reaches the far end at 6.83 s, 109.6 cm along the board, weaving 3.6 cm side to side:
            energy: of what the push gave it, 18.9% went to rolling resistance, and 0.681% otherwise (rolling without
    slipping, it should lose none)
  Oloid     reaches the far end at 9.02 s, 108.8 cm along the board, weaving 5.1 cm side to side:
            its centre 4.2426 to 4.5000 cm up (theory: 4.2426 to 4.5000);
```

OL_STEPS=9600:

```
            energy: of what the push gave it, 6.6% went to rolling resistance, and 0.046% otherwise (rolling without
    slipping, it should lose none)
            energy: of what the push gave it, 18.9% went to rolling resistance, and 0.126% otherwise (rolling without
    slipping, it should lose none)
            energy: of what the push gave it, 15.9% went to rolling resistance, and 0.054% otherwise (rolling without
    slipping, it should lose none)
```

The README's sample console block matches this default run line for line (times 3.93/9.02 s, distances, 0.229 %).

Known departures and honest caveats

- The height check partly tests the script against itself. The "theory" range comes from the same arc-support function the contact uses, scanned numerically, not from a closed-form derivation. It does agree with the closed-form values $R/\sqrt{2} = 4.2426$ and $3R/4 = 4.5000$.
- The min/max height is recorded once per frame (60 Hz), not every physics step.
- The contact model is the script's own sequential-impulse solver with Baumgarte stabilisation, not exact rolling constraints. The residual energy loss of 0.05-0.7 % is step-size error and shrinks about 5× from 2400 to 9600 steps/s.
- The rolling resistance (20 µm) is a chosen parameter. It takes 7-19 % of the push energy over the ~110 cm run.
- The setting-down yaw angles (12-26° off the axle) were tuned by hand so the rollers run straight on average.
- The board is level and the rollers are pushed. README: on a slope, past about 40 cm/s, one arc's normal force nearly vanishes and they skid. That limit was not tested here.
- Fixed 15 Oct 2026: README, header comment and help text said all three weave "4 to 6 cm"; the two-circle roller weaves 3.6 cm. They now give each body's measured weave (3.6 / 5.1 / 5.8 cm; help text 3.6 to 5.8 cm). The Cabinet's placard and catalogue text for this exhibit do not mention the weave. Text only; physics and console unchanged (hall vs standalone IDENTICAL, 900 frames).
- The oloid's centre height is not perfectly constant (it rises and falls by 0.26 cm). A real oloid on a slope therefore rolls unevenly; the exhibit shows this only as the pen trace.

References (as cited in README and code)

- Paul Schatz, the oloid, 1929.
- A. T. Stewart, the two-circle roller, 1966 (no title or journal given).
- Sphericon (no source cited).
- Mass properties: `reconstruction/gen-meshes.py` (NumPy/SciPy).

4.6 Euler's disk

Agrees with theory. Precession rate within 0.08-0.18% of theory from 25° down to 1°; with losses off, energy steady to the eighth figure. Its finite-time end follows its rolling-resistance law (now said), not Moffatt's air.
Folder: demo/WyomingWill/EulersDisk/

What it shows

A heavy steel disc is set rolling on its edge on a mirror at a 30° lean. It rolls round in place and leans further over as it loses energy. The flatter it gets, the faster its axis goes round, until it suddenly stops, flat. This is Joseph Bendik's 1990s toy; a spinning coin does the same.

Model

Bullet is not used for the dynamics. The README says Bullet's contacts "added energy about as fast as the rolling resistance took it". The script integrates the exact rolling-without-slipping equations: Newton's and Euler's equations for the disc, with the contact force solved each instant so that the contact point is stationary: $(I + m(r\cdot r - r\, r^{T}))\ \dot{\omega} = -\omega\times I\omega + m\, r\times(g\cdot up - \omega\times\dot{r}) +$ moments. Integration is RK4 at 9600 steps/s (`ED_STEPS`), with Gram-Schmidt re-orthonormalisation and the centre height reset from the contact geometry every step. The disc is steel (ρ = 7.85 g/cm³), a = 3.8 cm radius, 1.2 cm thick, rolling on the edge of its bottom face (h = 0.6 cm). Run values: m = 427 g, I1 = 1.594, I3 = 3.085 kg cm². Dissipation has two parts: (1) rolling resistance, a moment $\delta\cdot m\cdot g$ against the level part of ω, with δ = 30 µm (`ED_DELTA`); (2) a drag of 10 /s on the centre's sideways motion (`ED_DRAG`; README: "the air, and the mirror's grip not being perfect"). The disc is set going at α = 30° with the steady-precession Ω for that lean, 7.47 turns/s. It counts as stopped when α < 0.3°.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Precession Ω at 24.92°	$\Omega^2 = m\, g\, a\, (a\cos\alpha - h\sin\alpha) / (\sin\alpha\,(I1\, a\cos\alpha - I3\, h\sin\alpha))$ → 8.06 turns/s	8.04	−0.18 %
Ω at 19.88°	same → 8.88	8.87	−0.14 %
Ω at 14.98°	same → 10.11	10.10	−0.11 %
Ω at 9.99°	same → 12.25	12.24	−0.09 %
Ω at 5.97°	same → 15.73	15.72	−0.08 %
Ω at 4.00°	same → 19.15	19.14	−0.08 %
Ω at 1.98°	same → 27.14	27.11	−0.09 %
Ω at 0.99°	same → 38.33	38.29	−0.11 %
Rate of fall of √α near the end	$\delta\sqrt{(m\, g\, a / I1) / (3a)}$ = 0.00832 /s	0.00818 /s (line fit to samples with α < 2°)	−1.7 %
Finite-time singularity constant	$\Omega\,(t0 - t) = 3a/\delta$ = 3800 (t0 from the fit: 90.6 s)	3861 (0.32°), 3864 (0.69°), 3867 (1.20°), 3872 (1.84°)	+1.6 to +1.9 %
Energy conservation with no losses (ED_DELTA=0, ED_DRAG=0, 19.8 s)	E constant	2523.0883 kg cm²/s² at all 77 logged samples	no change at the logged 8 significant figures (< 4e-8 relative), with or without the drag; README now says "to the eighth figure over 20 s"
Stopping lean / time	(not theory) README: 0.30°, 81.73 s, 67 turns/s	0.30°, 81.73 s, 67 turns/s	identical to README

Run used

```
cd demo/WyomingWill/EulersDisk && HOLD_S=0 timeout 1500 bpp -f eulers-disk.lua -n 5400
cd demo/WyomingWill/EulersDisk && HOLD_S=0 ED_DELTA=0 ED_DRAG=0 ED_LOG=disk_e.csv timeout 1500 bpp -f eulers-disk.lua
    -n 1200
```

```
--- set going: leaning 30 degrees, going round 7.47 times a second; rolling resistance 30 microns ---
  the disc: 427 g, 7.6 cm across, 1.2 cm thick; I1 1.594, I3 3.085 kg cm^2
  t=  8.72 s  leaning 24.92 degrees: going round   8.04 times a second (theory, going round in place:   8.06; -0.18%)
  t= 51.17 s  leaning  5.97 degrees: going round  15.72 times a second (theory, going round in place:  15.73; -0.08%)
  t= 74.63 s  leaning  0.99 degrees: going round  38.29 times a second (theory, going round in place:  38.33; -0.11%)
  t= 81.73 s  it stops, flat, leaning 0.30 degrees, going round 67 times a second
              sqrt(alpha) falling at 0.00818 a second (theory, near the end: delta sqrt(m g a / I1) / (3 a) = 0.00832);
              so it would lie flat at t0 = 90.6 s, going round infinitely fast
              t= 77.17 s, leaning 0.69 degrees: W (t0 - t) = 3864 (theory: 3 a / delta = 3800)
              t= 68.63 s, leaning 1.84 degrees: W (t0 - t) = 3872 (theory: 3 a / delta = 3800)
```

Every number in the README's console sample (10.10/10.11, 15.72/15.73, 38.29/38.33, 81.73 s, 0.00818 vs 0.00832, t0 = 90.6 s, 3864 vs 3800) was reproduced exactly.

Known departures and honest caveats

- **The dissipation law is a modelling choice.** It is a constant rolling-resistance moment δ·m·g. That gives √α falling linearly, so $\alpha \propto (t0 - t)^2$ and $\Omega \propto 1/(t0 - t)$. Moffatt's finite-time-singularity analysis (Nature 2000) uses viscous dissipation in the air layer instead, giving $\alpha \propto (t0 - t)^{1/3}$. The exhibit's "theory" is the theory of its own dissipation model, so it cannot discriminate between real-world dissipation mechanisms. Since 15 Oct 2026 the README and lua header say this plainly: the finite-time end comes from the chosen rolling-resistance law, Moffatt's air law would give $\alpha \propto (t0 - t)^{1/3}$, $\Omega \propto (t0 - t)^{-1/6}$, and the exhibit can't tell them apart (README also cites Easwar, Rouyer and Menon 2002, vacuum experiments).
- The near-end law is a small-angle asymptote. The 1.7 % gaps (√α rate, Ω(t0 − t)) are consistent with the finite angles of the fit window (α < 2°) and with the extrapolated t0. Note t0 = 90.6 s, while the disc actually stops at 81.73 s.
- The extra sideways drag (10 /s) was added to keep the centre still. Without it, the README says, the centre drifts into mm-sized circles and Ω runs up to 20 % below the in-place theory. The 0.1 % agreement therefore depends partly on this tuned drag.
- The disc never slips and never bounces. The README admits the real toy stops around 0.3° when bouncing and slipping take over, which this model does not include. The 0.3° stop is an imposed threshold, not an emergent result.
- Fixed on 15 Oct 2026: the README and lua header said "430 g"; they now say 427 g, as the console prints.
- The energy test (no losses): re-run 15 Oct 2026 with ED_DELTA=0 alone and with ED_DELTA=0 ED_DRAG=0, 1200 frames (19.8 s): 2523.0883 kg cm²/s² at all 77 logged samples in both. The README and header used to say "a few parts in a million"; they now say "to the eighth figure over 20 s".

References

- Joseph Bendik, Euler's Disk toy (1990s), as named in README.
- H. K. Moffatt (2000), Nature: Euler's disk and its finite-time singularity (cited in README and lua header since 15 Oct 2026, as the contrasting air-viscosity law).
- K. Easwar, F. Rouyer, N. Menon (2002), vacuum experiments on the spinning disc (README, since 15 Oct 2026; journal not given there).

4.7 The rattleback

Agrees with theory. Spun the wrong way it reverses (0.67 s), the right way never; the reference model's growth rates (8.9/s and 1.7/s at 1 turn a second) and the rolling-resistance window now stated as measured. New README.
Folder: demo/WyomingWill/Rattleback/

What it shows

A rattleback (celt) is a boat-shaped body whose mass axes are turned a few degrees from the lines of its hull. Spun one way, it spins smoothly. Spun the other way, it starts to rock, stops, and spins back. Tapped on the end from rest, it starts spinning its preferred way. Three identical rattlebacks are shown: left, spun clockwise (the wrong way); middle, spun anticlockwise (its own way); right, "restless", spun clockwise on an almost lossless table, so it reverses more than once.

Model

The body is a solid half-ellipsoid with half-axes 6 × 1.5 × 1.2 cm, plastic at 1.2 g/cm³, with the hull turned 8° about the vertical from the principal axes. From `rattleback-data.lua`: m = 27.14 g, I = (0.01454, 0.20765, 0.19775) kg cm². Collision uses Bullet's own contact (unlike the other three exhibits here): a btConvexHullShape of 3073 surface points, 0.1 mm margin, contact-processing threshold 0.001 cm, friction 0.6, restitution 0. Bullet's gyroscopic term is off (`setFlags(0)`). Instead, Euler's torque-free equations are advanced with RK4 before every Bullet step. Rolling resistance is a moment δ·m·g against the horizontal part of ω about the contact, which is taken as directly below the centre of mass. Defaults: δ = 50 μm (left and middle) and 5 μm (restless). The run uses 1200 steps/s, no damping. Initial conditions: 1 turn/s about the vertical plus a 1 rad/s "hand wobble" about the long axis; the tap is 3 rad/s of pitch. Documentation: README.md (written 15 Oct 2026: what it shows, keys, model, theory and console, the reference model and its checks, files, headless env vars), the lua header, and `reference/` (rb_ref.py, an independent rolling-without-slipping ODE model; rb_scan.py; rattleback-test.lua).

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Mass properties	rb_ref.py: m = (2/3)π A B C ρ; I from the half-ellipsoid formulas	data file 0.0271434 kg; I = 0.0145353, 0.2076467, 0.1977530; rb_ref gives 0.02714336, 0.01453527, 0.20764671, 0.19775295	identical
Linear growth rate, wrong way (clockwise)	eigenvalues of linearised rb_ref (recomputed 15 Oct 2026); header/README now "8.9/s at 1 turn/s; 8.5/s at 6 rad/s" (fixed; header said 8.5 at 1 turn/s)	8.493 /s at 6 rad/s; 8.882 /s at 2π rad/s (1 turn/s)	matches
Growth rate, own way (anticlockwise)	code comment / README: 1.7 /s at 1 turn/s (README: 1.66 at 6 rad/s)	1.661 /s at 6 rad/s; 1.728 /s at 1 turn/s	OK
Skew 0 control	no instability	growth 0.0000 /s both ways (rb_ref); rb_ref integration: no reversal in 30 s	OK
Lossless reference reverses both ways	code: "unstable in BOTH directions"	rb_ref, ±6 rad/s, skew 8: first reversal at 4.68 s (+spin) and 1.07 s (−spin); energy drift −1.4e-9 / −1.5e-9	OK
Exhibit: wrong-way spin reverses	expected	Left REVERSED at 0.67 s (rocking up to 7°)	qualitative OK
Exhibit: own-way spin does not reverse	expected (δ = 50 μm damps the weak instability)	Middle never reversed in 60 s	OK
Restless (δ = 5 μm) reverses repeatedly	"again and again"	3 reversals: 0.65, 4.22, 6.23 s; then ~+0.02 turn/s creep to 60 s	qualitative OK; only 3 reversals
Tap from rest spins anticlockwise	expected	all three anticlockwise at 0.03 s; restless later reverses at 3.10 and 5.52 s	OK

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
δ window (left and middle; RB_ROLL, 30 s runs, exhibit start)	header/README now state the measured window (fixed 15 Oct 2026; header said "20-100 µm one-way; at 200 neither way reverses")	0, 10 µm: both reverse repeatedly. 20: Left reverses 3x (0.55, 5.35, 7.62 s). 30: Left 3x (0.52, 3.93, 5.38 s), Middle once at 8.73 s. 40, 50, 75, 100: one-way (Left once at 0.52-0.68 s, Middle never). 150, 300: neither reverses. 200: Left reverses late, 4.57 s	one-way window 40-100 µm; documented as such
Bullet harness vs reference, no losses (rattleback-test.lua, RB_EULER=1, RB_ROLL=0, skew 8, ±6 rad/s, 0.02 rad/s rock)	rb_ref first reversal: 1.07 s (−6, wrong way), 4.68 s (+6)	0.70 s (−6), 3.15 s (+6)	both ways, same order; Bullet reverses ~1/3 sooner

Run used

```
cd demo/WyomingWill/Rattleback && HOLD_S=0 timeout 1500 bpp -f rattleback.lua -n 3600
HOLD_S=0 RB_START=tap ... -n 1800 ;  HOLD_S=0 RB_ROLL={0,20,100,200} ... -n 1200
python3 reference/rb_ref.py (run via runpy, PYTHONDONTWRITEBYTECODE=1); linearised growth: rb_scan-style Jacobian on
    rb_ref
reference/: RB_SKEW={8,0} RB_SPIN={-6,6} RB_EULER=1 RB_ROLL=50 RB_LOG=... bpp -f rattleback-test.lua -n 1200
15 Oct 2026 re-runs: default -n 3600; RB_START=tap -n 1800; RB_ROLL={0,10,20,30,40,75,100,150,200,300} -n 1800;
  rb_ref.py and the growth rates again; rattleback-test.lua as above plus RB_ROLL=0 (skew 8, ±6), -n 600
```

```
  Restless REVERSED at  0.65 s: now spinning anticlockwise (rocking up to 8 deg)
  Left  REVERSED at  0.67 s: now spinning anticlockwise (rocking up to 7 deg)
t=  2.0s  turns/s: Left +0.62  Middle +0.79  Restless +0.71   (+ = anticlockwise)
  Restless REVERSED at  4.22 s: now spinning clockwise (rocking up to 16 deg)
  Restless REVERSED at  6.23 s: now spinning anticlockwise (rocking up to 16 deg)
skew 8 spin +6:  ... first reversal 4.68 s  rock 28.7 pitch 6.2  energy drift -1.4e-09
skew 8 spin -6:  ... first reversal 1.07 s  rock 28.7 pitch 6.1  energy drift -1.5e-09
skew 0 spin +6:  ... first reversal none  rock 0.0 pitch 0.0  energy drift -4.4e-14
(RB_ROLL=200)  Left  REVERSED at  4.57 s: now spinning anticlockwise (rocking up to 5 deg)
```

Known departures and honest caveats

- The theory comparison is **qualitative in the exhibit itself**. The console reports reversals and spin rates, never comparing them with a theory number. The quantitative checks (growth rates, the both-ways instability) live in `reference/rb_ref.py`, an independent rigid-ellipsoid, no-slip, energy-conserving model, not in the exhibit.
- The exhibit uses Bullet's general convex-hull contact, with friction, possible slip and a hull faceted into 3073 points. The reference model assumes no slip on a smooth ellipsoid. The two are not compared directly on reversal time or spin after reversal.
- One-way behaviour is **tuned via rolling resistance**. With no losses, this shape reverses both ways. The real asymmetry depends on table losses, a modelling choice. Fixed 15 Oct 2026: the header's window ("20-100 µm one-way; at 200 neither") is replaced by what runs from the exhibit's start show: one-way for 40-100 µm; at 20 and 30 µm the wrong-way one turns round and back (at 30 the own-way one also reverses once, late); at 150 and 300 neither reverses, but at 200 the wrong-way one still does, late (4.57 s), so above 100 it is near the edge. The window is narrow and not monotone; the default 50 µm sits inside it.
- The rolling-resistance contact point is approximated as directly below the centre of mass.
- After the restless one settles, it keeps a ~0.02 turn/s spin out to 60 s rather than coming to rest. This is likely contact-solver creep.
- Corrected 15 Oct 2026 (this dossier's earlier reading was wrong): in the reference harness (`rattleback-test.lua`, RB_ROLL=50) the body sits on the table from the start (height 0.7600 cm is the rest height plus the 0.01 cm margin). It spins at ±6.00 for ~10 s because the 50 µm rolling resistance at once removes almost all of the 0.02 rad/s starting rock (0.01 deg -> 0.0009 deg at the first frame); the rock then regrows slowly, and skew 8 at −6 rad/s reverses at 11.1 s; +6 does not, skew 0 neither,

in 20 s. With RB_ROLL=0 it reverses both ways (0.70 s and 3.15 s; rb_ref 1.07 and 4.68 s).

- Fixed 15 Oct 2026: the data-file fallback path was `demo/rattleback/`; it is now `demo/WyomingWill/Rattleback/`, and the exhibit runs from the repository root (checked).
- Fixed 15 Oct 2026: the header quoted 8.5 /s "at 1 turn/s"; that is the 6 rad/s value. It now says 8.9 /s at 1 turn/s (8.5 at 6 rad/s). The header's key list also said R spins "left clockwise, right anticlockwise" and T taps "both"; now left and right clockwise, middle anticlockwise, and all three.
- The restless one reverses 3 times (0.65, 4.22, 6.23 s), not "again and again" as the help text says; the header now gives the three times; the help text was left as it is.
- Text and a fallback path only; physics and console unchanged (hall vs standalone IDENTICAL, 900 frames; a fresh 600-frame standalone run matches the start of the earlier 3600-frame run line for line).
- Fixed 17 Oct 2026: **the start pause changed the console.** The `t=` lines every 2 s were printed on bpp's frame count (`N % 120`), which counts the pause's frames too, so with the pause they came at t = 1.5, 3.5, ... s instead of 2.0, 4.0, ... rattleback.lua now counts only the frames it runs (the same for the log file). With HOLD_S=0 the output is byte-identical to before (1800 frames); HOLD_S=0 for 1800 frames is now a prefix of the default for 1830 (also at 900 and 2400); hall against standalone (cmp2.sh, 1800 frames): IDENTICAL (18 run lines).

References

None are cited in code or reference files. The classical analyses (Walker 1896; Bondi 1986, Proc. R. Soc. A; Garcia & Hubbard 1988, Proc. R. Soc. A) are the standard theory for the spin-direction-dependent instability. They are added here for context and are not used by the exhibit.

4.8 The spinning eggs

Agrees with theory. The left egg rises at 0.30 s and stands at 0.68 s, then rolls and leans as spin friction slows it, back on its side at 24.40 s, at every step rate from 2400 to 19,200; an exact rolling integration gives 11.97° at 3 s and 24.39 s (the exhibit: 11.99°, 24.40 s). Rise from 60° to 30° within 9% of Moffatt and Shimomura (2.5% started in balance); spin falls at 0.934 turns/s² against 0.928 from Jellett's quantity; energy budget closes exactly.
Folder: demo/WyomingWill/SpinningEgg/

What it shows

A hard-boiled egg spun fast on its side rises onto its end and spins there like a top. Sliding friction at the table does it, as with the tippe top. Up on its end it rolls, and as spin friction slows it, it leans further and further over until it is back on its side. Spun faster, it also jumps briefly off the table as it rises. A raw egg spun the same way stays down: its liquid does not turn with the shell, the egg slows quickly, and after a finger stops it, it starts turning again by itself.

Model

Each egg is a Bullet rigid body; Bullet only integrates position and orientation (x += h v, q = exp(h w) q). The egg is a prolate spheroid with half-axes a = 2.85 cm and b = 2.15 cm, mass 58 g, with $A = m(a^2+b^2)/5$ across and $C = 2mb^2/5$ about the long axis. Before every step the script does the rest:

- gravity;
- Euler's equations, with Bullet's gyroscopic term off;
- the exact contact at the spheroid's lowest point: a unilateral normal impulse (so the egg can leave the table) and Coulomb friction, μ = 0.3, by sequential impulses over 6 iterations with Baumgarte bias 0.2;
- spin friction (a moment κ·N about the vertical) and rolling resistance (δ·N against the level part of w), both in the same solve, with κ = δ = 100 μm. The dish rim is a contact too.

Since 17 Oct 2026 the step is second-order and time-symmetric. The velocities it leaves belong to the middle of Bullet's move, so:

- no-slip and no-penetration are imposed at the lowest point half a step on, and the impulses act at the lowest point now;
- the gyroscopic term (implicit midpoint) and the sliding-friction direction are taken at the mean of the step's start and end, so friction opposes the mean slip;
- the contact is solved twice: a predictor with RK4 Euler, then a corrector with the predicted end state;
- the work tally takes each impulse times the step-mean velocity at its point, which is the exact energy identity of this step.

The run uses 4800 steps/s and no damping. Raw egg:

- A 0.35 mm shell (10 % of the mass) is the rigid body, with the potential-flow added inertia $m_f/5\cdot(a^2-b^2)^2/(a^2+b^2)$.
- The rest of the liquid is a second rotor, $I_{liq} = 0.1194$ kg cm², coupled to the shell by a linear drag with time constant 1.5 s.

Spins: left and right 25 turns/s, middle 40 turns/s. Each egg starts tipped 6° from horizontal, spun about the vertical.

Theory checks (default run, 4800 steps/s, unless noted)

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Spin when upright, left 25 t/s (Jellett, Moffatt & Shimomura 2002)	W = W0·A b/(C a) = 26.00 t/s; corrected for the J removed by spin friction and rolling resistance: 25.15	25.22 t/s	J changed +0.60 %
Same, middle 40 t/s (jumps)	41.60; corrected 40.26	40.76	+1.29 %
Same, κ = 0 (EG_KAPPA=0), left; and EG_SPIN=30	25.85; 31.02	25.94; 31.25 (identical at EG_STEPS 4800/9600/19200)	+0.58 %; +0.83 %
Rate of rise, left, 60° → 30° from upright	Moffatt-Shimomura gyroscopic balance, sliding: J dθ/dt = −μ M g h², J from the egg at 60°, κ taking J: 0.239 s	0.261 s, the same at EG_STEPS 2400 /4800/9600/14400/19200/38400	+9.1 %. Started in the balance (EG_BALANCED=1): 0.247 vs 0.241 s, +2.5 % (+2.3 to +2.7 % over 2400-19200)
Same, middle (jumps)	0.384 s	0.471 s at every step	+22.7 %; EG_BALANCED=1: 0.407 vs 0.385, +5.7 %
Energy budget, left: lost vs friction + landing work	equal (exact identity of the step: impulse · step-mean velocity)	41.88 vs 41.88 μJ	0.00 %
Energy budget, middle	equal	109.15 vs 109.15 μJ	0.00 %
Energy budget vs step (EG_STEPS 2400-38400)	→ 0	within ±0.01 % at every step, both eggs	closes. Before 17 Oct: −2.56 / −1.29 / −0.69 / −0.40 / −0.30 % (left), shrinking with the step
Rolling egg keeps energy (EG_KAPPA=0 EG_DELTA=0; Chaplygin's rolling body of revolution is conservative and integrable)	E constant, tilt bounded	1 → 10 s: E spread 0.0025 / 0.0019 / 0.0023 μJ (of 161.15) at 2400 / 4800 / 9600; tilt 9.4-10.6°	conserved. Before 17 Oct the tilt crept to 0.4-0.8° (nearly upright) by 10 s

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Rise onset vs step (2400-38400)	should not depend on step	left: rising at 0.30 s, standing at 0.68 s at every step; middle 0.50 s and 1.22 s	converged
Standing phase vs step, left (EG_STEPS 2400/4800/9600/19200/38400, 1800 frames)	should not depend on step	tilt at 3 s: 12.01 / 11.99 / 11.97 / 11.97 / 11.89°; at 15 s: 22.64 / 22.65 / 22.66 / 22.66 / 22.53°; falling back (15°) 7.97 s (18.5 t/s) at 2400-19200, 8.15 s at 38400; back on its side 24.40 s at all	**converged** (fixed 17 Oct 2026). 2400-19200 agree to 0.05° and 0.01 s; 38400 is off by single-precision rounding (see caveats)
Standing phase vs exact rolling, left (reconstruction/rolling-check.py: rolling-without-slipping equations + κ, δ, DOP853 rtol 1e-11, from the exhibit's state at 1 s)	tilt 11.97° (3 s), 17.10° (10 s), 22.69° (15 s); past 15° at 8.05 s (console rule); 60° at 24.39 s; friction ≤ 0.855 μN (rolling holds)	11.99°, 17.12°, 22.65°; 7.97 s; 24.40 s	≤ 0.04°; −0.08 s; +0.01 s
Spin-down standing: J falls at the moments' rate; standing J = C a W, so dW/dt = (dJ/dt)/(C a)	κ only, κ M g h/(C a): 0.83 t/s²; with the reference's dJ/dt = −1.782/s: 0.928 t/s²	dJ/dt −1.782/s, dW/dt −0.934 t/s² (2-15 s; identical at 2400/4800/19200)	0.6 % vs the J rate; κ-only estimate 11 % low (it leaves out δ and the tilt)
Middle egg standing (EG_STEPS 2400-38400)	should not depend on step	tilt at 3 s 4.84 / 4.88 / 4.91 / 4.90 / 4.73°; falling back 25.78 / 25.78 / 26.05 / 26.50 / 27.23 s	close, not tight (see caveats)
Jump threshold (Shimomura et al. 2006 observed jumps; README: "about 30 turns a second or faster")	—	29 t/s: none; 30: 4 jumps (16.0 ms, 0.10 mm); 32: 9 (≤ 0.45 mm); 34: 14 (≤ 0.62 mm); same counts at 19200	OK
Jumps at 40 t/s (README: 25 jumps, ≤ 0.6 mm, 21 ms)	measured real eggs: fractions of a mm, up to a few 10 ms	25 jumps, 0.29-1.09 s, longest 21.3 ms, highest 0.59 mm	matches README
Jumps vs friction at 35 t/s (README: still jumps for μ 0.2-0.5)	—	μ = 0.2: 22 jumps; μ = 0.5: 9 jumps	OK
Raw egg: shared spin of shell + liquid	W0·I_s/(I_s+I_liq) = 5.49 t/s (I_s 0.0336, I_liq 0.1194)	κ = 0 at 1.2 s: shell 5.87, liquid 5.27 (moment-weighted 5.40, my calculation); default κ: 4.49 / 4.75	κ = 0: −1.6 %. README says "nearly do", with these numbers
Raw egg re-spin after finger (κ = 0)	liquid's L shared: 3.40 t/s	3.37 t/s (same at 9600 and 19200)	−0.9 %
Raw egg re-spin, default κ	README: "about half"	1.43 vs 2.78 t/s	OK
Falls back (README: past 15° after 8 s, back on its side after 24 s under 3 t/s)	—	7.97 s (18.5 t/s); 24.40 s (2.6 t/s)	OK

Run used

```
cd demo/WyomingWill/SpinningEgg && HOLD_S=0 EG_STEPS={2400,4800,9600,19200,38400} EG_LOG=... timeout 3000 bpp -f
    spinning-egg.lua -n 1800   (also EG_STEPS=14400 -n 120; 76800 -n 600)
HOLD_S=0 EG_BALANCED=1 [EG_STEPS=2400|19200] ... -n 120;  EG_KAPPA=0 [EG_STEPS=9600|19200] -n 300;  EG_KAPPA=0
    EG_SPIN={20,30} EG_STEPS={4800,9600,19200} -n 300
EG_KAPPA=0 EG_DELTA=0 EG_STEPS={2400,4800,9600} EG_LOG=... -n 600;  EG_FAST={26,28,29,30,32,34} -n 150 (30-34 also at
    19200);  EG_FAST=35 EG_MU={0.2,0.5} -n 240
EG_TILT={3,3.5,4,5} EG_STEPS={2400,4800,19200[,9600,38400 for 3]} -n 300;  EG_SPEED={0.5,2} with EG_LOG (speed check)
HOLD_S=0 EG_STEPS={4800,19200} EG_STATE=1.0:state.txt -n 61; python3 reconstruction/rolling-check.py state.txt [0 0
    10]
```

```
  Left   rising at  0.30 s (tilt 31 deg, spinning 25.2 turns/s)
  Left   standing on its end at  0.68 s (tilt 11 deg, spinning 25.2 turns/s)
         spinning 25.22 turns/s; by Jellett's J it should be 25.15 (J 48.59, from 49.87, 48.30 with spin friction:
    +0.60%)
         energy lost 41.88 microjoules; friction took 41.88, landings -0.00: 41.88 (0.00% apart)
         it rose from 60 to 30 deg in 0.261 s; by J dtheta/dt = mu M g h^2 (sliding) it should take 0.239 (+9.1%)
  Middle standing on its end at  1.22 s (tilt 4 deg, spinning 40.8 turns/s)
         it rose from 60 to 30 deg in 0.471 s; by J dtheta/dt = mu M g h^2 (sliding) it should take 0.384 (+22.7%)
         it jumped 25 times as it rose (0.29 to 1.09 s): longest 21.3 ms, highest 0.59 mm off the table
  Left   falling back at  7.97 s (tilt 15 deg, spinning 18.5 turns/s)
  Left   back on its side at 24.40 s (tilt 63 deg, spinning 2.6 turns/s)
  Middle falling back at 25.78 s (tilt 15 deg, spinning 18.7 turns/s)
  rolling-check.py: past 15 deg (as the console counts): 8.05 s, spinning 18.7 turns/s; back at 60 deg at 24.39 s;
    most friction needed: 0.855 of mu N
```

The default-run lines match the README's console sample digit for digit. Hall and standalone output IDENTICAL (cmp2.sh, 1800 frames, 22 run lines); HOLD_S=0 for 1200 frames is an exact prefix of the default 0.5 s pause for 1230 (28 lines). Speed: the hard-boiled eggs' EG_LOG rows are bit-identical at EG_SPEED 0.5, 1 and 2 at every common time; console lines are sampled on frames (and the finger acts on frames), so their times and values move by up to a frame with speed, as before. Cost (CPU, 1800 frames, measured while two other processes held both cores): 8.4-8.6 ms/frame standalone (the old step: 7.6), 13.6 ms/frame in the Cabinet with the egg; frame budget 16.7 ms. Default steps unchanged (4800).

Known departures and honest caveats

- The egg is a symmetric spheroid; a real egg is fatter at one end (README admits this).
- **Fixed on 17 Oct 2026: the standing phase depended on the step.**
  - Before, at 2400 / 4800 / 9600 / 19200 / 38400 steps/s: tilt at 3 s 1.8 / 4.5 / 7.3 / 9.3 / 10.5°; falling back at 23.4 / 22.3 / 20.6 / 17.6 / 13.9 s; on its side 26.5 / 26.1 / 25.7 / 25.2 / 24.8 s.
  - Cause: Bullet's update makes the step a leapfrog. The velocities after the kick live at the half step, but no-slip (and the normal velocity) was imposed at the start-of-step lowest point. Rolling, the contact runs round the egg at the spin rate, so each step left an O(h) slip.
  - Two further first-order pieces showed up once that was fixed. First, the gyroscopic term was taken at the pre-step w. With κ = δ = 0 the energy then grew +8.7 / +4.4 / +2.2 μJ in 9 s at 2400 / 4800 / 9600; the implicit midpoint removes this. Second, spin friction and rolling resistance were applied after the no-slip solve. Moving them into the solve was harmless but is kept for consistency.
  - The earlier attempt (contact at the mid-step orientation, lever arm also there) was reproduced: 60° → 30° in 0.238 s at 4800, 0.243 at 2400. It is not time-symmetric; the adopted step puts the lever arm at the start of the step.
  - Converged by Richardson-style comparison and by an independent double-precision rolling integration (table).
- **Changed friction law (17 Oct 2026).** The old solver clamped the stick impulse radially to μN, so in the limit h → 0 friction pointed along $-K^{-1}\cdot$slip, not −slip, with K the anisotropic contact mass matrix. That is not Coulomb's law. Friction now opposes the slip, taken at the time of the kick. The rise check moved from 0.259 vs 0.240 (+8.2 %) to 0.261 vs 0.239 (+9.1 %), and the middle egg from +22.0 % to +22.7 %. The rise is converged with the new law: tilt at 0.5 s is 36.48 / 36.54 / 36.60 / 36.61 / 36.62° at 2400 / 4800 / 9600 / 19200 / 38400.
- **The energy budget is now an exact identity** (work = impulse · step-mean velocity, and the gyroscopic step conserves energy exactly). Its 0.00 % shows the integrator makes or loses no energy

besides friction; it no longer measures the step error as it did before.

- **Single-precision floor.** Bullet stores the state as floats. Above about 20000 steps/s the rounding accumulates with the number of steps. Against the double-precision reference (17.08-17.10° at 10 s) the tilt is off by +0.01° at 19200, −0.05° at 38400 and −0.15° at 76800. It is not the position: the same with all dishes at x = 0. So 38400 gives falling back at 8.15 s instead of 7.97.
- **Middle egg standing phase** agrees less tightly. Falling back is 25.8 s at 2400-4800, 26.1 at 9600, 26.5 at 19200 and 27.2 at 38400. Its 25 inelastic landings leave the standing state slightly different (tilt at 1 s 28.8-29.3°), and the float floor adds to it at the finest steps. Not investigated further.
- **"Falling back" is a 15° threshold on a slow lean.** The egg never stands still and then falls. It stops rising at 11° (rolling: the rise needs sliding) and leans steadily as it slows. The console's 15° and 60° crossings, debounced 0.25 s on frames, carry the nutation (±0.5°) in them: the reference crosses at 8.05 s against the exhibit's 7.97.
- The upright rolling egg would be gyroscopically stable above about 5.8 t/s. That is my calculation, bisecting simulations of the same rolling equations from upright with a 0.5° kick. The frictionless-table condition $C^2n^2 > 4\,A\,M\,g\,(a - b^2/a)$, from memory, gives 9.5 t/s. The egg never stands upright, though, so neither is a fall-back prediction. The lean is a slow drift of the rolling motion as J falls, which the rolling reference reproduces.
- Similar consequences of rolling once standing:
  - at 20 t/s with $\kappa = 0$ the left egg stops rising at about 18° and never "stands" (< 15°), identically at 4800 and 19200 (before: 1.25 s at 4800, 2.13 s at 9600, never at 19200);
  - at $\mu = 0.5$ the left egg stops at about 17° (before: "standing" 0.72 s at 13°).
- Start tilt (fixed 15 Oct, now re-measured): from 3° the left egg never rises at any step from 2400 to 38400 (converged; before it rose at 0.50-2.45 s depending on the step). At 3.5° the onset is still step-sensitive, 0.38 / 0.38 / 0.42 s at 2400 / 4800 / 19200, because 3.5° is at the stick/slip edge. From 4°: 0.35 s at every step. The default 6° rises at 0.30 s.
- The rise-rate check: the left egg rises 9 % slower than Moffatt-Shimomura's leading-order balance. The gap is the nutation from starting with pure spin about the vertical; started in balance it is 2.5 %. The middle egg is 23 % slower (6 % from balance); it jumps throughout its rise.
- Jellett's J is only approximately conserved (0.6-1.3 % change). The prediction is corrected for the J removed by spin friction, rolling resistance and the rim (the console calls the sum "spin friction"). The bare prediction is 26.00 t/s.
- Spin friction κ and rolling resistance δ (100 µm each) are chosen parameters. κ sets how fast the egg slows and so when it is back on its side.
- The contact is a sequential-impulse solver, not an exact unilateral constraint; landings are perfectly inelastic. The dish rim's contact still uses the lowest point at the step's start (first-order). The left and middle eggs never touch it in the default run.
- The raw egg's liquid is a two-rotor model (added inertia plus a linearly coupled rotor, $\tau = 1.5$ s, assumed). Its finger acts on frames, so its numbers move slightly with EG_SPEED.

References

- Cited in the README:
  - H. K. Moffatt and Y. Shimomura, "Spinning eggs — a paradox resolved", Nature 416, 385 (2002).
  - Y. Shimomura and colleagues, Proc. R. Soc. A (2006), on spinning eggs jumping; reported in Science News, 23 May 2006, "Well-spun egg also jumps".
  - R. McLachlan and M. Perlmutter, "Integrators for nonholonomic mechanical systems", J. Nonlinear Sci. (2006), doi 10.1007/s00332-005-0698-1: reversible integrators for rolling-type constraints, with energy error that random-walks rather than drifts.
  - K. Modin and O. Verdier, "What makes nonholonomic integrators work?", Numer. Math. 145, 405-435 (2020): reversibility, through reversible KAM theory, explains the near-conservation.

- Not cited in the README:
  - The rise equation $J\, d\theta/dt = -F\, h^2$ and the balance $C\, n = A\, \Omega \cos\theta$, as restated in K. Sasaki, "Spinning eggs — which end will rise?", arXiv:physics/0310163 (eqs. 7, 9-11).
  - From memory, not checked: the integrability of a rolling body of revolution (Chaplygin), e.g. A. V. Borisov and I. S. Mamaev, Regul. Chaotic Dyn. 7, 177 (2002); and the frictionless-plane stability condition quoted above.

4.9 The tippe top

Agrees with theory, with a numerical or open caveat. Inversion criterion right for all three tops; centre of mass rises 1.057 to 2.545 cm as theory says; stem threshold about 80 rad/s, within the sleeping-top rule's bounds. Open: the green top's spin collapses after 40 s without rolling resistance.
Folder: demo/WyomingWill/TippeTop/

What it shows

A ball with a short stem, its centre of mass a little below the ball's centre. Spun on its ball, stem up, it turns itself over within a few seconds and spins on its stem, raising its centre of mass; sliding friction at the table does the work. Three tops: red, spun fast (40 turns/s), turns over and stands on its stem; blue, the same top spun slowly (15 turns/s), only gets as far as lying on ball and stem; green, with its centre of mass almost at the ball's centre, never turns over. All end stem up again as they slow.

Model

- Top: one Bullet rigid body, 15 g, hollow ball R = 1.5 cm; I3 = 2/3 m R^2 = 0.0225 kg cm^2 about the stem, I1 = 0.9 I3 across. Centre of mass a = 0.45 cm below the ball's centre (alpha = a/R = 0.3); green top a = 0.06 cm (alpha = 0.04).
- Contact shape: Bullet compound of a true sphere (the ball) and a capsule (the stem, round tip radius 0.6 cm standing 0.6 cm out). Sliding friction 0.2 (Bullet), restitution 0, contact processing threshold 0.001.
- Bullet's gyroscopic term is off; before every physics step the script advances the body-frame angular velocity by Euler's equations with RK4 (as in the rattleback exhibit).
- Script-applied losses: "spin friction", a deceleration 0.5 rad/s^2 of the spin component about the vertical; rolling resistance, a constant moment delta*m*g against the horizontal angular velocity, delta = 30 microns, with the matching linear-velocity change about the contact.
- 4800 physics steps/s (80 per 1/60 s frame), no damping, g = 981 cm/s^2. Initial state: stem up tilted 12 deg (random azimuth), spin about the stem.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Inversion possible, red/blue (alpha = 0.3)	1 - alpha < I1/I3 < 1 + alpha: 0.7 < 0.9 < 1.3 (README)	red inverts (on its side 2.5 s, up on its stem 3.5 s); blue reaches 124-127 deg (ball and stem)	qualitative, agrees
Inversion impossible, green (alpha = 0.04)	0.9 outside 0.96-1.04 (README)	max tilt 1 deg (2 s), 0 deg thereafter; never turns over in 120 s	agrees
Centre-of-mass height, start	R - a cos(12 deg) = 1.06 cm	1.057 cm (TT_LOG)	agrees
CoM height standing on stem	stem-tip contact 2.55 cm below the CoM: 2.55 cm	max 2.545 cm, 2.49 cm at 10 s (tilt 166 deg)	agrees (CoM rises ~1.5 cm)
Green top CoM height	R - a = 1.44 cm	1.441 -> 1.440 cm	agrees
Time to turn over (red)	README now: "about 3.5 seconds", example "Left up on its stem at 3.5 s" (fixed 15 Oct 2026; was "about 3", "3.3 s")	on its side 2.5 s, up on its stem 3.5 s (re-run identical)	matches

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Spin needed to stand on the stem	README/lua now: "about 80 rad/s or more" with the rule's bounds (fixed 15 Oct 2026; was "about 100"). w > 2 sqrt(I1' m g d)/I3, d = 1.95 cm: 68 rad/s with I1' about the CoM (tip sliding freely); 132 (I about the tip-sphere centre) to 163 rad/s (I about the contact) for a tip that can't slide	TT_LOG: up on its stem with ~91 rad/s; steady (tilt 171-173 deg) while >= 83 rad/s; tilt first below 170 deg at 24.0 s with 81.8 rad/s; onto ball and stem at 38.7 s; blue never above 44 rad/s after 8.3 s, never stood	between the bounds, nearer the slippery one
Red returns stem up	README: sinks back "twenty-odd seconds later"; all three stem up in ~1.5 min	sinking from ~24-26 s; ball+stem 38.7 s, on ball 66.6 s, 0 deg by 92 s; blue 0 deg by 76 s	agrees
Without spin friction	README: red "spins on its stem for ever"	TT_SPINF=0: 175 deg, 14.7 t/s still at 60 s	agrees (60 s run)
Rolling resistance, effect (TT_ROLL sweep, 15 Oct 2026)	README/lua now give the measured effect (fixed; was "100 microns stops the top turning over at all")	red up on its stem at 3.4-3.7 s for every value 0-300 µm; back onto ball and stem at 47.0 s (0), 22.4 (10), 38.7 (30), 37.1 (50), 33.8 (75), 30.0 (100), 23.1 (150), 12.8 s (300). Blue: reaches ball and stem at 30 µm (and 10), stays within 8 deg (2-s reports) at 50-300. At 0 nothing settles: blue rocks 16-65 deg, green's tilt grows from ~40 s to 71 deg at 78 s with its spin collapsing (36 -> 0.4 t/s)	documented
Determinism	(repeat run)	identical event times on a second run	reproducible

No conservation check is printed by the exhibit; Jellett's integral (the quantity conserved during sliding on the ball) was not logged and was not checked here.

Run used

```
cd demo/WyomingWill/TippeTop
HOLD_S=0 bpp -f tippe-top.lua -n 7200           # main run, 120 s
HOLD_S=0 TT_LOG=tt.csv bpp ... -n 600           # CoM heights
HOLD_S=0 TT_ROLL=100 bpp ... -n 1200            HOLD_S=0 TT_SPINF=0 bpp ... -n 3600
# 15 Oct 2026: HOLD_S=0 TT_LOG=tt.csv bpp ... -n 7200 (same events as above);
#   TT_ROLL={0,10,50,75,100} -n 4800; TT_ROLL={150,300} -n 1800
```

```
  Left   on its side at   2.5 s (stem 83 deg from up)
  Left   up on its stem at   3.5 s (stem 156 deg from up)
t=  4.0s  stem from up, spin: Left 161 deg 14.6 t/s | Middle  10 deg 13.9 t/s | Right   1 deg 37.2 t/s
  Middle on its ball and stem at   8.3 s (stem 124 deg from up)
t= 26.0s  stem from up, spin: Left 160 deg 12.6 t/s | Middle 113 deg  5.0 t/s | Right   0 deg 35.4 t/s
  Left   on its ball and stem at  38.7 s (stem 127 deg from up)
  Middle spinning on its ball at  49.5 s (stem 59 deg from up)
  Left   spinning on its ball at  66.6 s (stem 59 deg from up)
t= 92.0s  stem from up, spin: Left   0 deg  1.2 t/s | Middle   0 deg  0.1 t/s | Right   0 deg 30.0 t/s
  Left   up on its stem at   3.4 s (stem 179 deg from up)        [TT_ROLL=100]
```

Known departures and honest caveats

- Several parameters were tuned to make the exhibit work (README): a hollow ball (a solid 2/5 m R^2 ball turned over but never stood up), a short round-ended stem (a long thin one stuck leaning), rolling resistance 30 microns (to settle the tops), spin friction 0.5 rad/s^2 (otherwise the red top never comes down).
- Spin friction and rolling resistance are phenomenological torques applied by the script, not contact physics; contact friction is Bullet's (Coulomb, 0.2).
- Event times depend on the random tilt direction (math.random); two runs here gave identical times, but other builds/Lua versions may differ.
- The stand-on-stem threshold is not a sharp prediction; the stem tip is a 6 mm sphere, and contact friction makes the effective criterion lie between the slippery and fixed-point bounds I estimated

(68-132 rad/s).

- Fixed 15 Oct 2026: README and lua said 100 microns of rolling resistance stops the (red) top turning over; it does not, at any value tried up to 300 µm. They now say what it does: settles the tops (with none they rock on), brings the red one down off its stem sooner (not monotonically: 10 µm 22 s), and from 50 µm keeps the slow blue one from turning over. Also fixed: the README's console example (3.3 s -> 3.5 s), "about 3 seconds" -> 3.5, the stem threshold "about 100 rad/s" -> about 80 with the run and the rule's bounds, and the header's env-var list (TT_SPINF, TT_ROLL added).
- Open: with no rolling resistance the green (centred) top, which cannot invert, starts rocking after ~40 s and loses its spin within ~10 s (25.6 -> 3.2 t/s between 38 and 46 s; the spin friction alone would take 0.6 t/s), far faster than the spin friction (0.5 rad/s^2) allows. Not investigated; likely a contact-solver effect that the rolling resistance otherwise damps. The README says it hasn't been looked into.
- Text only; physics and console unchanged (hall vs standalone IDENTICAL, 900 frames).
- The green "centred" top spins on for a long time (27.7 t/s at 120 s), slowed only by spin friction; it is "stem up" throughout.
- The state labels (on its side, on ball and stem, up on its stem) come from tilt and height thresholds held for 0.33 s; reported times lag the actual change by that much.
- Fixed 17 Oct 2026: **the start pause changed the console.** The `t=` lines every 2 s were printed on bpp's frame count (`N % 120`), which counts the pause's frames too, so with the pause they came at t = 1.5, 3.5, ... s instead of 2.0, 4.0, ... tippe-top.lua now counts only the frames it runs (the same for the log file). With HOLD_S=0 the output is byte-identical to before (1800 frames); HOLD_S=0 for 1800 frames is now a prefix of the default for 1830 (also at 900 and 2400); hall against standalone (cmp2.sh, 1800 frames): IDENTICAL (19 run lines).

References

None cited in the README or code (the lua header mentions Bohr and Pauli watching one). The inversion condition 1 - alpha < I1/I3 < 1 + alpha is the standard result from the tippe-top literature, not referenced in the exhibit.

4.10 The Dzhanibekov effect

Agrees with theory. Flip period 3.3129 s against 3.3131 s from the exact elliptic-function solution (-0.01%), 36 flips in 120 s. Energy and angular momentum are held by projection, and the README now says so. New README.
Folder: demo/WyomingWill/Dzhanibekov/

What it shows

Three identical aluminium T-handles shaped like a wing nut float without gravity, each spun at 1 turn/s about a different principal axis with the same 1% extra wobble. The one spun about its intermediate axis (the stem) flips end over end every few seconds, again and again, as Vladimir Dzhanibekov's wing nut did on Salyut 7 in 1985; the two spun about the smallest (bar) and largest (across both) axes just spin steadily (the "tennis racket theorem").

Model

(Facts from the lua header, `dzhanibekov-meshes/t-handle-data.lua`, and README.md, written 15 Oct 2026.)

- Body: solid aluminium (2.7 g/cm^3) T-handle, bar r = 0.5 cm x 10 cm along body x, stem r = 0.8 cm x 4 cm along -y; mass 42.92 g; principal moments I = (0.102132, 0.184989, 0.277521) kg cm^2 about (bar, stem, across). Stem = middle moment.
- Each handle is a Bullet rigid body (convex-hull shape used only for mouse hover; no gravity, no contacts), damping 0. Bullet integrates only the orientation from the angular velocity.

- Bullet's gyroscopic term is off (setFlags(0)). Before every physics step the script advances the body-frame angular velocity by Euler's free-body equations with classical RK4, then projects it back onto the initial kinetic energy and |L| (two Newton steps along the gradients of E and L^2), and writes it back to Bullet.
- 600 physics steps/s by default (10 per 1/60 s frame; slider/DZ_STEPS 120-2400). Initial state: w = W about the chosen axis plus eps*W about the next axis, W = 2pi rad/s, eps = 1%.
- Console: each sign change of the spin component along the chosen axis is a "flip"; every 10 s it prints relative energy and angular-momentum error and the maximum tilt of the spin axis from vertical.
- Companion script `bullet-rotation-test.lua` (I = 1:2:3, w0 = (0.01, 10, 0) rad/s) tests Bullet's own gyroscopic options against Euler's equations.

Theory checks

Exact values computed now from Euler's equations: closed form T = 4K(k)/w_p with w_p = sqrt((I3-I2)(L^2-2E I1)/(I1 I2 I3)), k^2 = (I2-I1)(2E I3-L^2)/((I3-I2)(L^2-2E I1)) (case L^2 > 2E I2, here k^2 = 0.99968), cross-checked by a high-precision ODE integration (scipy DOP853, rtol 1e-11) that also gives first-flip times and axis tilts.

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Time between flips, stem (middle axis)	T/2 = 2K(k)/w_p = 3.3131 s	mean of 35 intervals 3.3129 s (each printed as 3.30 or 3.32, frame-quantised)	-0.01%
Time of first flip	ODE: 1.657 s (1.6566)	1.67 s (reported at frame end, 1/60 s resolution)	within one frame
Flips in 120 s	ODE: 36	36 (last at 117.62 s; ODE 117.615 s)	exact
Bar (smallest axis): steady	ODE: 0 flips, axis tilt max 2.08 deg	0 flips, "axis wandered up to 2 deg"	agrees (console rounds to 1 deg)
Across (largest axis): steady	ODE: 0 flips, axis tilt max 0.60 deg	0 flips, "up to 1 deg"	agrees to the console's rounding
Energy conservation (all three, 120 s)	E constant	worst printed relative error 4.3e-7	~single-precision level
Angular momentum conservation	L constant	worst printed relative error 2.2e-7	~single-precision level
Flip period claim (lua header)	header now: "3.3129 s between flips (mean of 35, over 2 minutes) against the exact 3.3131 s" (fixed 15 Oct 2026; was "to 0.2%, with energy and angular momentum exact")	3.3129 s (re-run 15 Oct 2026: 36 flips, last 117.62 s)	-0.01%
Bullet gyro OFF, no Euler step (test, 60 s)	header: "nothing flips"	0 flips (exact: 21)	confirms
Bullet "explicit" gyro (flag 2)	header: energy +59%	E +59.1%, L +26.1%; mean flip interval 2.34 s vs exact 2.873 s	confirms
Bullet default "implicit" (flag 8)	header/README now: energy -31.9%, L -17.5%, flips on time (fixed 15 Oct 2026; header said -44% / -25%)	re-run 15 Oct 2026: E -31.9%, L -17.5%; first flip 1.433 s vs exact 1.437 s; same with flag 4	matches
Implicit at 16x steps (9600 Hz)	header now: "-3.3% and -1.7% a minute"	E -3.3%, L -1.7% (re-run)	confirms

Linearised instability rate (for context, not printed by the exhibit): lambda = W sqrt((I2-I1)(I3-I2)/(I1 I3)) = 3.27 /s (e-folding 0.31 s).

Run used

```
cd demo/WyomingWill/Dzhanibekov
HOLD_S=0 DZ_LOG=dz.csv bpp -f dzhanibekov.lua -n 7200
DZ_FLAGS={0,2,4,8} DZ_LOG=brt_N.csv bpp -f bullet-rotation-test.lua -n 3600   (plus DZ_FLAGS=8 DZ_HZ=9600; DZ_FLAGS=0
    DZ_EULER=1)
15 Oct 2026 re-runs: HOLD_S=0 bpp -f dzhanibekov.lua -n 7200; DZ_FLAGS={0,2,4,8} and DZ_FLAGS=8 DZ_HZ=9600, -n 3600
    (all as before)
```

```
--- spin 1.0 turns/s, nudge 1.0% ---
  Left   flipped at   1.67 s  (flip 1)
  Left   flipped at   4.98 s, 3.32 s after the last  (flip 2)
  Left   flipped at   8.28 s, 3.30 s after the last  (flip 3)
t=   10s  Left: flips 3, axis wandered up to 90 deg, energy +8.0e-08, ang. mom. +4.0e-08 | Middle: flips 0, axis
    wandered up to 2 deg, energy +1.3e-07, ang. mom. +6.7e-08 | Right: flips 0, axis wandered up to 1 deg, energy
    +2.3e-07, ang. mom. +1.2e-07
t=   70s  Left: flips 21, axis wandered up to 90 deg, energy -8.3e-08, ang. mom. -4.1e-08 | Middle: flips 0, axis
    wandered up to 2 deg, energy +3.1e-07, ang. mom. +1.5e-07 | Right: flips 0, axis wandered up to 1 deg, energy
    +4.3e-07, ang. mom. +2.2e-07
  Left   flipped at 117.62 s, 3.30 s after the last  (flip 36)
```

The bullet-rotation-test figures come from its CSV logs (the script prints nothing), analysed now.

Known departures and honest caveats

- The rotational dynamics are the script's, not Bullet's: Bullet's own gyroscopic options were found inadequate (above), so Euler's equations are integrated in Lua with RK4 plus an explicit projection onto constant E and |L|. Conservation is therefore enforced, not emergent; the flip period is the real test, and it matches. Since 15 Oct 2026 the header and README say so plainly (the header said energy and L were "exact" and the console "checks" them; the console's E/L figures, at most 4.3e-7 and 2.2e-7, show only single-precision rounding).
- Bullet is single precision: E and L hold only to ~1e-7. The header admits that with a vanishingly small nudge rounding noise starts the wobble sooner. Seen clearly in the test script's own Euler mode (energy-only projection, 0.1% nudge): mean flip interval 2.52 s vs exact 2.873 s (-12%), because the energy noise (about +/-1.5e-6 relative) exceeds the orbit's distance from the separatrix (~2.5e-7). At the exhibit's 1% nudge this does not matter (0.01%).
- Fixed 15 Oct 2026: the header's figures for Bullet's default implicit integrator (-44% energy, -25% L per minute) were not reproducible; re-run gives -31.9% / -17.5% at 600 Hz (same with flag 4), and header and README now quote those.
- No step-size convergence study is reported; only the default 600 steps/s was run here.
- Which axis is "middle" depends on proportions (header: with a 7 cm stem the bar becomes the middle axis); the 4 cm stem is a design choice.
- Fixed 15 Oct 2026: the header's headless-test path and the data-file fallback pointed at `demo/dzhanibekov/`; both now `demo/WyomingWill/Dzhanibekov/`, and the exhibit runs from the repository root (checked: first flip 1.67 s as from the folder).
- Fixed 15 Oct 2026: README.md written (what it shows, keys, theory and console, physics with the Bullet test results, files, env vars).
- Not changed: `bullet-rotation-test.lua`'s own comment says its DZ_EULER mode holds E and L, but the code projects energy only (the README says energy only).
- Text and paths only; physics and console unchanged (hall vs standalone IDENTICAL, 900 frames).

References

None cited beyond the historical note (Dzhanibekov, Salyut 7, 1985) in the lua header. Elliptic-function solution of Euler's equations: standard (e.g. Landau & Lifshitz, Mechanics, sec. 37) — added here for the check, not cited by the exhibit.

4.11 The gyroscope

Agrees with theory, with a numerical or open caveat. Precession 2.58 rad/s against m g r/(I ω) = 2.59 at the spin it has, and to 0.0-0.5% with the tilt term as it sags; README now right (2.56 rad/s at 100 turns/s). Rocked 1.5° it slides further than aimed for, cause not separated.
Folder: demo/WyomingWill/Gyroscope/

What it shows

Three identical toy gyroscopes spun to 100 turns/s. Left: held only by one end of its axle on a pedestal point, it does not fall but precesses with the axle level. Middle: the same, hung by a string. Right: axle upright, standing on a tightrope on a forked foot, balancing like a top on its point; an attendant rocks the wire 1 degree each way so it slides to and fro, upright, about 9 cm each way. As the rotors slow, the left two precess faster and sag; the tightrope one wobbles wider and finally comes off.

Model

- Gyroscope: one Bullet rigid body (hoop 50 g, axle 10 g, rotor 100 g; M = 160 g). Rotor r = 3 cm, I_rotor = 0.65 m r^2 = 0.585 kg cm^2; body moments (code): axial 1.069, in-hoop-plane 0.849, across-hoop 1.333 kg cm^2. Holding balls at r = 6.0 cm from the centre; the foot's wire contact 6.38 cm (PIV + ball + wire radius).
- Rotor spin is a scalar angular momentum h = I_rotor*S inside the body (gyrostat); Bullet never spins a body at 100 turns/s. Bullet's gyroscopic term is off; the script applies -w x (I w + h) as an external torque each step, 2nd-order Adams-Bashforth (README: a one-step torque grows the wobble; a direct velocity update makes it sag steadily).
- Pedestal: btPoint2PointConstraint at the ball's centre; released when the axle sags 40 deg below level. String: a 2 g, 12 cm rigid rod with ball joints at both ends. Wire: kinematic capsule, radius 0.06 cm, 30 cm long; foot = sphere + two capsule prongs; foot-wire friction 0.0025 at the contact: the friction slider (GY_MU, 0.005) sets the wire's, the frame keeps Bullet's default 0.5, and Bullet multiplies the two. It slides on slopes > 0.14 deg (still wire: holds at 0.1 deg, 0.4 cm in 8 s; slides at 0.2 deg, 13 cm in 8 s; checked 16 Oct 2026). Frame on table or posts: 0.5 × 0.6 = 0.3. Restitution 0 everywhere.
- Losses: rotor bearing 1 turn/s per second; pivot friction 0.5e-3 N m against any turning at the pedestal (a quarter at the string's knot). Rocking: wire tipped +/-1 deg about its middle in 0.6 s (cosine ramp), always up on the gyroscope's side, judged 0.3 s ahead, with timing adapted to reach ~9 cm each way.
- 2400 physics steps/s, 30 solver iterations, no damping, g = 981 cm/s^2.

Theory checks (theory values computed now from the code's constants unless marked README)

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Precession rate at 100 turns/s	Omega = m g r / (I_rotor w) = 0.16*981*6/(0.585*628.3) = 2.562 rad/s (2.45 s per turn); README/lua now say the same (fixed 15 Oct 2026; they said 2.67 rad/s, 2.35 s)	2.582 rad/s over 0.5-2 s (mean spin ~99 t/s; theory 2.588); 2.656 rad/s over 0-8 s at mean spin 96 t/s (theory 2.669)	-0.2% / -0.5%
Turns round in first 8 s (100 -> 92 t/s)	integral of m g r/(I w(t)) = 3.40 turns; README now "3.35 turns ... (theory 3.40; the console rounds it to 3.4)"	pedestal 3.35, string 3.36 (from GY_LOG); console "3.4 times round"	-1.5%
Precession as it sags (pedestal), with tilt term	m g r = Omega h + $Omega^2$ (I_pivot - I_axial) sin(theta), I_pivot = 7.093, I_axial = 1.069 kg cm^2	8-16 s 2.878 vs 2.877; 16-24 s 3.132 vs 3.132; 24-32 s 3.413 vs 3.430; 32-36 s 3.692 vs 3.685	0.0 to -0.5% (first-order formula alone: -1% to -5%)
Sag with no pivot friction (dip from the wobble)	m g r I / (I_rotor w)^2 (I = 6.6-7.1 about pivot): 2.6-2.8 deg at 100 t/s, 7.3-7.9 at 60, 16.5-17.7 at 40. README: "3 deg at 100, 16 at 40"	GY_PIVF=0: pedestal 8.3 deg at 60 t/s, 17.3 deg at 40 t/s; 0.1-3.4 deg sampled at 92-98 t/s	agrees (within the I range)
Sleeping-top criterion on the wire	I3 w > 2 sqrt(I m g l), l = 6.38 cm, I about the foot 7.36-7.85 kg cm^2: 46.7-48.2 t/s. README: "above 46 (46.7 to 48.2 ...)"	still wire, rough (GY_MU=0.3, 0.15 at the foot): off at 52 t/s (48.2 s)	README: 52, matches

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Still wire, slippery foot (slider 0.005, 0.0025 at the foot)	README: off at 45 t/s; with I about the CoM (ice rule) the bound is 15.9-19.9 t/s	off at 45 t/s (55.0 s)	matches README
Rocked 1 deg (default)	README: lasts 62 s, off at 38 t/s	off the wire at 62.0 s, 38 t/s	matches
Rocked 1.5 deg	README/lua now: 49 s, at 51 t/s, shorter than the still wire (fixed 15 Oct 2026; they said 55 s, "as long as on a still wire, or longer")	off at 49.0 s, 51 t/s	matches
Rocked 2 deg	README: thrown over in 13 s	off at 12.7 s	matches
Rocked 3 deg	README: 6 s	off at 5.9 s	matches
Slide range when rocked 1 deg	README: "about 9 cm each way (5 to 12 cm; the first time 14 ...)"	turn points (scratch copy logging the attendant's turn detection, 0-55 s): 13.9 first, then 4.8-12.5 cm	matches
Slide range when rocked 1.5 deg	README: stops 1 to 15 cm out; 14-15 cm in its last 6 s, attendant at its 4 cm limit	turn points 1.2-14.7 cm; 14.1, 13.9, 14.5, 14.7 cm at 42-47 s with the timing offset at its clamp (-4 cm)	matches (documented, not fixed)
Pedestal slides off / string stops	README now: pedestal off "after 37 seconds"; tightrope off "after a minute, at 38 turns a second"; string rotor stops at 100 s	pedestal off at 37.2 s (63 t/s); tightrope off 62.0 s (38 t/s); string spin 0 t/s at 100.0 s	matches

Run used

```
cd demo/WyomingWill/Gyroscope
HOLD_S=0 GY_LOG=gy.csv bpp -f gyroscope.lua -n 7200          # defaults: rock 1 deg
HOLD_S=0 GY_ROCK=0 bpp ... -n 7200     HOLD_S=0 GY_ROCK=0 GY_MU=0.3 bpp ... -n 5400
HOLD_S=0 GY_ROCK=1.5 bpp ... -n 5400   HOLD_S=0 GY_ROCK=2 ... -n 3600   HOLD_S=0 GY_ROCK=3 ... -n 1800
HOLD_S=0 GY_PIVF=0 bpp ... -n 3600
# re-run 15 Oct 2026 (after the text fixes): GY_ROCK = 1, 1.5, 2, 3, 0 with GY_LOG, -n 4800; GY_ROCK=0 GY_MU=0.3 -n
    4200
# 1.5 deg investigation: a scratch copy printing the attendant's turn points and timing offset (rock.late), GY_ROCK=1
    and 1.5, -n 3300
```

```
t=  8.0s  Pedestal below level  9.5 deg,  92 t/s, 3.4 times round | String below level  5.4 deg,  92 t/s, 3.4 times
    round | Tightrope from upright  4.1 deg,  92 t/s,  +0.8 cm along the wire, tipped -0.63 deg
t= 14.0s  ... Tightrope from upright  5.0 deg,  86 t/s,  +9.2 cm along the wire, tipped -1.00 deg
  Pedestal  off the pedestal at  37.2 s (spin 63 turns/s)
  Tightrope off the wire at  62.0 s (spin 38 turns/s)                        [rock 1]
  Tightrope off the wire at  55.0 s (spin 45 turns/s)                        [rock 0]
  Tightrope off the wire at  48.2 s (spin 52 turns/s)                        [rock 0, mu 0.3]
  Tightrope off the wire at  49.0 s (spin 51 turns/s)                        [rock 1.5]
  Tightrope off the wire at  12.7 s (spin 87 turns/s)                        [rock 2]
  Tightrope off the wire at   5.9 s (spin 94 turns/s)                        [rock 3]
t= 60.0s  Pedestal below level 17.3 deg,  40 t/s, 36.6 times round | ...      [pivot friction 0]
```

(The t=14.0 s line is shortened with "..."; precession rates are least-squares fits to the logged axle heading.)

Known departures and honest caveats

- The rotor is not simulated as a spinning body; the gyroscopic torque is applied explicitly (Adams-Bashforth). The README says the more obvious alternatives either grow the wobble or make it sag steadily.
- Sagging comes from a chosen pivot-friction torque (0.5e-3 N m), not from a contact model; the 40 deg release from the pedestal is a scripted rule; the string is a rigid 2 g rod, not a flexible string.
- The foot friction ("as if it ran on a little wheel") is chosen so the gyroscope slides on a 1 deg tilt. It is 0.0025 at the contact, not the 0.005 the slider shows (see the next point). The attendant is a feedback controller (lead 0.3 s, adaptive timing), not physics.
- Fixed 15 Oct 2026: the precession rate in README and lua header (was 2.67 rad/s, 2.35 s per turn, at 100 t/s; that is the value at about 96 t/s) now reads 2.56 rad/s, 2.45 s, from the stated m = 160 g, r = 6 cm, I = 0.65 m r^2; the "3.4 turns in 8 s" now says 3.35 (theory 3.40, console rounds to 3.4).

- Fixed 15 Oct 2026: rocked 1.5 deg, README and header now give the measured 49.0 s (51 t/s), shorter than the still wire's 55.0 s; only 1 deg (62.0 s) outlasts the still wire. The "about 40"/"about 50" turns/s wording is replaced by the run values: rough still wire 52, slippery still wire 45, rocked 1 deg 38; pedestal off at 37 s.
- Still open, now documented in the README: rocked 1.5 deg the attendant does not hold 9 cm. Its turn points scatter from 1.2 to 14.7 cm (at 1 deg: 4.8-12.5 after a first 13.9). The attendant's correction is an integral on the reach error (gain 0.3, offset clamped at +/-4 cm); at 1.5 deg the same timing error gives 1.5x the distance, and in the last 6 s before it falls it overshoots to 14-15 cm with the offset at its clamp. The wobble is wider (9-17 deg from upright vs 2-8 at 1 deg, 17-28 deg in the last swings); the README says that most likely the wobble pushes the foot along too, and that this has not been separated from the slope's push. Nothing in the physics was changed.
- The sleeping-top threshold is only approximate here: the foot is a sphere sliding on a thin capsule, not a fixed point; the README itself explains the slippery foot shifts the relevant I toward the centre of mass.
- Found 16 Oct 2026, documented, physics not changed: **the foot-wire friction was half what was stated.** Bullet multiplies the two bodies' friction; the slider (0.005) is set on the wire, the frame was left at Bullet's 0.5, so the foot has 0.0025 (and "rough", GY_MU=0.3, is 0.15). The README and lua said 0.005, sliding on slopes over 0.3 deg; it slides over 0.14 deg (checked above). Making the stated 0.005 true (frame 1.0, table and posts 0.3 so that the frame on the table stays 0.3) changed the validated tightrope results a lot: rocked 1 deg off at 53.9 s, 46 t/s (was 62.0 s, 38 t/s); still slippery wire 53.6 s, 46 t/s (55.0, 45); still rough (true 0.3) 43.0 s, 57 t/s (48.2, 52); rocked 1.5 deg still up at 60 s (49.0); rocked 2 deg 18.2 s (12.7); rocked 3 deg still up at 10 s (5.9). With a true 0.005, rocking 1 deg no longer outlasts the still wire. Since 0.005 was a chosen number, not a measured one, the physics is kept and the README, lua comments, slider and help text now give the values the foot really has (0.0025; 0.15). The pedestal and string gyroscopes do not touch anything and are unchanged (pedestal off at 37.2 s in every run).
- No step-size convergence study is documented; all runs at 2400 steps/s.
- Fixed 17 Oct 2026: **the start pause changed the console.** The `t=` lines every 2 s were printed on bpp's frame count (`N % 120`), which counts the pause's frames too, so with the pause they came at t = 1.5, 3.5, ... s instead of 2.0, 4.0, ... gyroscope.lua now counts only the frames it runs (the same for the log file). With HOLD_S=0 the output is byte-identical to before (1800 frames); HOLD_S=0 for 1800 frames is now a prefix of the default for 1830 (also at 900 and 2400); hall against standalone (cmp2.sh, 1800 frames): IDENTICAL (15 run lines).

References

None cited in the README or code. Formulas used are the textbook gyroscope precession rate and the sleeping-top (Maxwell/Klein-Sommerfeld) stability criterion.

4.12 The bicycle wheel

Agrees with theory, with a numerical or open caveat. Stool's turning matches angular-momentum conservation exactly with friction off; L kept to 0.093% over 120 s. A guard now reports and recovers from a NaN blow-up, whose cause (seen once) is still unknown.
Folder: demo/WyomingWill/BicycleWheel/

What it shows

A wooden lay figure on a swivel stool, feet off the floor, holds a spinning bicycle wheel with its axle upright; the stool is still. Turning the wheel over makes the stool turn the other way; turning it back stops the stool. Pushing the wheel out to arm's length slows the stool, pulling it in speeds it up again (the figure-skater effect). Nothing pushes on the floor: it is angular momentum about the stool's axis being shared between wheel and figure.

Model

Three Bullet rigid bodies (code, README):

- Stool seat + figure: one body, 75 kg, I = 15000 kg cm^2 (1.5 kg m^2) about the stool axis, on a vertical btHingeConstraint to a fixed anchor; bearing friction 0.02 N m (slider `stool friction`, applied by the script as a saturated torque).
- Hands: 1 kg, I = 100 kg cm^2, on a btSliderConstraint straight out in front, motor-driven between 34 cm (close) and 50 cm (out) in 1.5 s (smoothstep path + proportional correction, gain 40).
- Wheel (axle + handles): 3 kg, I = 1150 kg cm^2 about the axle (0.115 kg m^2), 600 across; hinged to the hands along the slide axis; a hinge motor turns it over in 1.2 s.
- The wheel's spin is not a Bullet rotation: it is a scalar angular momentum h = I3·S carried inside the body (a gyrostat). Bullet's gyroscopic term is off (setFlags(0)); the script applies the gyroscopic torque -w x (I w + h) each step, 2nd-order Adams-Bashforth (1.5·T_n - 0.5·T_(n-1)). The hinge reaction to that torque is what turns the stool.
- Wheel bearing friction 0.02 turns/s per second; the lost spin is handed to the axle as torque.
- 1200 physics steps/s (20 per 1/60 s frame), 30 solver iterations, no damping, gravity 981 cm/s^2 (irrelevant to the physics shown).
- The "should" value printed after each move is computed in Lua: (L0 - I3·S·(axle·up)) / (I_P + (m_w+m_hands)·d^2 + I_w,eff + I_hands), with L0 = total angular momentum about the stool axis when the move started.

Theory checks

Spin 4 turns/s: h = 0.115 x 2pi x 4 = 2.890 kg m^2/s. I_close = 1.5 + 4 x 0.34^2 + 0.115 + 0.010 = 2.087 kg m^2; I_out = 1.5 + 4 x 0.50^2 + 0.115 + 0.010 = 2.625 kg m^2 (hand calculation from code constants; README quotes 2.09 and 2.63).

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Stool rate, wheel turned over, held close (no friction)	w = 2h/I_close = 2 x 2.890/2.087 = 2.769 rad/s	2.769 (first round); 2.770 later rounds	0.000 / +0.001 rad/s (<0.04%)
Stool rate, pushed out (no friction)	w = 2h/I_out = 2.202 rad/s	2.202	0.000
Stool rate, pulled back in (no friction)	2.769 rad/s	2.769 (2.770 later)	<0.04%
Stool rate, axle up again (no friction)	0	0.000 (0.001 in rounds 5-6)	<=0.001 rad/s
Angular momentum about stool axis (no friction, 120 s, 6 rounds)	conserved; README: "within 0.1%"	max deviation 0.093%, end +0.050% (from BW_LOG)	within README bound

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Stool rate vs console "should", defaults (friction on), 24 moves	console theory from L at move start	turned over 2.719 vs 2.736 ... 1.136 vs 1.152; out 2.099 vs 2.114; in 2.565 vs 2.584; up -0.021 vs -0.018	sim 0.015-0.020 rad/s below (0.6-1.4%), as README predicts (~0.02 rad/s friction during move)
README friction sequence, turned over & close	README: 2.72, 2.46, 2.19 rad/s	2.719, 2.456, 2.192	matches
README frictionless table	README: 2.769 / 2.202 / 2.769 / 0.000	2.769 / 2.202 / 2.769 / +0.000	matches

Run used

```
cd demo/WyomingWill/BicycleWheel
HOLD_S=0 BW_STOOLF=0 BW_WHEELF=0 BW_LOG=bw0.csv bpp -f bicycle-wheel.lua -n 7200   # frictionless
HOLD_S=0 BW_LOG=bw.csv bpp -f bicycle-wheel.lua -n 7200                            # defaults
```

Frictionless:

```
t=  3.7s  wheel turned over (34 cm out): the stool turning at +2.769 rad/s (0.44 turns/s); it should, by its angular
    momentum: +2.769
t=  8.5s  wheel pushed out (50 cm out): the stool turning at +2.202 rad/s (0.35 turns/s); it should, by its angular
    momentum: +2.202
t= 13.0s  wheel pulled in (34 cm out): the stool turning at +2.769 rad/s (0.44 turns/s); it should, by its angular
    momentum: +2.769
t= 17.3s  wheel axle up again (34 cm out): the stool turning at +0.000 rad/s (0.00 turns/s); it should, by its
    angular momentum: +0.000
t=118.1s  wheel turned over (34 cm out): the stool turning at +2.770 rad/s (0.44 turns/s); it should, by its angular
    momentum: +2.770
```

Defaults:

```
t=  3.7s  wheel turned over (34 cm out): the stool turning at +2.719 rad/s (0.43 turns/s); it should, by its angular
    momentum: +2.736
t= 17.3s  wheel axle up again (34 cm out): the stool turning at -0.021 rad/s (0.00 turns/s); it should, by its
    angular momentum: -0.018
t= 22.8s  wheel turned over (34 cm out): the stool turning at +2.456 rad/s (0.39 turns/s); it should, by its angular
    momentum: +2.472
t= 41.8s  wheel turned over (34 cm out): the stool turning at +2.192 rad/s (0.35 turns/s); it should, by its angular
    momentum: +2.208
```

Angular-momentum figures (0.093%, 0.050%) were computed from the L_up column of the BW_LOG csv; no NaN appeared in either 120 s run.

Known departures and honest caveats

- The "theory" column the console prints is computed by the script from the simulated state at the start of each move; it is an internal consistency check (angular momentum bookkeeping), not an independent prediction. The independent hand calculation (2.769, 2.202 rad/s) agrees with it in the frictionless case.
- With friction on (defaults) the stool lags the console "should" by 0.015-0.020 rad/s because friction during the 1.2-1.5 s move is not accounted for (README says so). Total angular momentum falls 60% over 120 s in the default run; that is the stool bearing and wheel bearing, by design (the spin drops from 4.0 to 1.6 turns/s).
- After "axle up again" with friction on, the stool shows -0.021 rad/s rather than 0: the friction-induced residual.
- Mass properties are idealised: the figure+stool is one rigid body with a fixed 1.5 kg m^2; the arms are drawn only (the 1 kg "hands" body stands in for them); the wheel's inertia (0.115 kg m^2) and 600 kg cm^2 transverse are chosen, not derived from the mesh. The wheel's spin is a gyrostat number, not a rotating Bullet body.

- The gyroscopic torque is explicit (Adams-Bashforth); the agreement shown is at the default 1200 steps/s; no step-size study is documented.
- Open issue (user report, not reproduced): a user saw the bicycle wheel go to NaN once in the GUI after visiting the solar system exhibit in the hall (Cabinet). It could not be reproduced headless; the 7200-frame default run on 15 Oct 2026 was NaN-free too. Cause still unknown (the Cabinet prefixes each exhibit's sliders, so the solar system's settings are not shared with this exhibit). Guard added 15 Oct 2026: after every frame the script checks the positions, velocities and angular velocities of the stool, wheel and hands bodies, and the wheel's spin, angle, arm angle and slide distance; if any is not finite it prints one console line ("t=...s ** NOT A NUMBER: the simulation broke down (going by itself, move N, last move: ...; wheel ... cm out, spinning ... rad/s, turned ... rad; steps, speed, step); stool at (...) moving (...) turning (...); wheel ...; hands ... -- starting again") and restarts the run cleanly. Tested by injecting a NaN into the wheel's angular velocity at t = 5 s in a scratch copy: the line was printed, the run restarted, and the restarted run's first report (t = 3.7 s, +2.719 / +2.736) was identical to a fresh start. Also: the bearing-loss term IW3·d/h is now skipped if the step h were ever 0 (it would be 0/0); this does not occur in normal use and is not claimed to be the cause.
- Fixed on 15 Oct 2026: pressing R mid-move used to print a stray "wheel turned over ..." line about 1.7 s into the new run (the cut-off move's pending report, with the restarted clock). restart() now clears the pending report; tested by forcing a restart at t = 2.5 s, mid-turn, in a scratch copy: no stray line.
- Checks after these changes: default 7200-frame run gives console output byte-identical to the previous version of the script (27 lines); hall identity cmp2.sh "the bicycle wheel" 2400 frames: IDENTICAL (8 run lines); pause check 2400/2430 frames: prefix OK (10 lines).

References

None cited in the README or code. The physics is the standard textbook "rotating stool and bicycle wheel" demonstration of angular momentum conservation.

5. Wing B: Swinging

5.1 Newton's cradle

Agrees with theory. First strike the same at every step rate from 1200 to 9600 steps/s (was: a third of the energy lost at 9600); momentum to 0.00%; ball 4 gets 13.6% (Herrmann & Seitz: ~12%); the row settles over minutes, as real cradles do.
Folder: demo/WyomingWill/NewtonsCradle/

What it shows

Five steel balls hang in a row on V-threads. Lift one, two or three and let go: the same number fly out at the far end. Because the balls touch, a strike is a compression pulse through the row rather than a chain of two-ball collisions, so the far ball does not get quite everything; the row breaks up and, over a few minutes, ends up swinging together. With a 0.1 mm gap (key `G`) each strike becomes a clean one-in, one-out sequence.

Model

Balls: rigid spheres, 25 mm diameter, 64 g (code: rho 0.00785 kg/cm^3, 64.2 g computed), centres 15 cm below the thread tops. Each ball is a Bullet rigid body on a `btHingeConstraint` to a static anchor (it swings in the row's plane and turns with its threads); Bullet collisions are off (`collides = false`), solver iterations 40, Bullet damping 0. Bullet steps at 2400 /s (slider 1200-9600). Strikes are done by the script before every physics step: any pair pressed together, or predicted to touch within the step, is integrated with each ball moving along its own swing (the contact force along the line of centres, projected on each ball's swing tangent; inertia along the swing m(1 + 2/5 r^2/L^2), since the ball turns with its threads), with Hertz forces F = K d$^{3/2}$, K = 4/3 E* sqrt(R/2), E* = E / (2(1 - nu^2)), E = 200 GPa, nu = 0.3, times a Hunt-Crossley factor (1 + 1.5 alpha v), alpha = 0.25 x 0.0012 s/cm (a quarter of Hunt-Crossley for steel). Integrator: velocity Verlet, sub-steps <= 0.25 us; each ball is handed back to Bullet on its hinge circle, turned with its threads, with velocity W along its swing tangent and angular rate W/L (so the hinge solver has nothing to correct), placed so Bullet's step ends where the row had it. A squeeze still held at the end of a step is carried into the next step through the positions and worked out there. Knocks with |dv| < 3 cm/s are not counted as strikes. Clack sound is pre-synthesised by `make-sounds.py` (same Hertz strike at 20 ns steps, rigid-sphere dipole radiation with the h2 spherical Hankel function, table image at 0.85, 48 kHz) — not run here because it rewrites `cradle-sounds/`.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Two-ball contact time at 62.7 cm/s	Hertz: 2.87 (m^2/(R E*^{2} v))$^{1/5}$, m,R reduced: 84.2 us	84.2 us (script's own 2-ball integration)	exact to 0.1 us
Restitution of two-ball strike, damping 0.25	design target 0.982 (README)	0.982 (1.000 with NC_DAMP=0)	matches
Time to first strike	quarter period of physical pendulum, I = m(L^2 + 2/5 r^2), elliptic K, 30 deg: 0.1979 s (I reproduced 0.19789 s independently)	0.1975 s (2400 steps); 0.1977 s (3000, 4800); 0.1978 s (7200, 9600); 0.1967 s (1200)	-0.4 ms = one step (README says so)
Energy in first strike, no damping	conserved: 100%	100.0% (touching); 100.0% (gap)	0.0%
Momentum along row, first strike	conserved	0.00% off	exact
Touching row, no damping: speeds after	README: -4.5, -1.9, -0.9, 8.0, 62.0 cm/s	-4.5, -1.9, -0.9, 8.0, 62.0	identical
Touching row, damping 0.25: speeds after	README: -4.1, -1.6, -0.6, 8.5, 60.4 cm/s; energy 95.3%	-4.1, -1.6, -0.6, 8.5, 60.4; 95.3%	identical

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Gap 0.1 mm, no damping	README: 62.7 out, others < 0.05 cm/s	0.0, -0.0, -0.0, -0.0, 62.7; energy 100.0%	identical
Gap 0.1 mm, damping 0.25	four two-ball strikes at e = 0.982: each striker keeps (1-e)/2 x 62.7 = 0.56 cm/s, out 62.7 x 0.991^4 = 60.5	0.6, 0.6, 0.6, 0.5, 60.5; energy 93.0%	matches
Ball 4's share of incoming momentum	Herrmann & Seitz 1982 five-ball simulation: ~12% (README)	8.5/62.7 = 13.6% (README says 14%)	+1.6 points; same pattern (1-3 back, 4-5 forward)
Row merges to swinging together	Hutzler et al. 2004: 1-2 min on real cradles (README)	1 ball, 30 deg, auto off, 7 min run: together swing 5.2 deg; largest ball's swing about the row's mean falls below 1/2 of it at ~80 s, 1/5 at ~3 min, 1/10 at ~6.5 min (1200 and 9600 steps/s agree to a few s); energy 32% at 15 s, 19% at 1 min, then 16.6% (the together swing). Damping 0: never merges (relative swing ~24 deg after 3 min), energy kept to 1.5%	same scale as real cradles, somewhat slower; set by the contact damping alone (README says so)
Clack peak at listener, 60 cm/s	README (make-sounds.py console): 2.7 Pa, 103 dB, peak ~ $\text{speed}^{1.48}$	not run	README only
Step-size convergence of first strike	should be independent of step	1200, 2400, 3000, 4800, 7200, 9600 steps/s: -4.1 -1.6 -0.6 8.5 60.4-60.5 cm/s, energy 95.3-95.4%; NC_DAMP=0: -4.5 -1.9 -0.9 8.0 62.0, energy 100.0% (100.2% at 1200, over a 1.7 ms window with gravity acting)	converged (fixed 15 Oct 2026)

Run used

```
cd demo/WyomingWill/NewtonsCradle && HOLD_S=0 timeout 1500 ../../../release/bpp -f newtons-cradle.lua -n 3600 | grep
    -v ALSA
  (plus -n 900 / -n 40 runs with NC_DAMP=0; NC_GAP=0.01; NC_GAP=0.01 NC_DAMP=0;
    NC_STEPS=1200,2400,3000,4800,7200,9600 each with
   and without NC_DAMP=0; settling: a scratch copy of the exhibit with the auto-lift off and each ball's angle and
    the total
   energy logged every frame, 1 ball, 25200 frames at 2400/s, 14400 at 9600/s, 7200 at 1200/s, 10800 with NC_DAMP=0)

    two balls meeting at 62.7 cm/s: they touch for 84.2 microseconds (Hertz: 84.2); they part at 0.982 of it
  strike  1 at  0.197 s: speeds before   62.7    0.0    0.0    0.0    0.0
                              after    -4.1   -1.6   -0.6    8.5   60.4 cm/s; energy 95.3%; momentum along the row
    0.00% off ...
                    0.1975 s after letting go; a quarter swing from 30 degrees takes 0.1979 s
  [NC_DAMP=0]  after    -4.5   -1.9   -0.9    8.0   62.0 cm/s; energy 100.0%; momentum along the row 0.00% off
  [NC_GAP=0.01 NC_DAMP=0]  after     0.0   -0.0   -0.0   -0.0   62.7 cm/s; energy 100.0%
  [NC_GAP=0.01]  after     0.6    0.6    0.6    0.5   60.5 cm/s; energy 93.0%
  [1 ball] strike 90 at 13.425 s: ... after    22.8    4.6    5.8    6.9   12.3 cm/s; energy 119.1%; momentum 7.52%
    off (over 14.2 ms)
  [3 balls] strike 240 at 14.775 s: ... after   -20.4  -41.1  -16.5  -43.7  -45.4 cm/s (still not swinging together
    at 15 s)
  [NC_STEPS=9600]  after    -4.1   -1.6   -0.6    8.5   60.4 cm/s; energy 95.3%; momentum along the row 0.00% off
  [NC_STEPS=1200]  after    -4.1   -1.6   -0.6    8.5   60.5 cm/s; energy 95.4%; momentum along the row 0.01% off
```

Known departures and honest caveats

- Fixed 15 Oct 2026: **step-size failure at 9600 steps/s** (first strike kept 66.8% of the energy, ball 4 left at 22.6 cm/s; 6000 steps/s with no damping also failed, 84.4%). Cause: the "touching" test only asked whether a pair would be pressed at the end of the step if it moved freely (gap - closing speed x h < 0). A contact still squeezed but already opening (at 9600/s the pulse straddles steps: ball 4-5 still squeezed 14 um at a step boundary, separating at 23 cm/s) was predicted "open", so the next step skipped the strike and Bullet simply let the balls fly apart through each other, throwing the stored squeeze away. Now a pair is worked out if it is pressed now or will be within the step. A second, smaller error came from the hand-back: balls were moved along x without turning them on their

hinges, and the hinge solver's correction shifted squeezes of ~15 um by 10-50 nm between steps (0.1-0.3% of a strike's energy); and strikes at a swing angle were worked out along x with mass m, then projected by the hinge onto the swing, which made or lost energy by ~angle^2/2 per strike (with damping 0 the energy wandered by +-10% over a minute after the first fix alone); now each ball is handed back on its hinge circle, turned, with velocity along its swing, and the row is worked out along each ball's swing with its inertia m(1 + 2/5 r^2/L^2). The default first strike moved from 60.477 to 60.450 cm/s for ball 5 (shown 60.5 -> 60.4; README updated).

- Fixed 15 Oct 2026: **settling time**. The same lost squeezes, through many later strikes, drained the energy: with NC_DAMP=0 the old code lost 59% of the energy by 5 s and 81% by 20 s, and the row "merged" in ~15 s whether damped or not. That fast merging was the bug. Now with damping 0 energy is kept to 1.5% over 3 min and the row never merges; with the default damping it merges over minutes (table). The README's "Why the balls part" now gives the measured times and says the rate is set by the contact damping alone (a quarter of Hunt-Crossley, chosen for the 0.982 two-ball restitution, not tuned to Hutzler); a real cradle also has air, thread and alignment losses that the model lacks, so the near-agreement with 1-2 min is not a test of the model.
- Late "strikes" in long runs report energy 58-126% and momentum up to ~105% off; these are windows of 10-40 ms of repeated contact in which gravity and the threads act, so the per-strike bookkeeping is not a conservation test there.
- By itself the exhibit relifts every 15 s, so the settling is only seen with `A` off.
- Ball 4 gets 13.6% of the incoming momentum versus Herrmann & Seitz's ~12%: same pattern, not the same number.
- Threads are a hinge, so out-of-plane swinging and thread stretch are absent; the threads are drawn only.
- Sound: README numbers only (not regenerated; files unchanged, md5 checked). Fixed 15 Oct 2026: `make-sounds.py`'s docstring said the loudest file was "four balls at 120 cm/s"; its speeds stop at 100 and the files show clack-n1-v100.wav peaks at 0.95 (n2-v100 0.62, n1-v60 0.56), so it now says "one ball at 100 cm/s".
- Checks after the fix: hall vs standalone (cmp2.sh newton, 1500 frames): IDENTICAL (59 run lines); pause: HOLD_S=0 -n 840 is a prefix of the default -n 870 (30 of 30 lines), and -n 1200 of -n 1260 across the 15 s relift (52 of 52).

References

- H. Hertz (contact law, contact-time formula 2.87(...)$^{1/5}$) — cited by name.
- K. H. Hunt and F. R. E. Crossley (contact damping) — cited by name.
- F. Herrmann and M. Seitz (1982), five-ball simulation of the cradle (README; journal not given).
- E. J. Hinch and S. Saint-Jean (1999), break-up of a line of balls (README).
- S. Hutzler, G. Delaney, D. Weaire and F. MacLeod, "Rocking Newton's cradle", Am. J. Phys. 72 (2004).

5.2 Huygens' clocks

Agrees with theory. All three starts lock in anti-phase at 180.0°, at 0.9985-0.9988 Hz against 0.9984 Hz from theory; at 9600 steps/s the same to 1e-4 Hz (one start, 45 s).
Folder: demo/WyomingWill/HuygensClocks/

What it shows

Two pendulum clocks hang from one beam that can slide a little on a spring. Started out of step (60, then 300, then 120 degrees apart, four minutes each), within about half a minute they swing in exact opposition and the beam stops: Huygens' 1665 "odd kind of sympathy". In opposition their pushes on the beam cancel, so the beam's friction takes no energy; any in-phase part is damped away.

Model

Bullet. The beam (1.5 kg including the clock cases) is a rigid body restricted to slide along x (linear factor (1,0,0), no rotation, no gravity). Each pendulum is one rigid body (brass bob 3.5 cm diameter, 191 g, plus a light aluminium rod; bob centre 25 cm from the pivot) on a `btHingeConstraint` to the beam. Done by the script before every physics step, as velocity changes: the beam's spring (k from 1.5 Hz with the beam alone) and friction (5 kg/s); each pendulum's air drag (a torque proportional to angular rate, Q = 150 alone); and the escapement, an angular-velocity kick at each zero crossing, sized to replace exactly the air loss over half a swing at 8 degrees. Bullet's own damping 0, solver iterations 40, 2400 physics steps/s (Bullet's semi-implicit Euler with the hinge solver). Clock hands advance by each clock's own swing count.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Anti-phase normal mode (no escapement, no friction)	$w^2 = m\,g\,d / I$ (beam still): 0.9996 Hz	not run as a free mode; used as reference	(my own 3x3 eigen-solve: 0.99964 Hz)
In-phase normal mode	root of $\det[k - (M+2m)w^2, -2md\,w^2; -md\,w^2, mgd - I\,w^2] = 0$: 0.9179 Hz	not compared with a simulated mode	(my eigen-solve: 0.91787 Hz)
Anti-phase at 8 deg amplitude	0.9996 Hz x pi / (2 K(sin(4 deg))): 0.9984 Hz	settled clock frequency 0.9987 Hz (start 60 deg), 0.9988 (300 deg), 0.9985 (120 deg)	+0.03%, +0.04%, +0.01% (README: escapement kicks account for the rest)
Final phase difference	180 deg (anti-phase locking, Bennett et al. 2002)	180.0 deg at t = 60 s for all three starts	exact to 0.1 deg
Beam amplitude decays	beam stops in anti-phase	5.335 -> 0.544 -> 0.062 -> 0.008 mm at 15/30/45/60 s (start 60 deg)	~x10 per 15 s; to 0.008 mm
Time to lock	README: "within a minute"	within 0.5 deg of 180 by 30 s for all three starts	consistent
Step-size check (my run, HC_STEPS=9600)	should not depend on step	161.9 / 179.5 / 180.1 deg; 1.0063 / 1.0005 / 0.9990 Hz; beam 5.338 / 0.543 / 0.062 mm	agrees with 2400 /s to 1e-4 Hz and 0.003 mm
README console listing	README t = 15..60 s lines	identical to this run, digit for digit	matches

No energy or momentum conservation check is performed by the exhibit (the system is driven and damped by design).

Run used

```
cd demo/WyomingWill/HuygensClocks && HOLD_S=0 timeout 1500 ../../../release/bpp -f huygens-clocks.lua -n 3600 | grep
    -v ALSA
  (also HC_PHASE=60, 300, 120 with -n 3700 to reach the t=60 s line; HC_STEPS=9600 -n 3600)

  stopped, they'd swing in opposition at 0.9996 Hz (the beam still), or together at 0.9179 Hz (the beam swinging
    against them)
  in opposition, swinging 8 degrees (a pendulum's slower, the further it swings: the elliptic integral): 0.9984 Hz
  t=   15 s  the clocks 161.9 degrees apart (in opposition: 180); swinging at 1.0063 Hz; the beam moving up to 5.335
    mm
  t=   30 s  the clocks 179.5 degrees apart (in opposition: 180); swinging at 1.0004 Hz; the beam moving up to 0.544
    mm
  t=   45 s  the clocks 180.1 degrees apart (in opposition: 180); swinging at 0.9989 Hz; the beam moving up to 0.062
    mm
  t=   60 s  the clocks 180.0 degrees apart (in opposition: 180); swinging at 0.9987 Hz; the beam moving up to 0.008
    mm
  [HC_PHASE=300] t=   60 s  the clocks 180.0 degrees apart ...; swinging at 0.9988 Hz; the beam moving up to 0.008 mm
  [HC_PHASE=120] t=   60 s  the clocks 180.0 degrees apart ...; swinging at 0.9985 Hz; the beam moving up to 0.004 mm
```

Known departures and honest caveats

- Locks in about 30 s, against Huygens' "within half an hour". The beam parameters (1.5 kg, 1.5 Hz spring, 5 kg/s friction) are chosen for a fast, visible demonstration, not to match Huygens' chairs and beam.
- The escapement is an idealised impulse at the bottom of each swing, sized exactly to the linear air loss at 8 degrees; real escapements (and Bennett et al.'s model) differ. The 0.03% excess of the settled frequency over the free anti-phase mode is put down to the kicks (README), not separately verified.
- Pendulums are confined to the plane (hinges); the beam only slides (no rocking).
- The in-phase mode frequency (0.9179 Hz) is printed but never compared against a simulated in-phase swing.
- Fixed on 15 Oct 2026: the lua header used to call the "together" mode "faster"; it now says "slower", matching its own console value (0.9179 Hz against 0.9996 Hz) and the README. Comment only; the console output is unchanged.
- A third normal mode (mostly the beam on its spring, about 1.63 Hz by my eigen-solve) exists and is not mentioned.
- The 15 s report window's "swinging at" frequency is the last single period of clock 1, so early values (1.0063 Hz) reflect the transient.

References

- C. Huygens, 1665 observation ("an odd kind of sympathy") — README.
- M. Bennett, M. F. Schatz, H. Rockwood and K. Wiesenfeld, Proc. R. Soc. A (2002) — README and lua header (title not given there).

5.3 Kapitza's pendulum

Agrees with theory. Stability threshold 0.783 cm bracketed (0.76 falls, 0.80 stands); rocking period 1.133 s against 1.135 s for its 7.6° swing (-0.2%), now at a converged 19,200 steps/s (was 4800, 1.7% short).
Folder: demo/WyomingWill/Kapitza/

What it shows

An upside-down pendulum stands on a pivot shaken up and down 40 times a second and does not fall; nudged, it rocks slowly about the upright as if gravity pointed up. Left: shaken 1.0 cm, it stays up. Right: shaken 0.6 cm, too little, it falls and ends up hanging. Both are stood up again, tilted 8 degrees, every 15 s.

Model

Bullet. Each pendulum is one rigid body (brass bob 3 cm diameter plus aluminium rod radius 0.25 cm; bob centre 20 cm from the pivot; 130.8 g; centre of mass 19.19 cm from the pivot; inertia from the composite) on a `btHingeConstraint` to a pivot block. The block is a 10^4 kg body moved kinematically by the script before every step: position y = y0 + a cos(w t) and velocity -a w sin(w t), w = 2 pi x 40 Hz. No friction, Bullet damping 0, solver iterations 40, 19200 physics steps/s (480 per shake; raised from 4800 on 15 Oct 2026, see below; steps slider 2400-19200). The slow motion is measured on the angle averaged over each 1/40 s shake; rocking period from upward zero crossings of that average.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Equivalent length	l = I / (m d): 19.761 cm	(input; independent recomputation 19.7614 cm)	exact
Stability threshold	Kapitza: a w > sqrt(2 g l) = 196.9 cm/s, a > 0.783 cm	1.00 cm stays up; 0.60 cm falls at 0.69 s	qualitatively correct
Threshold bracket (KP_A1=0.80 KP_A2=0.76)	0.80 cm (a w = 201.1) should stand; 0.76 cm (191.0) should fall	0.80 cm "stays up (as it should)"; 0.76 cm falls at 1.39 s	correct on both sides, within +/-3% of threshold
Rocking period, a = 1.0 cm, small-rock theory	2 pi / W, $W^2 = a^2 w^2 / (2 l^2) - g / l$: 1.124 s	1.133 s over 6 rocks, up to 7.6 deg	+0.8% (expected: the rock is 7.6 deg, not small)
Rocking period for its size (new console term)	averaged eq. to next order $\phi'' = -W^2 \phi + c \phi^3$, $c = 2K/3 - g/(6l)$, $K = a^2 w^2/(2 l^2)$; Lindstedt $T = (2\pi/W)(1 + 3 c A^2/(8 W^2))$, A = 7.6 deg: 1.135 s (+0.97%)	1.133 s	-0.2%
Exact (non-averaged) pendulum equation, integrated independently (Python RK4, 400 steps per shake, same shake-averaged measurement)	$\theta'' = (g - a w^2 \cos wt) \sin\theta / l$, from 8 deg: 1.1341 s; from 2 deg: 1.1237 s	1.133 s (8 deg); KP_TILT=2: 1.122 s	-0.1% / -0.15%
Step-size convergence of the rocking period (KP_STEPS)	should converge to the exact equation's 1.134 s	2400: 1.073; 3600: 1.103; 4800: 1.115; 7200: 1.125; 9600: 1.129; 14400: 1.132; 19200: 1.133; 38400: 1.134 s	converged to 0.1% at the new default
Repeatability over 4 restarts (3600 frames)	—	1.133 s and 7.6 deg each time	stable (fixed: see caveats)
Hall identity / pause	identical in the Cabinet; HOLD_S=0 output a prefix of the default-pause output	cmp2.sh "Kapitza" 1500 frames: IDENTICAL (12 run lines); pause check 1500/1530 frames: prefix OK (18 lines)	pass

No energy check (the shaker does work by design).

Run used

```
cd demo/WyomingWill/Kapitza && HOLD_S=0 timeout 1500 ../../../release/bpp -f kapitza.lua -n 3600 | grep -v ALSA
  (also KP_A1=0.80 KP_A2=0.76 -n 1200; KP_TILT=2 -n 600; KP_STEPS=2400..38400 -n 600)

  the pendulum: 130.8 g, its centre 19.19 cm from the pivot, equivalent length l = I/(m d) = 19.761 cm;
  it stays up if a w > sqrt(2 g l) = 196.9 cm/s, i.e. a > 0.783 cm
  Left   shaken 1.00 cm: a w = 251.3 cm/s > 196.9 -- it should stay up, rocking with a period of 2 pi / W = 1.124 s
  Right  shaken 0.60 cm: a w = 150.8 cm/s < 196.9 -- it should fall
  Right  falls over at 0.69 s
  Left   stays up, rocking up to 7.6 degrees either side: a period of 1.133 s over 6 rocks (theory, 2 pi / W: 1.124
    s, +0.8%; for its size: 1.135 s, -0.2%)
  [KP_STEPS=4800, the old default] Left   stays up, rocking up to 7.6 degrees either side: a period of 1.115 s over 6
    rocks (theory, 2 pi / W: 1.124 s, -0.8%; for its size: 1.135 s, -1.7%)
  [KP_TILT=2] Left   stays up, rocking up to 1.9 degrees either side: a period of 1.122 s over 6 rocks (theory, 2 pi
    / W: 1.124 s, -0.2%; for its size: 1.125 s, -0.2%)
  [KP_A1=0.80] Left   stays up (as it should)
  [KP_A2=0.76] Right  falls over at 1.39 s
```

Known departures and honest caveats

- Fixed on 15 Oct 2026, step size: at the old default of 4800 steps/s the rocking period was 1.7% short of the converged value (1.115 s against 1.134 s), and that error happened to cancel the finite-amplitude lengthening, so "-0.8% from 2 pi / W" agreed for the wrong reason. The default is now 19200 steps/s (slider max raised to 19200), where the period is within 0.1% of both the 38400 run and an independent integration of the exact pendulum equation. Cost: about 6 ms of CPU per frame headless.
- Fixed on 15 Oct 2026, comparison: the console now also gives the period for the rock's measured size (Lindstedt correction of the averaged equation, about +1%), so the comparison is no longer between a finite rock and a small-rock formula only. The remaining -0.2% is half the averaging's next terms (the exact equation gives 1.1341 s vs 1.135) and half step error.
- Fixed on 15 Oct 2026, restart bug: after each 15 s restart the first shake-average bin included the previous run's stale bin plus the new 8-deg start angle, so later runs reported "rocking up to 8.0 degrees" (the start tilt) instead of 7.6. Now every run reports 7.6.
- Kapitza's formulas are first-order averages (fast shaking, small rocking); the README says so.
- Near threshold (0.80 cm) the rocking period (4.3 s theory) is too long for the 8 s report window, so only "stays up" is checked, not a period.
- No upper stability limit is checked (Kapitza's averaging cannot see it; the DoublePendulum exhibit's Floquet analysis does that for the two-rod chain).
- The pivot is driven exactly (kinematic), i.e. an ideal shaker; the hinge is frictionless and the motion is planar.
- Fixed on 15 Oct 2026, documentation: the lua header listed a non-existent `KP_LOG=file.csv`; it now lists `KP_A1`, `KP_A2`, as the README does. The README's "131 g" now reads 130.8 g, as the console.

References

- A. Stephenson (1908), first observation — README / lua header.
- P. L. Kapitza (1951), averaged-potential explanation — README / lua header (titles and journals not given).

5.4 The double pendulum

Agrees with theory, with a numerical or open caveat. Energy to 2e-11; half-step twin 64 fm off at 1 s; Lyapunov 3.42/s; normal modes to 0.000% and 0.135%; Floquet window 0.728-1.356 cm. The throw-over time of the too-hard chain (not the fact) isn't converged, and the README says so.
Folder: demo/WyomingWill/DoublePendulum/

What it shows

Three tables. The butterfly: seven double pendulums let go from 120 degrees, each a billionth of a radian further round (bottom bobs 2 Å apart), swing as one for about five seconds and then fly apart: chaos. The two pure swings: the slow normal mode (rods together, lower rod sqrt 2 times as far), the fast one (rods opposed, 2.4 times as often), and a mix of both that never repeats, each with a chart recorder. Upside down: double pendulums on pivots shaken at 40 Hz; 0.6 cm falls, 1.0 cm stands (Kapitza/Acheson), 1.42 cm is thrown over by a parametric sway.

Model

Bullet only draws. The script integrates Lagrange's exact equations for equal point bobs on massless rods. Butterfly (l = 20 cm) and pure swings (l = 12 cm): Bulirsch-Stoer (Gragg modified midpoint with 2, 4, 6, 8, 10 substeps, Neville extrapolation in h^2, tenth order), 240 steps/s. Shaken chains (l = 10 cm): classical RK4, 4800 steps/s (120 per shake), pivot acceleration added to g. No friction anywhere. Extra hidden copies: a "twin" of butterfly pendulum 1 at half step (for a step-halving check) and a "shadow" offset by 1e-8 rad, renormalised every 0.25 s (Benettin's method for the largest Lyapunov exponent; norm in angles and angular rates / sqrt(g/l)). Upright stability: linearised equations integrated over one shake (RK4, 400 steps) to the 4x4 monodromy matrix; Floquet multipliers from its traces; window edges by 40-step bisection. Averaged theory: extra stiffness $(a^2\ w^2 / 2)\ D\ M^{-1}\ D$. Normal-mode amplitude correction: Lindstedt / normal-form average of the quartic Hamiltonian terms.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Butterfly energy (BS, 30 s)	conserved	max drift 2.1e-11 of m g l over all seven	conserved to 2e-11
Step-halving twin, bottom bob at t = 1 s	should be << 2 Å start spacing	64.4 fm (2 Å / 64.4 fm ~ 3100x)	integration error negligible
Twin reaches 1 cm	later than the seven	8.3 s vs 6.3 s for the neighbour	arithmetic error lags by 2 s
Lyapunov exponent (Benettin, 30 s)	no closed form	3.42 /s, doubling every 0.203 s	—
Separation 2 Å -> 1 cm	ln(1 cm / 2 Å) / lambda = 5.2 s	neighbour reached 1 cm at 6.3 s	observed 1.1 s (21%) later; order-of-magnitude check only
Slow mode, small swings	T = 2 pi / sqrt((g/l)(2 - sqrt 2)) = 0.90796 s (mine 0.907959)	—	—
Slow mode at 7 deg	Lindstedt: $x(1 + 0.0849\ A^2)$: 0.90911 s	0.90911 s over 21 swings	-0.000%
Slow mode shape	lower/upper = +sqrt 2 = 1.4142	1.4091	-0.36%
Fast mode, small swings	2 pi / sqrt((g/l)(2 + sqrt 2)) = 0.37609 s (mine 0.376089)	—	small-swing theory alone 2.7% off
Fast mode at 7 deg	Lindstedt: $x(1 + 1.8526\ A^2)$: 0.38678 s	0.38626 s over 51 swings	-0.135%
Fast mode shape	-sqrt 2 = -1.4142	-1.4112	-0.21%
Mixed swing (first 10 s)	half of each mode, each at its amplitude-corrected period	max deviation 0.1247 deg (1.78% of 7 deg); 5.11 deg with small-swing periods	good with correction

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Lower stability edge, upright chain	averaged: a w > sqrt((2 + sqrt 2) g l): a > 0.728 cm (mine 0.7282); Floquet: 0.728 cm	0.60 cm falls at 0.79 s; 1.00 cm stands	correct
Upper stability edge	Floquet: a < 1.356 cm (averaging cannot see it)	1.42 cm thrown over at 3.96 s (4800/s); 4.46 s (my 9600/s run)	correct; timing step-dependent
Rocking period, a = 1.0 cm	Floquet exact 0.879 s; averaged 0.881 s	0.881 s over 12 rocks, up to 3.8 deg	+0.3% / +0.1% (same at 9600 steps/s)
README console listing	README lines	identical to this run (incl. 1.42 cm at 3.96 s)	matches

Run used

```
cd demo/WyomingWill/DoublePendulum && HOLD_S=0 timeout 1500 ../../../release/bpp -f double-pendulum.lua -n 2400 |
    grep -v ALSA
  (also DP_STEPS=9600 -n 1200 as a step-size check of the shaken chains)

    t=1 s  the bottom bobs: the next one 1.0 nm apart, the farthest 6.3 nm  (the first, stepped twice as finely: 64.4
    fm)
    exactly (Floquet), it stands if 0.728 cm < a < 1.356 cm
    shaken 1.42 cm: the shaking pumps up a sway and throws it over at 3.96 s
    shaken 1.00 cm: it stands, rocking up to 3.8 degrees: a period of 0.881 s over 12 rocks (exact 0.879 s, +0.3%;
    averaged 0.881 s, +0.1%)
  Slow swing: a period of 0.90911 s over 21 swings (theory 0.90911 s for its size, -0.000%; small swings 0.90796 s);
  Fast swing: a period of 0.38626 s over 51 swings (theory 0.38678 s for its size, -0.135%; small swings 0.37609 s);
    to within 0.1247 degrees (1.78% of its 7; with the small-swing periods, 5.11 degrees)
  The butterfly, over 30 s: energy steady in all seven to 2.1e-11 of m g l;
    nearby swings part at 3.42 per second (Lyapunov, Benettin's method): doubling every 0.203 s,
    so 2.0 angstroms grows to 1 cm in about ln(1 cm / 2.0 angstroms) / 3.42 = 5.2 s; the next one got there at 6.3 s
    and the first, stepped twice as finely, only at 8.3 s
  [DP_STEPS=9600] shaken 1.42 cm: the shaking pumps up a sway and throws it over at 4.46 s
```

Known departures and honest caveats

- Idealised: point bobs, massless rods, no friction or air; Bullet plays no part in the physics.
- The Lyapunov estimate is a finite-time value from one 30 s trajectory; the predicted 5.2 s vs observed 6.3 s differ by 21%. Since 15 Oct 2026 the README says so itself: an order-of-magnitude check, not a precise test (the early growth is slower, before the gap aligns with the fastest-growing direction, and the bob gap is not the Benettin norm).
- Fast mode remains 0.135% off after the A^2 correction; the README attributes this to the next order.
- Mode-shape ratios (1.4091, -1.4112) are amplitude ratios of the measured maxima of each rod, slightly off sqrt 2 at 7 deg.
- The 1.42 cm chain's throw-over time is not converged in step size (3.96 s at 4800/s, 4.46 s at 9600/s, re-run 15 Oct 2026): it grows from a tiny seed that the step errors are part of, so only "is thrown over" is robust. Since 15 Oct 2026 the README states this. Its other numbers (0.881 s rocking, 0.79 s fall) were unchanged at 9600/s.
- The rocking period is measured on shake-averaged angles from a 4-degree tilt along the slow mode.
- "the next one 52.6 cm apart, the farthest 25.8 cm" at t = 7 s: "farthest" means pendulum 7 (the largest initial offset), not the maximum separation; after saturation it can be the smaller.
- Fixed on 15 Oct 2026: the lua header used to say energy holds "to about 1e-9" and the twin is "a few femtometres off after a second, a hundred-thousandth of the atom's width"; it now says 2e-11 of m g l and 64 fm, some three thousand times less than 2 Å, as the run gives (2.1e-11, 64.4 fm). The README's "start a hundred thousand times smaller" sentence was corrected the same way. Comments/README only; console output unchanged.

References

- D. J. Acheson (1993), stabilisation of an inverted chain of pendulums by vibration — README / lua header (journal not given).

- P. L. Kapitza (averaging), via the neighbouring Kapitza exhibit — README.
- Benettin's method (Lyapunov exponent), Lindstedt's method / normal form, Floquet theory, Bulirsch-Stoer with Gragg's modified midpoint — cited by method name only.

5.5 Pendulums with a twist

Agrees with theory. No flip below 70.53°; at μ = 3 the hidden integral stays 0.7500000000 (wanders at 2.8 and 3.2); spring's growth 0.526/s against 0.527 (Floquet).
Folder: demo/WyomingWill/PendulumsTwist/

What it shows

Three tables of pendulums that defy intuition. A "flip row" of nine equal-bob double pendulums, released from rest with both rods at the same angle (60 to 120 degrees), shows that energy forbids the lower rod from going over the top below 70.53 degrees, while being *allowed* above that line does not mean it *happens* (72 and 76 degrees never flip). Three swinging Atwood machines (mass ratios mu = 2.8, 3.0, 3.2) look alike, but only at mu = 3 is there a hidden second conserved quantity, shown on a dial whose needle stays still. Two swinging springs show 2:1 autoparametric resonance: tuned bounce:swing = 2:1, the bounce drains into a swing that grows to about 11 degrees and back; tuned 2.4:1 it just bounces.

Model

Every pendulum is integrated by the script from its exact Lagrangian equations of motion; Bullet only draws (all objects are non-colliding, mass-0 display bodies; cords and springs are drawn as beads). Integrator: Bulirsch-Stoer (Gragg modified midpoint with 2, 4, 6, 8, 10 substeps, Richardson-extrapolated to zero step: tenth order), 240 macro-steps per simulated second (4 per 1/60 s frame); the Atwood machines take two half-steps, i.e. 480 per second. No damping or friction anywhere. Units cm, kg, s; g = 981 cm/s^2.

- Flip row: rods l = 7 cm, equal point bobs, angles 60, 66, 70, 72, 76, 80, 90, 100, 120 degrees.
- Atwood: (1 + mu) r'' = r theta'^{2} - g (mu - cos theta), r theta'' = -2 r' theta' - g sin theta; start r = 18 cm, theta = 60 degrees, from rest; zero-size pulleys 8 cm apart.
- Springs: stretched length l = 20 cm at rest, released Z = 2 cm lower with a 0.05-degree swing; k chosen so sqrt(k/m) = 2.0 or 2.4 times sqrt(g/l). The exact (Floquet) growth rate is computed in the script by RK4 (2000 steps) over one bounce period of the linearised swing equation with prescribed r(t).
- Everything restarts every 60 s (PT_CYCLE); the report prints at the end of each cycle.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Lower-rod flip threshold	E = -3mgl cos a must reach -mgl: cos a <= 1/3, a >= 70.53 deg	Never flipped: 60, 66, 70 (below line); also 72, 76 (above line, allowed but not seen in 60 s). Flipped: 80 (10.86 s), 90 (2.81 s), 100 (1.26 s), 120 (1.82 s)	Consistent: no flip below 70.53 deg
Upper-rod flip threshold	needs +mgl: cos a <= -1/3, a >= 109.47 deg	Only 120 deg flips its upper rod (4.87 s); 100 deg does not	Consistent
Flip-row energy conservation	E constant	max abs(E - E0) = 1.1e-08 of m g l	Conserved (README: 1.1e-08, identical)
Atwood mu = 3 second integral	I = p_xi^2/8 - E xi^2 + 2 xi^4, p_xi = 4 xi'(xi^2 + eta^2), constant only at mu = 3 (Casasayas-Nunes-Tufillaro 1990)	I between 0.7500000000 and 0.7500000000	Constant to 10 printed digits
Same quantity, mu = 2.8	not conserved	I between 0.6151989479 and 1.0500000000	Wanders, as expected

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Same quantity, mu = 3.2	not conserved	I between 0.3572701538 and 0.8542564850	Wanders, as expected
Atwood energy (relative)	E constant	3.1e-13 (2.8), 2.8e-13 (3.0), 2.1e-13 (3.2)	Conserved to round-off
Atwood closest approach to pulley	(no theory; README value)	7.0 / 4.5 / 3.7 cm; no pulley hit	Matches README exactly
Spring 2:1 growth rate	s = (3/4)(Z/l) sqrt(g/l) = 0.525 /s (first order); Floquet multiplier of one bounce: 0.527 /s	fitted from swing peaks 0.2 to 3 deg: 0.526 /s	-0.2% vs Floquet, +0.2% vs first order
Spring 2:1 peak swing	(README: "grows to 11 degrees")	at most 11.4 degrees	Matches README
Spring 2.4:1	Floquet: no growth (multipliers on unit circle)	never grew; at most 0.0 degrees	Exact
Spring energy (relative)	E constant	4.9e-14 (2.0:1), 3.6e-14 (2.4:1)	Conserved to round-off

The threshold angles 70.53 and 109.47, and the theoretical rates 0.525 and 0.527, are computed by the script and printed at each release.

Run used

```
cd demo/WyomingWill/PendulumsTwist && HOLD_S=0 timeout 1500 ../../../release/bpp -f pendulums-twist.lua -n 3620
```

```
    Tuned 2.0:1, the swing grows as exp(s t): s = (3/4)(Z/l) w = 0.525/s to first order, 0.527/s exactly (Floquet);
    flip row, let go at 80 degrees: the lower rod goes over the top at 10.86 s
    the flip row: the lower rod over the top -- 80: 10.9 s, 90: 2.8 s, 100: 1.3 s, 120: 1.8 s (upper 4.9 s); never:
    60, 66, 70, 72, 76 degrees (below 70.53 it can't; above, it may)
      energy steady to 1.1e-08 of m g l
    Atwood, 2.8 times: I between 0.6151989479 and 1.0500000000 (no such constant here); energy steady to 3.1e-13;
    closest to its pulley 7.0 cm
    Atwood, 3.0 times: I between 0.7500000000 and 0.7500000000 (constant, as at 3 it must be); energy steady to
    2.8e-13; closest to its pulley 4.5 cm
    Atwood, 3.2 times: I between 0.3572701538 and 0.8542564850 (no such constant here); energy steady to 2.1e-13;
    closest to its pulley 3.7 cm
    spring, 2.0:1: the swing grew at 0.526/s, from 0.2 to 3 degrees (theory 0.527 exactly, 0.525 to first order); at
    most 11.4 degrees; energy steady to 4.9e-14
    spring, 2.4:1: the swing never grew; at most 0.0 degrees; energy steady to 3.6e-14
```

Every number in the README's console excerpt was reproduced digit for digit by this run.

Known departures and honest caveats

- The README says plainly that the mu = 2.8 and 3.2 Atwood machines are **not visibly chaotic** when released from rest at 60 degrees. The author reports (README; not re-run here) that two copies started 1e-6 rad apart separate at no measurable exponential rate for release angles from 60 to 170 degrees and ratios from 1.3 to 3.2. So the exhibit does not claim to show chaos. What sets mu = 3 apart is only the hidden constant, shown on the dial (needle turns 120 deg per 0.5 change in I).
- The README also states (not re-run here) that from 90 degrees at mu = 2.5 the light weight is whipped into its pulley within 16 s. The script stops integrating a machine if r < 3% of 18 cm. The pulleys have zero size.
- Flip times near the threshold (72, 76 deg "never"; 80 deg at 10.9 s) cover a 60 s window only. "Never" means "not within 60 s", not proven never. The energy line is a necessary condition, not a sufficient one.
- The flip-row energy drift (1.1e-08 of mgl) is about five orders larger than the Atwood and spring drifts; it is still small. Fixed on 15 Oct 2026: the Lua header comment used to say "steady to 1e-10"; it now says "steady to 1e-8 of m g l", matching the console and README (1.1e-08). Comment only; console output unchanged.
- The growth rate is fitted to swing peaks between 0.2 and 3 degrees only, where the linear (exponential) phase holds. The later energy exchange (11.4 deg maximum) is nonlinear and is not compared with theory.

- Idealisations: point bobs, massless rigid rods and inextensible massless cords, frictionless pulleys of zero radius, massless linear springs.

References

- S. Casasayas, A. Nunes, N. Tufillaro (1990): the integrable swinging Atwood machine at mu = 3 (cited in code and README; no journal given).
- Pujol, Tufillaro and group, arXiv:2507.00058: the form of the integral I used here (README).
- Mathieu's equation and Floquet theory, for the parametric growth rate (README; no specific source).

5.6 The chain fountain

Agrees with theory, with a numerical or open caveat. Now starts from a poured heap of ball chain of Biggins and Warner's size: h2/h1 = 0.140 against 0.12-0.14 measured, matching every link simulated within the pour-to-pour spread (an independent bead-chain model agrees), at 14 ms a frame at 1/8 speed. The old laid coil (H) stands at 0.06-0.07: the pile's arrangement was what was missing.
Folder: demo/WyomingWill/ChainFountain/

What it shows

A ball chain (4.5 mm balls 6.5 mm apart, as Biggins and Warner used), poured into a jar 1.5 m above the floor as a jumbled heap, is drawn over the rim by an unseen hand and let go. It then siphons out on its own, and rises in an arch above the rim instead of sliding over it. This is the "Mould effect", filmed by Steve Mould in 2013 and explained by Biggins and Warner in 2014. A link picked up by one end rotates and pushes its other end into the pile, and the pile's upward reaction throws the chain up. A rope, with no links, would just slide over. Since 16 Oct 2026 the default is this poured heap; H (or CF_PILE=coil) gives the coil of 1.5 cm links the exhibit had before, kept as the comparison.

Model

- Default (since 16 Oct 2026): the poured heap. Chain: 1500 Bullet capsule links (9.75 m), 0.65 cm joint to joint, radius 0.225 cm, the mass at the ends ($I = m L^2/4 + 0.4\ m\ r^2$, as balls on light rods), cone-twist joints with a 30 deg swing limit, true friction 0.3 (0.548 a body; Bullet multiplies), no restitution, no damping. Drawn with chain-fountain-meshes/bead-link.obj.
- Heap: poured once offline (CF_POUR=1 CF_SEED=2 CF_SAVEHEAP; every link live; a hand 6 cm over the jar floor wandering within 2 cm of the axis, 1 m/s), settled, stored as joint points from the jar floor up (chain-fountain-meshes/heap.txt). In the exhibit its links are fixed bodies except a live zone: every link within 4 cm across of the pick-up and no more than 0.5 cm below it (CF_ZONE, CF_ZDEPTH), plus the 10 links after the pick-up; live heap links 1 cm outside the zone and slower than 1 cm/s are fixed again; a live link next to a fixed one is held to it by a cone-twist joint. Each link keeps one fixed body, parked below the floor while it moves (hall identity).
- Run: a scripted hand draws the end up 4 cm over the rim, out to x = 12 cm, down at 1.2 g to the floor, and lets go (0.64 s). 1800 steps/s (slider min 1800 in heap mode), 20 iterations, joint ERP 0.5 (Bullet's 0.2 is not step-converged), touching contacts only (CPT 0). Starts at 1/8 speed (Z: 1/8 <-> 1).
- Sliders in heap mode: bending held at 30 deg (the heap was poured so; moving it puts it back and the console says so once); layers inert; friction restarts; height works.
- Both piles: links below y = 2.5 cm are frozen (joint removed, drawn only), so b = 0. Wake, freeze and measurement run after every physics step; output is the same at any display speed.
- The coil (H, CF_PILE=coil), as before:
- Chain: Bullet rigid capsule links, 1.5 cm joint-to-joint, radius 0.3 cm, mass 0.5 g, no damping, no restitution. Neighbours are joined by btConeTwistConstraint with a swing limit equal to the bending angle (default 30 deg; CF_BEND; slider 20-90 deg, was 10-90) and free twist. Friction 0.3 set on links, glass and disc; Bullet multiplies the two bodies' values, so every chain contact has 0.09 (found 16 Oct 2026; with a true 0.3, CF_MU=0.548, the fountain is unchanged: 8.6-9.8 cm at 30 deg, 5.1-5.7 at 90).

- Pile and jar: a rigid disc stands in for the coil, lowered one layer (0.65 cm) when the first link of a new layer is woken. The jar is 24 glass slabs (7 cm inside radius, 8 cm wall).
- Initial coil (changed 15 Oct 2026): flat spirals, outside in then inside out, all turning the same way. Every link is 1.5 cm and no joint is laid bent past its limit. The inner radius is max(3 cm, the tightest the chain can turn while tipping down) = 3.1 cm at 30 deg, 4.8 cm at 20. At a layer's end the chain tips down 0.65 cm over 3 links, going on round. 10 layers: 15.8 m (1053 links) at 30 deg, 9.6 m at 20, 16.1 m at 60/90. Waiting links are drawn, not simulated.
- Simulation window: links are woken so that 10 links ahead of the pick-up still lie where they were laid, at most 1200 per simulated second. The last link woken is held at its far end by a cone-twist joint to a fixed body in the next link's place. Links below y = 2.5 cm are frozen: their joint is removed and they become drawn-only, non-colliding meshes. About 130-200 links are active.
- Stepping: exactly 1/1800 s at every display speed. Each frame takes as many whole steps as its share of simulated time. Wake, freeze and measurement run after every step. 20 solver iterations.
- Bullet does all dynamics and contacts. The script does wake, anchor, freeze, the pile disc, and (read-only) measurement.

Theory checks

Biggins and Warner (2014): pile push a m v^2 up on the link being lifted, floor pull b m v^2. Steady flow gives v^2 = g h1/(1 - a - b) and h2 = a v^2/g (narrow arch). Here b = 0 by construction (frozen links, no joint, no collisions). Fresh runs, 10 layers, CF_SPEED=1, steady window t = 1.00 s to the last coil link woken. v = rate links land x 1.5 cm. "Push" = disc contact force minus the weight of links touching it, divided by m v^2 (CF_PUSH=1). "Carried" (new, CF_PUSH=1) = -(T - m v_i^2 - m g y)/(m v^2) on the freely falling chain 20-100 cm below the pick-up. T is the tension worked out from the links' accelerations, summed up from the free lowest link. A steady B&W chain keeps this equal to a from pile to floor.

Quantity	Theory / source	Simulation (this run)	Agreement
Heap (default): h2/h1	B&W 0.14; Pantaleone 0.12 +- 0.01 (ball chains)	0.140 with the shipped heap (h2 20.7 cm, h1 148.3 cm); 0.110-0.176 (mean 0.147) over 5 heaps; every link simulated: 0.117-0.160 (0.140)	**yes**
Heap: speed	experiments 0.69-0.71 sqrt(2 g h1)	v 431 cm/s = 0.80 sqrt(2 g h1) (no air drag here, b = 0)	faster
Heap: a from height vs speed	equal in a steady narrow arch	0.11 vs 0.22: window 0.64 s, links land at 447 then 416 cm/s as the heap runs out; rms landing 389 vs 431 cm/s (wavy)	**no**: not steady; the console says so and the README reads a from the height
Heap: rod model	a = (1 - I/(m b^2/4))/2	gives no push for balls on light rods (console says so; B&W's three-ball link: 1/6)	push here comes from the pile
Heap: same run at any speed, hall, pause	identical	1x = 1/4 = 1/8 (18 lines); cmp2 IDENTICAL (16); pause prefix exact (18)	**yes, exact**
Heap: cost per frame at 1/8 (budget 16.7 ms)		flow 14.0 / whole run 11.2 ms standalone; Cabinet 14.8 / 12.1	fits (at 1/4 it would be 28-30)
The coil (H, CF_PILE=coil): the rows below			
Same run at any display speed	must not depend on frame rate	0.25x with pause = 1x (30 deg); 0.5x = 1x (60 deg, CF_PUSH)	**yes, exact**
Speed bound	ideal chain sqrt(g h1) = 384 cm/s	397 / 450 / 425 / 432 cm/s (30/20/60/90)	faster, as a > 0 requires
Pile push a (direct)	B&W rod model a = (1 - I/(m b^2/4))/2: 1/6 for their chain, 0.31 for these links	0.31 / 0.27 / 0.19 / 0.17; 0.05-0.10 lower (0.22/0.17/0.13/0.12) if every link within 7 mm of the pile is counted as lying on it	push present, same order as theirs

Quantity	Theory / source	Simulation (this run)	Agreement
a from speed, 1 - g h1/v^2	= push if nothing else acts	0.06 / 0.27 / 0.18 / 0.21	60, 90: within the push's uncertainty. 30: **no**, the rim holds the chain up with 0.31 m v^2 (20: 0.14)
a from height, g h2/v^2	= push (narrow steady arch)	0.07 / 0.05 / 0.07 / 0.07; h2 = 10.5 / 10.7 / 12.8 / 13.2 cm	**no**: fountain about 1/3 of a v^2/g with the measured push (35 cm at 60 deg)
30 deg: height vs speed	equal	0.07 vs 0.06	agree only by chance (push 0.31)
Energy at landing	lossless chain: 1/(2(1 - a)) of the drop = 60-72%	42 / 44 / 50 / 49%; rms landing speed 352 / 361 / 383 / 380 cm/s, 11-20% below v	**no**: 11-30% of the power lost on the way
a the falling chain carries	= push (steady, lossless)	-0.13 / -0.07 / 0.04 / 0.03	**no**: a third or less of the push reaches the falling chain; below zero where the rim takes more (20, 30)
Stationary pick-up (CF_LAYOUT=flake, CF_JAR=0, 40 layers; 45/60/90 deg)	h2 = a v^2/g	push 0.28/0.23/0.13; h2 18.2/17.3/9.1 cm = 0.36/0.40/0.36 of push v^2/g; spiral on the same table 0.30/0.29 (60/90)	**no**: helps a little (60 deg 17.3 vs 10.6 cm), not the cause
Flake: theory with the carried a (0.10/0.08/0.06)	h2 = a v^2/g; v = sqrt(g h1/(1 - a)); energy 1/(2(1 - a))	h2 ratio 1.01/1.16/0.79; rms landing speed 0.979/0.972/0.962 of the formula; energy 53/51/49% against 56/54/53%	**yes, within 21% (height), 4% (speed), 4 points (energy)**
One thing at a time (flake 60 and spiral 60): steps 3600, 7200; iterations 50, 100; friction 0; joint stops off (CF_STOP=179); 5, 15 woken ahead; links half size (CF_LINK=0.5)		h2/(push v^2/g) 0.25-0.50 (flake), 0.24-0.35 (spiral); energy 43-53%	no model channel restores the fountain
Shortcut vs every link simulated (CF_LIVE=1; 6 layers, 5 runs each at jar 146-154 cm, CF_PUSH=2)	the shortcut should not matter	h2 every link / exhibit's way: 8.6±0.4 / 10.6±1.5 (30), 6.3±0.5 / 10.8±0.6 (60), 5.5±0.4 / 11.2±0.8 (90) cm. Unmoved-pile push 0.49/0.23/0.20 vs 0.20/0.17/0.15. Carried -0.13/-0.19/-0.16 vs -0.01/0.01/0.01. Rim 0.26/0.32/0.28 vs 0.13/0.03/0.03 m $v^{2.}$ v 399/379/397 vs 420/418/423 cm/s	shortcut **not** the cause of the lost push (every-link chain carries none either), but it **raises the fountain** 1.2x (30) to 2x (60, 90) and keeps the chain off the rim
Reference convergence		every link, 3600 steps/s: h2 8.5, 9.4 (30); 5.7, 6.0 (90). Settled 0.3 s first: 9.6 (30). Landed links left on the floor: 8.5 (30)	within the 5-run spread
Cheaper match (CF_WAKE=layer)	= every link	4 live layers at 30: v 406±17, h2 8.7±0.6, push 0.48, carried -0.09, rim 0.20; 3 layers match at 60; at 90 neither (v 356-359 vs 397, rim 0.57-0.62 vs 0.28)	matches at 30 and 60
Cost per frame, 0.25x, standalone (Cabinet run, 1200 frames)	16.7 ms at 60 fps	exhibit 4.4 ms (12.8 s); 4 live layers 21 ms (30.3 s); every link 35 ms (46.9 s)	faithful ways not affordable: default unchanged

Quantity	Theory / source	Simulation (this run)	Agreement
Cheaper faithful modes (cost per frame, 10 layers, 30 deg, quarter / real speed; physics 5 runs vs every-link)	budget 16.7 ms	exhibit 5.4 / 22 ms; every link 52 / 209; every link + sleep 53 / 208 (one island, no gain); 4 live layers 32 / 129; + only touching contacts (CF_CPT=0) 27 / 109, physics unchanged at 30; + 10 iterations 25 / 99, changes it (v 372 vs 399, rim 0.52); + 1200 steps 21 / 86, unchanged at 30; 3 live layers + CF_CPT=0 + 1200 steps 15.0 / 62 (Cabinet 15.1 at quarter speed)	only that last one fits quarter speed, and it matches at 30 deg only (60: v 352 vs 379; 90: 354 vs 397, rim 0.61 vs 0.28); nothing fits real speed
Independent reference: ideal bead chain (reconstruction/bead-chain.c; B&W's geometry, 4.5 mm balls 6.5 mm apart, 30 deg; rigid rods; geometric bending limit; inelastic contacts; PBD, 2e-5 s), same jar, coil and drop, 3 layers	experiments: h2/h1 0.14 (B&W), 0.12-0.14 (Pantaleone), a 0.12 / b 0.11 (Biggins EPL fit)	h2/h1 0.038-0.063 in every variant: friction 0.3 / 0.09 / rim 0, step 2e-5 / 1e-5, 4 / 8 sweeps, bend 20, pile above rim, jar 2 cm lower. Pile push 0.19-0.29; carried -0.02 to -0.67; rim 0.09-0.68	**reference = Bullet**: with friction 0.09 it gives v 404, h2 7.5, rim 0.09, carried -0.07, against Bullet all-live at 30 deg v 399 ± 15, h2 8.6 ± 0.4, rim 0.26, carried -0.13. The low fountain is not Bullet's links or joints. (bead-chain.c applied friction in its position passes and over-gripped; longer coils and heaps jammed in it. bead-pour.c replaces it for the heap test, see above)
Heap test (16 Oct 2026): ball chain of B&W's size (0.65 cm links, 4.5 mm, mass at the joints, 9.75 m) poured into the jar (CF_POUR=1) vs laid as the exhibit's coil (CF_POUR=2), every link live, end drawn over the rim by a scripted hand, h1 = 148 cm, friction 0.3 per contact	experiments 0.12-0.14 (B&W; Pantaleone), v = 0.69-0.71 sqrt(2 g h1)	Bullet heap, 5 pours: h2/h1 0.109-0.147 (mean 0.124), v 0.69-0.80, pile push 0.16-0.21, carried -0.21 to 0.12, rim 0-0.43; 3600 steps/s (3 pours) 0.132-0.159 (0.142 vs 0.131). Bullet coil: chain on the rim (2.6-10.9 m v^2), slows and stalls, h2/h1 0.055-0.057; stalls at 3600 steps and with friction 0. Reference (reconstruction/bead-pour.c, rods 1 mm solid, XPBD 4 passes per 1e-5 s) heap, 3 pours: 0.123-0.183 (0.151), half step 0.174/0.157, 8 passes 0.190; coil 0.058 (rim 2.16), half step 0.060	**yes: the pile was the missing ingredient.** Heap = experiments' range in both engines; coil rides the rim at 0.06. 1 of 5 Bullet and 1 of 3 reference pours (and some reloads) caught the rim (0.06-0.12)
Heap with the exhibit's 1.5 cm links (450 links; CF_T0=0.5)		pour forms rings round the wall, not a heap; h2/h1 0.078-0.106 (4 pours) vs coil 0.057-0.077 (3 heights); 700 links fill the jar and siphon out while pouring	the exhibit's coarse links stay low however they lie in this jar

Quantity	Theory / source	Simulation (this run)	Agreement
Heap shortcut vs every link of the same 5 heaps (second round: ERP 0.5, CPT 0, true friction 0.3)	= full, at 1800 and 3600 steps/s	every link: h2/h1 0.140 (0.117-0.160) at 1800, 0.137 (0.093-0.167) at 3600; pile push 0.17-0.29. Zone 3 cm/0.5 deep: 0.161 / 0.168. Zone 3/1 deep: 0.155. **Zone 4 cm/0.5 deep (CF_PILE=heap): 0.147 (0.110-0.176) / 0.145 (0.069-0.177, one run on the rim)**; v 416-445, rim 0-0.17, carried 0-0.13, pile push 0.13-0.20. (First-round option, heap fixed + links lying on the chain woken: 1.6x too tall.)	**yes** for the 4 cm zone (h2/h1, v, rim, carried within the pour-to-pour spread at both steps; pile push a little low); the 3 cm zone 15% too tall
Step convergence of the full heap (every link, 5 heaps)	must not depend on the step	Bullet ERP 0.2: h2/h1 0.131 (1800), 0.160 (3600), 0.150 (7200, 3 heaps). ERP 0.5: 0.148 (1800), 0.145 (3600); ERP 0.8: 0.135 (1800); ERP 0.5 at 1200: 0.118	1800 converged only with ERP 0.5 (the joints give too much at 0.2); 1200 not
Cost per frame during the flow (t 0-1.75 s) / whole run, standalone (Cabinet)	16.7 ms	exhibit at 1/4: 4.8 / 4.8 (5.4 / 5.4). CF_PILE=heap at 1/4: 27.5 / 21.8 (30.0 / 24.0); at 1/8 (its default): 14.1 / 11.4 (14.3 / 11.7). Zone 3 cm at 1/4: 22.5 (24.4) flow. Every link of the heap at 1/4: 36.4 / 48.8. No zone at all (10 ahead): 13.5 flow. Solver iterations 5 instead of 20: -12%. GUI under xvfb (software GL): heap 186 ms a frame, exhibit 77	**fits only at 1/8 speed**; nothing faithful fits at 1/4
h2/h1	B&W measured 0.14 (a = 0.12 by their formula with b = 0)	0.070 / 0.071 / 0.085 / 0.088 (spiral), up to 0.13 (flake 45)	fountain lower
Chaos spread (jar 146, 148, 152, 154 cm)		v 414-448 cm/s, push 0.15-0.41, height-a 0.05-0.07	
Step 1/3600 s	converged?	20 deg: v 477, push 0.39, h2 19.8; 30: 437, 0.32, 14.3	**not converged** at 20 deg
Fountain above rim (CF_LOG, t >= 0.5 s), median (IQR), highest; 14 layers in brackets	README	20: 4.7 (3.5-5.8), 11.9 [4.5 (3.0-5.8)]; 30: 5.4 (3.4-8.0), 12.4 [4.1 (1.9-6.7)]; 60: 7.7 (6.0-8.3), 10.1 [6.3]; 90: 7.8 (6.6-8.6), 12.4 [7.3]	README replaced
Stiffer chain, higher fountain	B&W; old README	no trend beyond chaos (medians at 4 jar heights: 6-8 / 5-7 / 4-8 / 5-8 cm)	old trend came from the bad coil
Time to empty	README "about 4 s"	4.25 s (30), 4.00 (60, 90), 2.50 (20, shorter chain)	yes

Run used

```
cd demo/WyomingWill/ChainFountain
HOLD_S=0 ../../../release/bpp -f chain-fountain.lua -n 1500    # the default: the heap at 1/8; CF_SPEED=1 -n 180 and
    CF_SPEED=0.25 -n 720 give the same lines
bash ../Cabinet/report/cmp2.sh chain ChainFountain chain-fountain.lua 1500 cf   # IDENTICAL (16); CF_PILE=coil 1200:
    IDENTICAL (22)
# the coil's runs below were made when it was the default: add CF_PILE=coil to repeat them
CF_PUSH=1 CF_SPEED=1 CF_BEND=<20|30|60|90> CF_LOG=d<b>.csv HOLD_S=0 ../../../release/bpp -f chain-fountain.lua -n 300
CF_PUSH=1 CF_SPEED=1 CF_LAYERS=14 CF_BEND=<b> ... -n 400 ; CF_HEIGHT=<146|148|152|154> ... ; CF_STEPS=3600 ... -n 400
CF_PUSH=1 CF_SPEED=1 CF_JAR=0 CF_LAYOUT=flake CF_LAYERS=40 CF_BEND=<45|60|90> HOLD_S=0 ... -n 400   # plus CF_STEPS,
    CF_ITER, CF_MU=0, CF_STOP=179, CF_AHEAD, CF_LINK=0.5 one at a time
timeout 3000 ../../../release/bpp -f chain-fountain.lua -n 1200        # 0.25x with pause: identical to 1x
bash ../Cabinet/report/cmp2.sh chain ChainFountain chain-fountain.lua 1200 cf   # IDENTICAL (22 lines); CF_PUSH=1:
    IDENTICAL (25); flake 60 CF_PUSH=1 (1300 frames): IDENTICAL (23)
CF_LIVE=1 CF_LAYERS=<6> CF_HEIGHT=<146..154> CF_BEND=<30|60|90> CF_PUSH=2 CF_SPEED=1 HOLD_S=0 ... -n 400   # and the
    same without CF_LIVE, and with CF_WAKE=layer CF_LIVELAYERS=<3|4>
CF_WAKE=layer CF_LIVELAYERS=3 CF_CPT=0 CF_STEPS=1200 bash ../Cabinet/report/cmp2.sh chain ... 1200 cfm   # IDENTICAL
    (23); pause prefix (25 lines) and 0.5x = 1x also hold
CF_LIVE=1 CF_LAYERS=6 CF_PUSH=2 bash ../Cabinet/report/cmp2.sh chain ... 1100 cfl   # IDENTICAL (21); CF_WAKE=layer
    CF_LIVELAYERS=4 CF_PUSH=2, 1200 frames: IDENTICAL (26)
B="CF_LINK=0.4333 CF_RADIUS=0.225 CF_BEADI=1 CF_NLINKS=1500 CF_MU=0.548 CF_PUSH=2 CF_SPEED=1 HOLD_S=0"
env $B CF_POUR=1 CF_SEED=<1..5> [CF_STEPS=3600] ../../../release/bpp -f chain-fountain.lua -n 1150   # heap;
    CF_POUR=2 [CF_HEIGHT=148|152]: coil
CF_POUR=1 CF_NLINKS=450 CF_SEED=<1..4> CF_T0=0.5 CF_PUSH=2 CF_SPEED=1 HOLD_S=0 ... -n 1150   # exhibit links;
    CF_POUR=2: coil
CF_PILE=heap CF_POUR=1 CF_SEED=<1..5> CF_SAVEHEAP=heap-k.txt CF_MU=0.548 CF_PUSH=2 CF_SPEED=1 HOLD_S=0 ... -n 1150
    # the five heaps
CF_PILE=heap CF_HEAP=heap-k.txt [CF_LIVE=1] [CF_ZONE=3|4] [CF_ZDEPTH=0.5|1] [CF_STEPS=1200|3600] [CF_ERP=0.2|0.5|0.8]
    CF_PUSH=2 CF_SPEED=1 HOLD_S=0 ... -n 900   # zone vs every link
(cd reconstruction; cc -O2 -o bead-pour bead-pour.c -lm; ./bead-pour NB=1500 D=0.45 ROD=0.05 SEED=<1|2|3>
    SAVE=heapk.txt; ./bead-pour ... LOAD=heapk.txt HS=5e-6 | ITERS=8; ./bead-pour ... LAYOUT=1 [HS=5e-6])
CF_PILE=heap bash ../Cabinet/report/cmp2.sh chain ChainFountain chain-fountain.lua 1500 cfz1   # IDENTICAL (15);
    default 1200: IDENTICAL (22)
```

```
--- 9.8 m of chain, 1500 links; the jar 150 cm up; joints bend 30 deg at most ---
drawn over and let go (0.64 s)
t= 0.25s  fountain 12.3 cm above the rim (average 11.3, highest 13.4); chain 529 cm/s; 564 links moving, 1.2 m out
t= 0.50s  fountain 16.4 cm above the rim (average 15.1, highest 16.5); chain 489 cm/s; 691 links moving, 2.4 m out
t= 0.75s  fountain 16.1 cm above the rim (average 16.3, highest 17.2); chain 435 cm/s; 580 links moving, 3.5 m out
t= 1.00s  fountain 17.3 cm above the rim (average 17.5, highest 18.2); chain 451 cm/s; 529 links moving, 4.6 m out
t= 1.25s  fountain 15.0 cm above the rim (average 14.9, highest 18.2); chain 444 cm/s; 482 links moving, 5.7 m out
t= 1.50s  fountain 11.5 cm above the rim (average 12.5, highest 18.2); chain 413 cm/s; 425 links moving, 6.8 m out
t= 1.75s  fountain 11.0 cm above the rim (average 11.1, highest 18.2); chain 379 cm/s; 302 links moving, 7.8 m out
t= 2.00s  fountain -108.1 cm above the rim (average -64.4, highest 18.2); chain 636 cm/s; 172 links moving, 8.6 m out
t= 2.25s  all of it out of the jar; the fountain stood up to 18.2 cm above the rim
  theory (Biggins and Warner 2014), steady part t=1.00-1.64s; h1 = 148.3 cm, the pick-up above where a link lands
    (the rim is 7.2 cm above the pick-up)
  chain 431 cm/s (the rate links land): an ideal chain (a = b = 0) would go sqrt(g h1) = 381 cm/s and not rise above
    the pick-up
  top of the chain 20.7 cm above the pick-up on average (h2): h2/h1 = 0.140 (their chain: 0.14); b = 0 here, so v =
    sqrt(g (h1 + h2)) = 407 cm/s
  a = g h2/v^2 = 0.11 from the height, 1 - g h1/v^2 = 0.22 from the speed (the rod model gives none for balls on
    light rods; 1/6 for their three-ball links)
  not steady: links land at 447 cm/s in the first half of that window and 416 in the second (the heap running out);
    the speed rule needs a steady chain, so its reading here is unreliable
  the links reach the floor at 389 cm/s (rms), with 52% of the energy of their drop (an ideal chain: 50%; one the
    pile pushes, more)
```

The coil (CF_PILE=coil):

```
t= 4.25s  all of it out of the jar; the fountain stood up to 12.4 cm above the rim
  theory (Biggins and Warner 2014), steady part t=1.00-3.68s; h1 = 150.2 cm, the pick-up above where a link lands
    (the rim is 5.3 cm above the pick-up)
  chain 397 cm/s (the rate links land): an ideal chain (a = b = 0) would go sqrt(g h1) = 384 cm/s and not rise above
    the pick-up
  top of the chain 10.5 cm above the pick-up on average (h2): h2/h1 = 0.070 (their chain: 0.14); b = 0 here, so v =
    sqrt(g (h1 + h2)) = 397 cm/s
  a = g h2/v^2 = 0.07 from the height, 1 - g h1/v^2 = 0.06 from the speed (rod model for one of these links: 0.31)
  the links reach the floor at 352 cm/s (rms), with 42% of the energy of their drop (an ideal chain: 50%; one the
    pile pushes, more)
  the pile's push, measured (its contact forces less the weight lying on it): a = 0.31; the rim holds the chain up
    with 0.31 m v^2
  with that push a steady narrow arch would stand h2 = a v^2/g = 50.4 cm above the pick-up (it stood 10.5)
  what the falling chain carries of it (its tension, from the links' accelerations, 20-100 cm below the pick-up): a =
    -0.13
[flake 60, table] the pile's push ... a = 0.23 ... h2 = a v^2/g = 42.9 cm (it stood 17.3) ... carries of it ... a =
    0.08
[60 deg] a = g h2/v^2 = 0.07 from the height, 1 - g h1/v^2 = 0.18 from the speed ... a = 0.19; the rim ... 0.02 m v^2
```

Pause: CF_SPEED=1, HOLD_S=0 for 300 frames is an exact prefix of the default hold for 330 frames (24 lines; with CF_PUSH=1, 27 lines). Default output (CF_PUSH off) is byte-identical to the committed b241d9c. 0.25x with pause = 1x with CF_PUSH=1; flake 60 at 0.5x = 1x. 16 Oct 2026 (heap test): default output, CF_PUSH=1 and CF_LIVE=1 CF_LAYERS=6 CF_PUSH=2 byte-identical to the committed f7027b4 (24, 27, 23 lines); Third round (heap default): CF_PILE=coil byte-identical to the previous default (with pause 1230 frames, 1x 300 frames, CF_PUSH=1 300 frames); heap: HOLD_S=0 1400 frames an exact prefix of the default 1500 (18 lines), 1x = 1/4 = 1/8 (18 lines); coil: prefix (24 lines), 1x = 1/4 = 1/8 (24); cmp2 heap IDENTICAL (16), coil IDENTICAL (22); the Cabinet loads 21 exhibits. Second round: default, CF_PUSH=1 and CF_LIVE=1 CF_LAYERS=6 CF_PUSH=2 byte-identical to 4b48858 (24, 27, 23 lines); CF_PILE=heap (final): HOLD_S=0 1400 frames an exact prefix of the default hold for 1430 (17 lines); 1x = 0.5x = 0.25x = 1/8 (17 lines); run to run identical.

Known departures and honest caveats

- Fixed 15 Oct 2026: frame-rate dependence (all work per physics step, fixed 1/1800 s) and the 60-degree stall (last woken link anchored; landed links don't collide).
- Fixed 15 Oct 2026: the coil was laid wrong. The spiral went in to 3 cm at every bending, so at 20 deg 304 of 1054 joints were laid bent past their stop (worst about 29 deg against 20). At every bending each layer began with a 0.65 cm upright link (90 deg bends) and the next layer wound the other way. When woken, the solver kicked these joints straight. The old "stiffer chain, higher fountain" (20 deg median 11.1 cm) came from those kicks: with the fixed coil it is 4.7. Only the lead's 3 corners, woken at t = 0, are still laid past the limit.
- Measurement changes (15 Oct): v is now the rate links land, not the rate they are woken. At 20 deg and 4800 steps/s the waking ran ahead (wake rate 592 cm/s, above the sqrt(2 g h1) = 542 energy bound). New lines: landing speed and energy, and with CF_PUSH=1 the directly measured push. CF_PUSH is off by default because eachContact makes the run 2.6x slower in the Cabinet (34.7 s against 13.2 s for 1200 frames).
- **Located 15 Oct 2026, mechanism still open: the fountain is low because the pile's push doesn't reach the moving chain.** The pile pushes up with a = 0.13-0.31, but the falling chain carries only a third or less of it. A pile whose pick-up stays put (flake on a table) raises the fountain only a little (ratio 0.36-0.40 against 0.29-0.30 for the spiral on the same table). No model channel tried (step x2 and x4, iterations x2.5 and x5, friction 0, joint stops off, link size x0.5, look-ahead 5/15) brings the ratio above 0.50. With the push the chain does carry, the flake obeys all three B&W relations: height within 21%, speed within 4%, energy within 4 points. So the earlier apparent 11-30% energy loss mostly came from using the contact push as a. The gap is in the pick-up zone. The chain is dragged and lifted over several links (still only about 90% of full speed 4-6 cm above the pile, from the inferred tension), not picked up a link at a time. This does not change with step or iterations, so it is not solver convergence. Without joint stops the push itself halves. On the spiral, and with half-size links, the falling chain carries 0.00-0.04 while the fountain stands higher than that would give. So between pile and falling leg those flows are not steady either. Whether real chains also carry only part of the contact push is unknown; B&W's own h2 = 0.14 h1 means a = 0.12 against their rod model's 1/6.
- Not step-converged at 20 deg (1/3600 s: v +6%, h2 +85%). The run is chaotic, with spreads as in the table.
- The waking scheme is not neutral. With 5 links woken ahead instead of 10 (scratch copy with CF_K), v = 476 / 452 cm/s at 30 / 60 deg (+20% / +6%) and a from speed 0.35 / 0.28, against a push of 0.21 / 0.16. The anchor's force is not measured. With 20 the woken links are dragged into a heap (276-446 links moving; at 60 deg the rim carries 1.3 m v^2). So v is known to about 20%.
- **Tested 15 Oct 2026: the wake/anchor shortcut does not lose the push, but it does make the fountain too tall.** With every link simulated, the moving chain still carries none of the pile's push (carried -0.13 to -0.19), so the lost push is how rigid links come off a pile in this model. The full chain drags the top of the pile and rubs the rim (0.26-0.32 m v^2). Its fountain is 8.6 / 6.3 / 5.5 cm

against the exhibit's 10.6 / 10.8 / 11.2 at 30 / 60 / 90 deg. The exhibit's fountain is therefore up to 2x taller than the full simulation of the same chain. A cheaper scheme (4 live top layers, no anchor) matches the full chain at 30 deg, but costs 21 ms/frame at quarter speed against 4.4 now (full: 35 ms; budget 16.7). The default is unchanged; the README says so. The flake pile could not be run all-live: on the bare table the live 25-layer stack is dragged off as a mass.

- **Cost vs fidelity (15 Oct 2026).** Neighbouring links already never collide (bpp adds joints with disableCollisionsBetweenLinkedBodies; there are 0 contacts between links 1 or 2 apart). A live pile's cost is its 900-1900 resting contact points (exhibit: about 20). Sleeping doesn't help: the jointed chain is one simulation island. No tested way is faithful at 30, 60 and 90 deg within 16.7 ms a frame. The default is unchanged. Possible C++ gains, in bpp's src/: expose btContactSolverInfo (solver mode, split impulse) and the contact breaking threshold to Lua; build against a BT_THREADSAFE Bullet with btDiscreteDynamicsWorldMt and btSequentialImpulseConstraintSolverMt (at most about 2x on the 2 cores here). None is expected to give the 3-4x the live pile needs at quarter speed, or 13x at real speed.
- **16 Oct 2026, heap test: the pile was the missing ingredient.** A ball chain of the experiments' size, poured into the jar, gives h2/h1 = 0.11-0.18 in Bullet and in an independent bead-and-rod reference (experiments 0.12-0.14); the same chain laid as this exhibit's coil rides the rim at 0.06 (reference) or stalls on it (Bullet). Not step or solver (both engines agree; halving the step moves the heap less than pour to pour). Still: about 1 pour in 4 catches the rim and runs slow; the chain runs at 0.69-0.84 of sqrt(2 g h1) against 0.70 measured (no air drag); windows are short (0.4-1 s, 9.75 m of chain against the experiments' 30-75 m). The exhibit itself is unchanged: its 1.5 cm links can't form a heap in a 7 cm jar (a pour lays rings round the wall; 0.08-0.11 vs coil 0.06-0.08). The first CF_PILE=heap (heap fixed, links lying on the chain woken) was 1.6x too tall and step-dependent; replaced, see the next item.
- **16 Oct 2026, second round: a faithful heap mode, but only at 1/8 speed.** (1) Bullet's full heap was not step-converged at its default joint ERP 0.2 (h2/h1 0.131 / 0.160 / 0.150 at 1800 / 3600 / 7200 steps/s); ERP 0.5 makes 1800 converged (0.148 vs 0.145 at 3600; 0.8 gives 0.135), 1200 is not (0.118). (2) A live zone round the pick-up, 4 cm across and 0.5 cm deep, matches every link simulated within the pour-to-pour spread at 1800 and 3600 (h2/h1 0.147 / 0.145 against 0.140 / 0.137; v, rim and carried push likewise; pile push a little low, 0.13-0.20 vs 0.17-0.29); 3 cm stands 15% too tall. (3) Cost: the falling 1.5 m of fine chain alone is 13.5 ms a frame at 1/4 speed and the zone adds 14: 27.5 ms during the flow (Cabinet 30.0). At 1/8 speed it fits: 14.1 (Cabinet 14.3). Solver iterations barely matter; it is per-body and per-joint work. (4) Two bpp traps: `.trans` of a moving body is the last frame's drawn transform (use body:getOrientation()); creating fixed bodies mid-run made the hall and standalone runs differ (bpp's per-address AABB memory), so each heap link keeps one fixed body and parks it. (5) Rendering not measured on a GPU; under xvfb the heap frame takes 186 ms, the exhibit's 77.
- **16 Oct 2026, third round: the heap is the default.** One line chooses it (setMode; CF_PILE=coil for the coil), H switches heap <-> coil and restarts (not one of bpp's S, D, R, P nor the hall's Tab). Bending is held at 30 deg in heap mode rather than pouring a heap per bending (each would need its own check against every link simulated); layers is inert; steps >= 1800; friction restarts. Console in heap mode: the meaningless rod-model figure (-0.10) replaced; a line says the window is not steady (447 then 416 cm/s), so the speed reading of a (0.22) is unreliable and the height's (0.11) is the one to read; the window rule is unchanged. The coil's output is byte-identical to the previous default; H in either direction gives exactly the run of that pile started on its own. Full findings: demo/WyomingWill/ChainFountain/FINDINGS.md.
- Fixed 16 Oct 2026 in the reference's own check: its groove friction check let the chain run off the end of the groove (it read 3.8x Coulomb at every setting). Fixed, it shows 1 pass per 1e-5 s over-grips 5x (the earlier ITERS=1 heap runs are void) and 4 passes give 1.09 (0.09 of it the groove's bumps, present with no friction). Also found in passing: removing a body takes its joints out of Bullet's solver order in pointer order, so a script must remove joints one by one first (the heap option does; the coil's waking never removes more than one at a time).

- **16 Oct 2026: a better chain model?** The literature's real ball chains give h2/h1 = 0.12-0.14, with v about 0.70 sqrt(2 g h1). Sources: Biggins and Warner 2014; Biggins, EPL 2014, fit a = 0.12, b = 0.11; Pantaleone, arXiv:1910.03125, three chains. An independent exact model of such a chain (beads, rigid rods, geometric limits) in the exhibit's jar and coil gives the same low fountain as Bullet (h2/h1 0.04-0.06, push not carried, the chain riding on the rim). Changing the link/joint model would therefore not fix it. The likeliest missing ingredient was the experiments' jumbled pile; heaps and long coils jammed in bead-chain.c, so it was tested later the same day with bead-pour.c and in Bullet (see the heap test above: confirmed). Default unchanged; nothing adopted.
- Experiment options added (env only; defaults unchanged, byte-identical output): CF_GLASSMU, CF_SLEEP, CF_CPT, CF_STATICLAYERS, CF_LIVE, CF_SETTLE, CF_NOFREEZE, CF_WAKE=layer, CF_LIVELAYERS, CF_PUSH=2; CF_LAYOUT=flake, CF_JAR=0, CF_ITER, CF_MU, CF_AHEAD, CF_STOP, CF_LINK; CF_PUSH=1 now also prints h2 = a v^2/g and the carried a. The flake is for 45 deg and more only (wider turns don't fit), a layer holds about 0.3 m, so a 40-layer pile is 26 cm tall and h1 drifts from about 150 to 125 cm in a run.
- The coil is a rigid disc, not a heap. When the disc drops a layer, the last ~10 woken links of the old layer fall 0.65 cm. The anchor is infinitely strong. The floor is a freeze at 2.5 cm (b = 0 by construction).
- Checked 16 Oct 2026 (contact audit): every contact the chain makes is 0.548 × 0.548 = 0.300 in the heap (links, fixed links, glass, disc) and 0.3 × 0.3 = 0.09 in the coil, as the README says; the floor (Bullet's 0.5) only meets links that are then taken out. Two stale texts fixed, nothing else: the lua header and README gave `CF_MU` as "friction, 0.3" (the heap's default is 0.548 a body); the coil's friction slider help now says its 0.3 is 0.09 at a contact.

References

- S. Mould, video, 2013 (README).
- J. S. Biggins, "Growth and shape of a chain fountain", EPL 106 (2014) 44001, arXiv:1401.5810: fit a = 0.12, b = 0.11; 1 L beaker, h1 = 1.72 m.
- J. Pantaleone, "A quantitative analysis of the chain fountain", arXiv:1910.03125: ball chains D 0.30 / 0.44 cm, L 0.44 / 0.63 cm, h_above/h_below 0.12 +- 0.01, V/sqrt(2 g h) 0.70 / 0.69; bar chain 0.14; drops 1.29-3.57 m; box 9 x 9 x 13 cm; max bending about 20 deg.
- M. Macklin et al., "Small steps in physics simulation", SCA 2019 (the reference's method; from memory).
- M. Mueller et al., "Detailed rigid body simulation with extended position based dynamics", SCA 2020 (bead-pour.c's friction; from memory). C. Ericson, Real-Time Collision Detection (2005), 5.1.9 (segment-segment nearest points; from memory).
- J. S. Biggins, M. Warner, "Understanding the chain fountain", Proc. R. Soc. A 470 (2014) 20130689, arXiv:1310.4056: v^2 = g h1/(1 - a - b), h2/h1 = a/(1 - a - b), rod model a = (1 - I/(m b^2/4))/2 = 1/6 for three beads per link, measured h2 = 0.14 h1, energy requires a + b <= 1/2.

5.7 The falling Slinky

Agrees with theory. Bottom stays put 0.2038 s against sqrt(M/3k) = 0.2018 s: 0.6% from having 86 separate coils (an exact 86-coil chain waits 0.2030 s) and 0.4% from Bullet's step, both now explained in the README.
Folder: demo/WyomingWill/Slinky/

What it shows

A Slinky hangs stretched under its own weight, with a steel ball hanging level with its bottom. When both are released, the ball falls at once, but the Slinky's bottom hangs motionless until the collapse coming down from the top reaches it. Meanwhile the Slinky's centre of mass falls exactly like the ball. Beside it, the same Slinky walks end over end down seven stairs.

Model

- Falling Slinky: M = 215 g, N = 86 coils (SL_COILS), each a Bullet rigid body of M/N = 2.5 g. The collision shape is a thin square plate, 7 cm across. Closed length is 5.6 cm, stiffness k = 1.76 N/m, with no pre-tension. Neighbours are joined by a btSliderConstraint along the axis, with a lower limit at the closed spacing, no restitution, and rotation locked. This gives inelastic coil-on-coil stops. The script applies the spring force kseg = k(N-1) per gap before every physics step, and only while the gap is stretched. Each coil's initial stretch is set from the weight below it, so nothing moves before release. Bullet steps at 4800 per simulated second with 60 solver iterations and no damping. Coil friction is 0.5 and floor friction 0.6. The display runs at 0.1x real speed (SL_SPEED). The hand releases after a 0.5 s "hold" slider. 0.05 s after landing, the pile is replaced by a frozen closed stack, because loose coils jostle in Bullet.
- Stairs walker: the same M and k, built as 43 capsule pieces of 2 coils each, constrained to the plane (x, y, rotation about z). The two sides of each piece connect to the next piece by pull-only springs of k(42)/2 each, with damping 0.05 N s/m. A shear spring of 10 k(42) keeps pieces in line. Forces are applied by the script as velocity kicks each step, and Bullet handles the contacts (friction 0.6, restitution 0). There are 7 steps, 12 cm rise and 13 cm tread, from 84 cm. It restarts 3 s after coming to rest.

Theory checks

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Time the bottom stays put	t* = sqrt(M / 3k) = 0.2018 s (script)	0.2038 s (86 coils)	+1.0% (README now: 0.2038 s)
Same, the discrete chain itself (new, 15 Oct 2026: reconstruction/slinky-chain.c, point masses + pull-only springs + sticking collisions, step 1e-6 s)	continuum limit 0.2018	43: 0.20412; 86: 0.20296; 172: 0.20238; 344: 0.20208 s (same at 5e-7 s)	+1.15 / +0.58 / +0.29 / +0.14 %: halves as coils double
Bullet vs step size, 86 coils (SL_STEPS, new env)	the chain's 0.2030	4800: 0.2038; 9600: 0.2033; 19200: 0.2031; 38400: 0.2030 s	converges to the chain; the step adds +0.4% at the default
Bullet, 43 / 86 / 172 coils at 4800 steps/s	should approach 0.2018 as N grows (old README)	0.2046 / 0.2038 / 0.2048 s	**does not**: the step error grows with N
Same at 9600 / 19200 steps/s	—	9600: 0.2043 / 0.2033 / 0.2030; 19200: 0.2042 / 0.2031 / 0.2026 s (38400: 43 coils 0.2042, 86 0.2030)	with small steps it does approach, as the chain does
Detection threshold (1 cm/s vs 10 cm/s, scratch copy)	should not matter	0.2038 s both	not the cause
Hanging stretch	M g / 2k = 59.9 cm	printed length 65.5 cm minus 5.6 cm closed = 59.9 cm	exact (README: "stretches 60 cm")

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Ball's fall when the bottom starts moving	g t^2 / 2 at t = 0.2038 s = 20.4 cm	20.4 cm	exact (README: 20 cm)
Bottom stationary before t*	displacement 0	bottom drop 0.000 cm at t = 0.100 and 0.200 s (SL_LOG)	exact
Centre of mass in free fall	y_cm drop = g t^2 / 2 (only gravity acts)	t = 0.1/0.2/0.3/0.4 s: CoM drop 4.915/19.641/44.176/78.521 cm vs ball 4.915/19.640/44.176/78.521 cm; g t^2/2 = 4.905/19.620/44.145/78.480	CoM = ball to 0.001 cm. The 0.01-0.04 cm offset from g t^2/2 is one 1/600 s logging-frame phase
Landing time	(no theory given)	0.492 s after release	(README gives none)
Stairs: time per step, 7 steps	README now: "about 0.4 s a step (2.6 to 3.1 s for a clean walk)"; sample walk 3.03 s	21 walks at native 0.1x speed: clean walks 2.56-3.07 s to the floor	matches the corrected README (the old "a third of a second", 2.64 s sample, was from an earlier run)
Stairs failure rate	old README: "one walk in three or four" goes wrong; README now: "nearly one walk in two"	native 0.1x speed, 21 walks in a row (HOLD_S=0, -n 84000): 12 clean; 4 reached the floor but skipped steps (3 4 5 F; 3 4 5 7 F; 3 4 5 6 F F; 3 4 5 F); 3 stopped on step 7; 2 never started (rest on step 2, no landings). Earlier SL_SPEED=1 sample: 4 of 9 clean	9/21 = 43% imperfect; README corrected

Run used

```
cd demo/WyomingWill/Slinky
SL_LOG=sl.csv HOLD_S=0 timeout 1500 ../../../release/bpp -f falling-slinky.lua -n 9000      # 15 s simulated at 0.1x
SL_COILS=172 HOLD_S=0 timeout 1500 ../../../release/bpp -f falling-slinky.lua -n 600
SL_COILS=43  HOLD_S=0 timeout 1500 ../../../release/bpp -f falling-slinky.lua -n 600
SL_SPEED=1   HOLD_S=0 timeout 2400 ../../../release/bpp -f falling-slinky.lua -n 3600      # 60 s, walk statistics
    (report)
HOLD_S=0 ../../../release/bpp -f falling-slinky.lua -n 84000                                 # 15 Oct 2026: 21 walks
    at 0.1x
SL_STEPS=9600/19200/38400 (SL_COILS=43/86/172) ... -n 400                                    # 15 Oct 2026: step
    study
cc -O2 -o slinky-chain reconstruction/slinky-chain.c -lm; ./slinky-chain {43,86,172,344} 1e-6  # 15 Oct 2026
```

(HOLD_S has no effect here. This exhibit uses its own "hold" slider, 0.5 s.)

```
--- hung up: 215 g, 86 coils, stiffness 1.76 N/m; 65.5 cm long hanging (5.6 cm closed up); the bottom should stay put
    for 0.202 s ---
t= 0.254s  the bottom starts to move, 0.2038 s after letting go (it should, by theory: 0.2018 s); the ball had fallen
    20.4 cm
t= 0.542s  the Slinky lands, 0.492 s after letting go
stairs t= 3.75s  the walking Slinky comes to rest, on the floor: 7 steps down in 3.03 s
stairs t= 3.64s  the walking Slinky comes to rest, on the floor: 7 steps down in 2.98 s
[43 coils]  t= 0.255s  the bottom starts to move, 0.2046 s after letting go (it should, by theory: 0.2018 s); the
    ball had fallen 20.6 cm
[172 coils] t= 0.255s  the bottom starts to move, 0.2048 s after letting go (it should, by theory: 0.2018 s); the
    ball had fallen 20.6 cm
[SL_SPEED=1] stairs t= 2.18s  the walking Slinky comes to rest, on step 6: 5 steps down in 1.40 s
[SL_STEPS=19200] t= 0.253s  the bottom starts to move, 0.2031 s after letting go (it should, by theory: 0.2018 s);
    the ball had fallen 20.2 cm
86 coils, step 1e-06 s: the bottom moves at 0.20296 s (its gap closes at 0.20296 s); sqrt(M/3k) = 0.20179 s
[0.1x, walk 9]  stairs t= 7.25s  the walking Slinky comes to rest, on step 7: 6 steps down in 2.80 s
[0.1x, walk 14] stairs t= 1.28s  the walking Slinky comes to rest, on step 2: 1 steps down in 0.00 s
```

Known departures and honest caveats

- Fixed on 15 Oct 2026, the hang-time explanation: the README used to say more coils bring the wait closer to sqrt(M/3k). At the default 4800 steps/s they do not (43/86/172 coils: 0.2046/0.2038/0.2048 s). Established cause: the +1.0% at 86 coils is +0.6% from the discreteness itself (an exact chain of 86 sticky point masses waits 0.2030 s; the gap halves as coils double) and +0.4% from Bullet's step (0.2038 at 4800 steps/s, 0.2031 at 19200, 0.2030 at 38400). The step error grows with N because the

collapse front then crosses each coil in fewer steps, which is why 172 coils are no better at 4800 steps/s. The detection threshold (1 vs 10 cm/s) makes no difference. README and lua header now say this; SL_STEPS was added for headless testing, and reconstruction/slinky-chain.c (the independent chain check the README cites) was added. Default steps not raised: 4800 steps/s already costs ~20 ms per frame here with the 86 coils and the stairs walker.

- The theory is for a continuous spring with **no pre-tension**. The model deliberately has none, although a real Slinky does.
- Coil collisions are perfectly inelastic slider limits, not true contact. After landing, the coils are replaced by a frozen pile, so post-landing dynamics are not physical.
- The walker is a planar, 43-piece reduced model with tuned damping. The README says 0.05 N s/m was chosen because less damping gallops and more stalls. The 12 x 13 cm stairs are the size it "walks best on". Walks are sensitive to tiny differences (the walker is rebuilt identically each time; walks differ through Bullet's internal ordering, so the "statistics" are of that sensitivity). At SL_SPEED=1 landings are detected once per 1/60 s frame instead of per 1/600 s (the physics step, 1/4800 s, is the same), so brief landings can be missed and a clean walk logged as skipping; the README now says to count walks at the native 0.1x speed. Fixed on 15 Oct 2026: the README's "one walk in three or four" is now "nearly one walk in two", with the 21-walk breakdown, and its sample walk listing is replaced by walk 1 of this run.
- The ball is a 50 g Bullet sphere released from a fixed constraint at the same instant.

References

- None cited by name in the code or README. The sqrt(M/3k) result is derived in the README from momentum M g t.

5.8 The Brazil-nut effect

Agrees with theory, with a numerical or open caveat. Rebuilt in real glass (bounce 0.95, friction 0.1), by a validated bead model ported into bpp bit for bit. Glued-bead walls give the right rolls (up the middle 5.9-7.5, down the walls 11.7-13.8 mm/s) and the solid big bead rises 7.6-10.6 s into the hard shaking and stays up; smooth glass walls reverse the rolls and it stays down, in all four pours. Floor push at Γ = 0.8: 0.19 of the weight against 0.20 rigid. A glass lid closes the box, as in real cells (open, beads flew to 67-84 cm over 44 cm walls). Beads 1/100 as stiff as glass (results unchanged 10x stiffer); about 20 ms a frame against 16.7 at 1/4 speed on the test machine.
Folder: demo/WyomingWill/BrazilNut/

What it shows

Two glass-fronted boxes of real glass beads, a layer one bead thick and 20 cm deep, are shaken vertically at 5 Hz: gently (peak 0.8 g) for 2 s, then hard (3 g). A 3 cm solid glass bead starts at the bottom of each. The boxes differ only in their side walls. The left box's walls are lined with glued 1-cm glass beads, as laboratories roughen theirs. There the bed turns over in two rolls, up the middle and down the walls (Knight, Jaeger and Nagel's mechanism), and the big bead comes up about 10 cm within seconds and stays in the upper part of the bed. The right box has smooth glass walls: the rolls run the other way, up beside the walls, and the big bead stays down. Each big bead carries a painted band that turns with its spin. A red bead in each, poured at the top by the left wall, trails a fading line of its last 6 s. The boxes are closed by a lid. Runs at 1/4 of real time, until R pours again.

Model

Not Bullet. Each box is a **DemBed**, bpp's C++ port of the reference model `reconstruction/bead-bed.c` (FINDINGS.md, GLASS-DESIGN.md), one thread per box. Bullet only draws display-only spheres.

- **Port.** bead-bed.c's code was lifted mechanically into `src/demcore.cpp`: globals became members and bodies are unchanged (`reconstruction/port/`, HOOK.md). Its frame loop is bead-bed.c's main loop cut into steps. Wrapper and Lua binding: `src/dembed.cpp`, registered in `viewer.cpp`.

- **Contacts.** Hertz normal force K $d^{1.5}$, K = 4/3 E* sqrt(R_eff), E* = E / 2(1 - nu^2), nu 0.22, with Tsuji's damping alpha sqrt(m_eff K) d^1/4 d'; alpha is bisected so that a two-body collision returns e = 0.95. Tangential: a Cundall-Strack spring kt = 2(1 - nu)/(2 - nu) x 1.5 K sqrt(d) (Mindlin), with the same damping ratio, capped by Coulomb at mu Fn.
- **Materials.** mu 0.1 bead-bead, bead-wall and bead to glued bead; e 0.95 everywhere (Foerster et al. 1994, Cornell table: glass e 0.92-0.97, mu 0.05-0.18).
- **Stiffness.** E = 0.63 GPa, 1/100 of glass, with NS 40 steps per contact time: DT = 5.0 us, 3326 steps per 1/60 s.
- **Beads.** 280 small beads, radius 0.5 cm x (0.95 + 0.1 x ((37 i) mod 100)/99), density 2.5 g/cm^3; big bead radius 1.5 cm. Held to the plane, spinning about its normal.
- **Box.** 15 cm wide and 44 cm high inside, closed by a flat lid that moves with it and is met like the walls (glass: mu 0.1, e 0.95). Without the lid, beads fly up to 67-84 cm (16 pours) and some bead is above 44 cm in 32-45% of frames: they would leave a real open box. The lid was added to bead-bed.c (`LID`, `TEST=lid`) and ported. Floor, walls and lid move as y = A(t) sin(wt). A is 0.8 g's (8.0 mm) for 0 to 2 s, ramps linearly to 3 g's (29.8 mm) over 2 to 3 s, then stays. The speed includes A'(t) sin wt.
- **Rough walls.** Glued spheres of radius 0.5 cm, centres 1 cm apart from the floor to the lid, half-embedded, 44 per wall. Each is a fixed sphere met by the full contact law.
- **Integration and pour.** Velocity Verlet with a cell list. bead-bed.c's pour (the old lua's rows and random-number generator, `BN_SEED` 7), then 0.5 s to settle.
- **Measurements, all in C++ on the 1/60 s frame clock** (so independent of the display speed):
  - the floor's push from the bed's momentum change, in 2-degree phase bins, over shakes 4-10 (gentle) and 26-75 (hard);
  - lift-off below 0.1 W, and landing when the CoM stops approaching the floor;
  - rise rates every 0.5 s for |x| < 3 cm and |x| > 5 cm, 8-45 s;
  - the big bead's centre height: when its 2 s mean first reaches 12 cm, then its mean and least height and its time in the upper half over 25-45 s and again over 45-120 s;
  - the red bead's lowest point, and the highest any bead reaches.
- **Drawing.** The big bead's band is turned by its spin in the model, integrated each step; the red beads' trails are their positions every 0.25 s. Both are display only.
- **No automatic re-pour** (it used to re-pour every 50 s): the boxes run until R, so the big bead's place after minutes can be seen. The last check prints at 2 minutes.

Theory checks (default run, `BN_SEED` 7; "reference" = the same boxes run headless for seeds 1, 2, 7 and 8 with `reconstruction/dembed-check`, the same code)

Quantity	Theory (formula / source)	Simulation, seed 7	Reference, 4 pours	Agreement
Amplitude	A = Gamma g / (2 pi f)^2	8.0 mm, 29.8 mm	same	exact (definition)
Gamma = 0.8: bed stays on the floor	least push (1 - Gamma) W = 0.20 W, rigid bed	never left; least 0.19 W, both boxes	0.18-0.19 W	0.01-0.02 W low: the soft bed rebounds a little on its contacts
Gamma = 3: lift-off	arcsin(1/Gamma) = 19.5 deg if landed by then; else as soon as landed	left 40-42, right 42-44 deg	40-44	still landing at 19.5 (CoM coming down at 14 cm/s); leaves once down, as in FINDINGS

Quantity	Theory (formula / source)	Simulation, seed 7	Reference, 4 pours	Agreement
Gamma = 3: landed	rigid inelastic bed lands at 1.4 deg (Mehta and Luck 1990)	46-48, 44-46 deg	44-48	**departs from the rigid model, as real layers do** (Bougie et al. 2002; Pastor et al. 2007)
Rolls, rough walls	up the middle, down the walls (Knight et al. 1993)	+6.7 / -13.3 mm/s	+5.4 to +6.7 / -14.0 to -11.4	right direction, every pour
Rolls, smooth glass walls	reversed at low wall friction (Gallas et al. 1992; FINDINGS §6)	-0.3 / +4.2 mm/s	-0.6 to +1.2 / +3.1 to +4.2	up beside the walls, every pour
Big bead rises, rough walls	carried up by the rolls (Knight et al. 1993)	2 s mean at 12 cm 7.5 s into the hard shaking (10.5 s)	7.5-9.5 s	yes, every pour
Big bead stays up, rough walls	trapped above the wall stream	25-45 s: mean 15.7 cm, least 10.6, upper half 85%; 45-120 s: 15.7 cm, least 10.4, 85%	15.5-16.1 cm, 83-91%; 45-120 s 15.3-16.0 cm, 78-89%	yes, for 2 minutes (bed 20.2 cm deep at rest)
Big bead, smooth walls	not carried up	25-45 s: mean 8.9 cm, least 5.0, upper half 0%; 45-120 s: 8.9 cm, least 2.5, 0%	7.6-10.4 cm, 0-3%; 45-120 s 8.1-10.9 cm, 0-3% (one pour's 2 s mean touched 12 cm once, at 41 s)	stays down
The lid	closed cell; must not change the bed's behaviour	touched in 0.06-0.14% of frames	open box (lining to 70 cm), same 4 pours: rough +6.2 to +7.4 / -12.5 to -15.0, up at 11.4-14.1 s, 15.2-16.0 cm; smooth -0.7 to +0.4 / +3.1 to +4.0, 7.9-9.2 cm; push, lift-off, landing the same	unchanged within the pour spread
Big bead's turn	rolling: a = 5/7 g sin(theta), slip from tan(theta) = 3.5 mu; turns clockwise rolling right	the band turns by the model's spin	bead-bed.c's incline test through the hook: identical	right magnitude and sense
The red bead	down the wall stream (rough)	left: lowest 7.9 cm at 45.0 s, by the right wall; right: 9.2 cm at 26.8 s, 2.3 cm from the left wall	not in the reference	qualitative only: one bead is slow and pour-dependent; its trail shows the direction
Stiffness (open box, before the lid)	results independent of E	0.63 GPa (seeds 7, 8): left +6.1, +7.5 / -11.7, -13.5 mm/s, up 7.6, 10.6 s into the hard shaking, late 15.7, 16.3 cm; right +0.4, -0.7 / +4.0, +3.9, never up, 7.9, 8.1 cm; gentle push 0.19	6.3 GPa, NS 80: left +6.0, +5.2 / -12.4, -11.3, up 7.7, 8.5 s, late 14.6, 15.6 cm; right +0.5, -0.2 / +2.9, +2.9, never up, 7.9, 8.0 cm; gentle push 0.199, 0.198	the same within the pour-to-pour spread; the gentle push nears 0.20 with stiffer beads (the 0.01 shortfall is the softening)
Port vs bead-bed.c	bit for bit	`hook-check.lua` through bpp: bead-bed.c's tests (drop, pair, incline, gdrop, ghollow) and lid give the same 100 lines; a 4 s design run gives the same settle, rolls, overlaps	`dembed-check` against `bead-bed`: 11 configurations (both laws, both boxes, glued walls, grit, plates, rolling, ETH, push report) and a low-lid run the same	identical

Cost — Two threads, one per box (`DemBed:start/finish`); the next frame's stepping starts as soon as a frame is reported, so drawing overlaps it. Measured headless (no drawing) on this 2-core Xeon at 2.1 GHz, idle, at the default 1/4 speed, 300 frames at a time: 23-31 ms a frame over the first 4 simulated s (the settled bed and the gentle opening: a packed bed has more contacts), then 20-21 ms in the hard shaking; in the Cabinet 23-30, then about 20 ms. Nearly all of it is waiting for the beads (19-20 ms). (The lid, band and trail changed nothing measurable.) The 60 fps frame has 16.7 ms, so here the exhibit runs at about 0.8 of its quarter speed once shaking hard (about 1/5 of real time), and about half to two thirds of it in the first 4 s; a machine about 1.2x faster per core holds 60 fps. A box costs 4.8 CPU-s per simulated second in bpp (`-O3`, plain x86-64); `-march=native` would cut that by about a quarter but changes the rounding (fused multiply-adds), so the numbers would no longer match `bead-bed.c` bit for bit. Real time (Z) needs four times as much: about 80 ms a frame here.

Run used

```
cd demo/WyomingWill/BrazilNut && HOLD_S=0 timeout 7000 ../../../release/bpp -f brazil-nut.lua -n 29000
  (and BN_SPEED=1 HOLD_S=0 -n 7260; the default pause -n 29030; cmp2.sh brazil BrazilNut brazil-nut.lua 29030 bn;
   the reference: reconstruction/dembed-check with the exhibit's keys, seeds 1 2 7 8, rough and smooth, to 120 s,
   with the lid (LID=44 H=42) and open (H=70), and seeds 3-6 open to 60 s; YM 6.3e8 NS 40; earlier, open with
   H=44, 6.3e9 NS 80 for the stiffness check; HOOK.md has the port checks)
```

```
--- poured: 280 glass beads a box, shaken at 5 Hz: gently (0.8 g, 8.0 mm) for 2 s, then hard (3.0 g, 29.8 mm) from 3
    s ---
  Gamma 0.8: below 1 -- the bed should never leave the floor; Gamma 3.0: it should leave it 19.5 degrees into each
    upswing (arcsin 1/Gamma), if it has landed by then
  Left  Gamma 0.8: the bed never left the floor (the floor's push never less than 0.19 of its weight; a rigid bed's:
    0.20; the reference: 0.18 to 0.19)
  Right Gamma 0.8: the bed never left the floor (the floor's push never less than 0.19 of its weight; a rigid bed's:
    0.20; the reference: 0.19 to 0.19)
  Left  the big bead is up: its middle 12 cm above the floor (over 2 s) at 10.5 s, 7.5 s into the hard shaking (the
    reference, 4 pours: 7.5 to 9.5 s)
  Left  Gamma 3.0: the bed leaves the floor 40 to 42 degrees into the upswing (a rigid bed: arcsin 1/Gamma = 19.5;
    the reference: 40 to 42);
        it can't till it has landed: at 18 to 20 degrees its middle is still coming down (14 cm/s); landed at 46 to
    48 (the reference: landed at 44 to 48)
  Right Gamma 3.0: the bed leaves the floor 42 to 44 degrees into the upswing (a rigid bed: arcsin 1/Gamma = 19.5;
    the reference: 40 to 44);
        it can't till it has landed: at 18 to 20 degrees its middle is still coming down (14 cm/s); landed at 44 to
    46 (the reference: landed at 44 to 48)
  Left  from 8 s to 45 s: beads in the middle rising 6.7 mm/s on average, beside the walls falling 13.3 mm/s (the
    reference: +5.4 to +6.7 / -14.0 to -11.4)
  Left  the big bead from 25 s to 45 s: its middle 15.7 cm above the floor on average, never below 10.6; in the bed's
    upper half 85% of the time (the reference: 15.5 to 16.1 cm, 83 to 91%)
  Left  the red bead, poured at the top by the left wall: lowest 7.9 cm above the floor, at 45.0 s, 14.1 cm from the
    left wall
  Right from 8 s to 45 s: beads in the middle falling 0.3 mm/s on average, beside the walls rising 4.2 mm/s (the
    reference: -0.6 to +1.2 / +3.1 to +4.2)
  Right the big bead from 25 s to 45 s: its middle 8.9 cm above the floor on average, never below 5.0; in the bed's
    upper half 0% of the time (the reference: 7.6 to 10.4 cm, 0 to 3%)
  Right the red bead, poured at the top by the left wall: lowest 9.2 cm above the floor, at 26.8 s, 2.3 cm from the
    left wall
  The walls decide: with rough walls the beads go down beside the walls and up the middle (-13.3 mm/s by the walls)
    and the big bead comes up (15.7 cm);
  with smooth glass walls they go up beside the walls (+4.2 mm/s) and it stays down (8.9 cm).
  Left  the big bead from 45 s to 2 minutes: its middle 15.7 cm above the floor on average, never below 10.4; in the
    bed's upper half 85% of the time (the reference: 15.3 to 16.0 cm, 78 to 89%)
  Right the big bead from 45 s to 2 minutes: its middle 8.9 cm above the floor on average, never below 2.5; in the
    bed's upper half 0% of the time (the reference: 8.1 to 10.9 cm, 0 to 3%)
```

The README's console block matches this run. Hall against standalone (cmp2.sh, 29030 frames): IDENTICAL (17 run lines: the first three are printed as the hall loads). HOLD_S=0 for 29000 frames is the default pause's output for 29030 frames, all 20 lines. BN_SPEED=1 for 7260 frames prints the same 20 lines as 1/4 speed, number for number: everything is measured in C++ on the 1/60 s simulated frame clock. The bead threads stepping while a frame is drawn changed nothing (speed 1, 2760 frames, before and after). An older bpp, without DemBed: the console says the exhibit needs bpp from 17 October 2026 and the boxes stand empty, with no Lua error.

Known departures and honest caveats

- **Softer than glass.** E = 0.63 GPa, 1/100 of glass, so that two boxes run at 1/4 speed. Against 6.3 GPa the design gives the same rolls, rise and height within the pour-to-pour spread (seeds 7, 8, both boxes; table), and a gentle-phase push of 0.198-0.199 instead of 0.18-0.19. GLASS-DESIGN.md §3.5 found the rise times and plateau unchanged from 0.63 to 63 GPa, with rolls about 10% faster at 0.63.
- **No front and back panes, no air.** In a real one-bead-thick cell the panes and the air dissipate energy, so a real bed would likely be calmer.
- **The lid is part of the design.** An open box of this height can't hold the bed: shaken hard, bouncy glass throws beads up to 84 cm. The lid is touched rarely and doesn't change the results beyond the spread (table), but it is a closed cell, not the open box of the earlier exhibit.
- **Glued beads assumed to bounce like free glass** (e 0.95); an epoxy-backed bead may bounce less.
- **The big bead rises to the upper part, not to the surface.** It levels off about 4 cm below the bed's packed top (20.2 cm). The shaken surface is lively, its top tenth of beads well above 20 cm, so the old "reaches the top" test is never met.
- **Not converged with bed depth or frequency.** The design was chosen in GLASS-DESIGN.md over Gamma, f, depth, width and wall type. A shallower bed (140 beads) is fluidised and its glass big bead stays down even with rough walls.
- **Mostly a model.** No experiment with 1-cm glass beads in a one-bead cell was found to compare with. The qualitative facts (rough walls give normal rolls and a rising intruder; smooth walls don't, or the patterns are irreproducible) are from Knight et al. 1993, Hejmady et al. (arXiv:1111.4484) and the MRI experiments.
- **The earlier exhibit (Bullet, until 17 Oct 2026)** used friction 0.64 / 0.8 and bounce 0.01: not glass. Its rolls and rise weren't step-converged (FINDINGS.md). Bullet can't run e 0.95: restitution pumps energy into a packed bed at any step rate, and raising its restitution threshold doesn't fix it (GLASS-DESIGN.md §4). So the exhibit was moved off Bullet.

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5.9 The solar system

Agrees with theory. Planets within 5 km of JPL's DE421 after 10 years, the Moon 39 km; Mercury's perihelion 42.981″/century against 42.981″; rotation checks hold. The checks now stop at their exact dates, and give the same lines at any rate.
Folder: demo/WyomingWill/SolarSystem/

What it shows

The Sun, the eight planets, Pluto, five asteroids (Ceres, Pallas, Juno, Vesta, Hygiea) and 21 moons, moving under gravity from 2026-01-01 0h TDB, starting from JPL's states for that day. An orrery shows heliocentric distances square-rooted (directions and ecliptic heights true), and vitrines show each planet with its moons. Every body wears a drawn map and turns by the IAU rotation model, so the Moon's librations and Mercury's 3:2 spin-orbit resonance can be seen. The console compares the integration with JPL every year (planets, Moon) and every month (moons, asteroids).

Model

All gravity is computed in Lua (`solar-engine.lua`). Bullet only places meshes: world gravity is set to 0 and every body's transform is written each frame, so Bullet runs no dynamics. Everything is in ICRF coordinates, AU and days.

- *Main integration*: 16 bodies, all pulling each other (Newton). They are the Sun, Mercury, Venus, Earth, the Moon, Pluto, five asteroids, and the system barycentres of Mars, Jupiter, Saturn, Uranus and Neptune. Two terms are added: the first-order Schwarzschild correction to the Sun's pull, $GM/(c^2r^3)[(4GM/r - v^2)r + 4(r\cdot v)v]$, and the Earth's J2 acting on the Moon (code comment: the Earth's pole is taken as ICRF z, "near enough"). The step is a fixed 1-day Bulirsch–Stoer step: Gragg modified midpoint with 2, 4, 6, 8 and 10 substeps, Richardson-extrapolated to 10th order, with no error control. Positions between steps come from cubic Hermite interpolation (for display; the checks are taken at whole days, the ends of steps). Alongside runs a massless "Mercury without Einstein" twin, pulled by Newton only.
- *Moon systems* (Mars, Jupiter, Saturn, Uranus, Neptune, Pluto): a Wisdom–Holman drift–kick–drift step. Each moon drifts on its Kepler orbit for half a step. The kick holds the other moons (with the planet's reflex wobble), the planet's J2 and J4, Mars's C22/S22 rotating with Mars, and the Sun's tide (the Sun taken from the main integration). Each system has 16 steps per orbit of its innermost moon (console, in days): Mars 0.020, Jupiter 0.111, Saturn 0.059, Uranus 0.088, Neptune 0.367, Pluto 0.399. Between steps a moon is carried on its Kepler orbit from the last step (less than one step).
- *Checks*: each frame first runs every check whose date falls before the frame's time, with the integration stopped exactly at that date (the yearly and monthly dates are whole days from the start, so the main integrator's 1-day steps land on them), then advances to the frame. The check figures are therefore the same at any SS_RATE (fixed 15 Oct 2026; before, checks ran on the first frame after the date and interpolated back).
- *Perihelion check*: a separate two-body run of Mercury plus the Schwarzschild term, stepped by RK4 at h = 0.01 d. The Laplace–Runge–Lenz vector is averaged over each perihelion-to-perihelion orbit and a straight line is fitted through its direction.
- *Data*: DE421 states and masses for the Sun, planets, Moon and Pluto. Moons and asteroids start from JPL Horizons. Moon-system masses are DE440's. Pluto's GM is 975.43 km³/s², the value Horizons' Charon orbit implies (README: DE421's 977.0 puts Charon 20,000 km off in a year).
- *Rotation*: IAU WGCCRE 2015 pole and prime-meridian models, keeping only the main periodic terms. Mars, its moons and the Moon's terms come from the 2009 report.

Theory checks

Values are from the fixed code (15 Oct 2026); the check lines are identical at SS_RATE=5, 60 and 100. The Phobos, Titania, Miranda and Ceres rates are differences between yearly Horizons lines (2026-01-01 → 2027-01-01 → 2031-01-01).

Quantity	Theory (formula / source)	Simulation (this run)	Agreement
Planets vs DE421 (heliocentric), 2027 / 2031 / 2036	JPL DE421	2027: all 0 km. 2031: Mars 1, Jupiter 2, others 0. 2036: Mercury 1, Earth 1, Mars 1, Jupiter 5, Saturn 1, others 0	≤ 5 km in 10 yr (README "within 5 km") ✓
Moon vs DE421 (geocentric)	DE421	4 / 20 / 39 km (2027 / 2031 / 2036)	39 km ≈ 0.05″ (README)
Mercury without GR, lag behind DE421	needs the Schwarzschild term	49 km (2027), 624 (2031), 1813 (2036); not monotonic (733, 624, 404, 286, 531, 1107, 1813 for 2030–36)	shows that GR is needed; README "49 / 1800 km", and now says the yearly figure is not monotonic ✓
Mercury perihelion advance	$\Delta\varpi = 6\pi GM/(c^2 a(1-e^2))$ per orbit = 42.981″/century	42.981″/century (over 3, 20 and 40 orbits)	0.000″ to 3 decimals
Energy (Newtonian)	conserved	max |ΔE/E| 1.4e-9 (2035)	README "about 1e-9" ✓
Angular momentum	conserved	max |ΔL/L| 3.0e-10 (2035)	README "3e-10" ✓
Laplace resonance L_I − 3L_E + 2L_G	180° (free libration 0.07°, README)	yearly means 179.87–180.12°, spread 0.15–0.23°; 20-day means 179.65–180.32°	±0.35° (README "179.7–180.3") ✓; these are frame-sampled and move by ~0.01° with SS_RATE
Neptune–Pluto 3:2	P_Pluto/P_Neptune ≈ 1.5; min separation > 17 AU	ratio 1.489–1.515 (osculating); closest 32.04 AU	consistent; 25 yr cannot test the 17 AU bound (README)
Moons, asteroids vs Horizons, 2026-02-01	JPL Horizons	Phobos 83, Io 5, Mimas 41, Titania 102, Titan 0, Ceres 1 km	start-up; README same line identical ✓ (was Io 78, Mimas 518 from a high-rate run)
Same, 2027-01-01	Horizons	Phobos 986, Mimas 575, Titania 858, Miranda 404, Ceres 218, Titan 31 km	README same line identical ✓
Same, 2031-01-01	Horizons	Phobos 4771, Titania 4502, Miranda 2060, Mimas 2794, Ceres 2010, Titan 90, Triton 108, Charon 209 km	README same line identical ✓
Phobos drift	Horizons	≈ 950 km/yr	README "950 km a year" ✓
Titania / Miranda drift	Horizons	≈ 900 / 400 km/yr	README ✓
Earth subsolar point, 2026-01-01 0h TDB	IERS Earth via Skyfield: 23.02 S, 178.87 W	23.01 S, 178.79 W	0.01° lat, 0.08° lon
Moon libration (sub-Earth point)	longitude up to 7.9°; latitude 1.5° + 5.0–5.3° = 6.7–6.8°	each year 7.5–8.0 W to 7.4–8.1 E; 6.8–6.9 S to 6.8 N	✓
Mercury 3:2 hot poles at perihelion	subsolar longitude 0° or 180°	0.0 to 0.2 from 0°/180° (e.g. 2036: −180.0, 0.0, −180.0, 0.0)	≤ 0.2°
WH vs Bulirsch–Stoer prototype (README)	BS reference	Mimas 3 km off after 20 d (README, not rerun)	—

Run used

```
cd demo/WyomingWill/SolarSystem
HOLD_S=0 SS_RATE=60  SS_YEARS=10 timeout 3000 ../../../release/bpp -f solar-system.lua -n 3660
HOLD_S=0 SS_RATE=100 SS_YEARS=10 timeout 3000 ../../../release/bpp -f solar-system.lua -n 2200
HOLD_S=0 SS_RATE=5|60|100 SS_YEARS=4 ... -n 13180|1100|662          # three years, rate comparison
```

All 120 check lines (DE421 blocks and Horizons lines, including the yearly Laplace means) are identical, character for character, at rates 60 and 100 over ten years, and at 5, 60 and 100 over three; the rate-60 lines are also identical to the pre-fix rate-60 run (at 1 d/frame the old sampling already landed on the dates). Only the frame-sampled lines (20-day Laplace means, the Moon's face extremes, Mercury's

perihelion longitudes) differ, by 0.01–0.2. Verbatim:

```
the Earth: on 2026-01-01 at 0h (TDB) the Sun is overhead at 23.01 S, 178.79 W (... through Skyfield: 23.02 S, 178.87
    W ...)
the moons: Wisdom-Holman, steps (days): Mars 0.020, Jupiter 0.111, Saturn 0.059, Uranus 0.088, Neptune 0.367, Pluto
    0.399
2026-02-01  against Horizons, km off: Phobos 83, Deimos 4, Io 5, Europa 11, Ganymede 19, Callisto 17, Mimas 41,
    Enceladus 6, ...
2027-01-01  against JPL's DE421, km off, from the Sun: Mercury 0, Venus 0, Earth 0, Mars 0, Jupiter 0, Saturn 0,
    Uranus 0, Neptune 0, Pluto 0;
            Mercury's perihelion turned by Einstein: 42.981" a century, over 3 orbits (theory, ... an orbit: 42.981")
2031-01-01  against Horizons, km off: Phobos 4771, ..., Mimas 2794, ..., Titania 4502, Oberon 937, Triton 108, Charon
    209, Ceres 2010, ...
2036-01-01  against JPL's DE421, km off, from the Sun: Mercury 1, Venus 0, Earth 1, Mars 1, Jupiter 5, Saturn 1,
    Uranus 0, Neptune 0, Pluto 0;
            the Moon, from the Earth: 39 km
            a Mercury without Einstein's correction: 1813 km off
            energy steady to -1.3e-09, angular momentum to -2.5e-10
```

Before the fix, the rate-100 run gave (2026-02-01) Io 945, Mimas 4688, Enceladus 2014 and (2036) Mercury 1220, Venus 11, Moon 122 km; now Io 5, Mimas 41, Enceladus 6 and Mercury 1, Venus 0, Moon 39, as at rate 60.

Checks: hall vs standalone (`SS_RATE=100 SS_YEARS=3 bash ../Cabinet/report/cmp2.sh solar SolarSystem solar-system.lua 700 ss`): IDENTICAL (101 lines). Start pause: HOLD_S=0 for 600 frames is a prefix of the default pause for 630 frames (SS_RATE=100, SS_YEARS=3).

Known departures and honest caveats

- **Fixed 15 Oct 2026: check sampling at high SS_RATE.** The checks used to fire on the first frame after their date. At 100 d/s (1.67 d a frame) the main bodies were then read off the Hermite cubic outside its step (extrapolated up to a day: Mercury 1220 km, the Moon 122 km in 2036) and the moons carried back from their last Wisdom–Holman step by a bare Kepler orbit (Mimas 4,688 km, Io 945 km on 2026-02-01). Now each frame first stops the integration at every check date it passes and runs the check there (`checks(tn)` in solar-system.lua), then goes on to the frame's time; the displayed positions are unchanged. A second rate dependence was removed in `E.advance` (solar-engine.lua): the moons' steps inside the current day are finished before the next day is stepped, so each kick takes the Sun from the same day's interpolant whatever the frame pattern. With both, the check lines agree to the km at rates 5, 60 and 100.
- **Fixed 15 Oct 2026: README sample output.** The 2027 block (Laplace 180.08, spread 0.18, with its 20-day-means line), the 2036 planets (Mercury 1, Earth 1, Mars 1, Jupiter 5, Saturn 1), the Horizons 2026-02-01 / 2027 / 2031 lines, "within 5 km", Phobos 950 km/yr, "20–210 km" and the Mimas wording now come from the fresh run. The README now says the no-Einstein distance is not monotonic year to year (49, 190, 494, 733, 624, 404, 286, 531, 1107, 1813 km for 2027–36): the orbit differs, and the distance depends on where Mercury is in it each Jan 1.
- Still frame-sampled (by design, not checks against theory values): the Laplace 20-day means and yearly means, the Moon's face extremes, Mercury's perihelion longitudes and Neptune–Pluto closest approach are taken once per frame, so they change slightly with SS_RATE (20-day means by ~0.01°, the yearly means agreed to 0.01° in these runs, perihelion longitudes by up to 0.2°; at 1.67 d/frame the perihelion is located to within a frame).
- Between Wisdom–Holman steps a moon's position is its Kepler drift from the last step (no kick); at the check dates this covers under one step (≤ 0.4 d for Pluto, 0.059 d for Saturn) and is the same at every rate.
- Physics left out (README):
  - The Moon's 39 km is the Moon's figure, tides and Earth-pole precession; the J2 pole is fixed at ICRF z.
  - Phobos (~950 km/yr) is Mars's higher-order gravity field. Adding C22/S22 cut it from 3,200 km/yr, and Deimos from 330 to 50.
  - Titania and Miranda (900 and 400 km/yr) are not explained by J2 or older masses.

  - Mimas: 575 km after a year, 2,794 after five, growing with a month-to-month wobble of 100–200 km (the README's old "500–2,400, rising and falling" is not what the monthly series shows; README updated). The README attributes it to its 2:1 resonance with Tethys (70-year swing); that cause is not tested here.
  - Asteroids (Ceres 2,000 km in 5 yr, ~1″) lack Horizons' 11 extra perturbing asteroids and DE441's barycentre.
- DE421 masses drive the main integration, but DE440 masses drive the moon systems. This mix is deliberate.
- The 1-day Bulirsch–Stoer step has no error control. Pluto–Neptune "> 17 AU" is a known fact, not shown in the 10 years run.
- Rotation models keep only the main periodic terms. Several moons' meridians (Titan, Mimas, Enceladus, Tethys, Dione, Iapetus) and Pluto/Charon are not cross-checked against a second source (README).
- Surface maps are drawn, not photographs. Variable features such as the Red Spot and sunspots are illustrative.
- The README's Lainey L1.2 alternative start (1.5° swing, period 2039 d vs 2070 d theory) was not rerun, since it needs the Horizons files removed.

References (as cited in the code and README)

- JPL DE421 ephemeris (via skyfield-data/jplephem); DE440/DE441 (masses, Horizons' asteroid barycentre); JPL Horizons; JPL satellite solutions.
- Archinal et al. 2018, "Report of the IAU WGCCRE: 2015", Celest. Mech. Dyn. Astr. 130:22; the 2009 WGCCRE report, CMDA 109:101; NAIF pck00008; the hapsira library.
- Wisdom & Holman mapping; Bulirsch–Stoer / Gragg modified midpoint; Lainey's L1.2 theory of the Galilean moons (via astronomy-engine).
- IERS Earth orientation via Skyfield; Natural Earth coastlines.